

Deep Learning for Computer Vision

114-1/Fall 2025

<https://cool.ntu.edu.tw/courses/53063> (NTU COOL)

<http://vllab.ee.ntu.edu.tw/dlcv.html> (Public website)

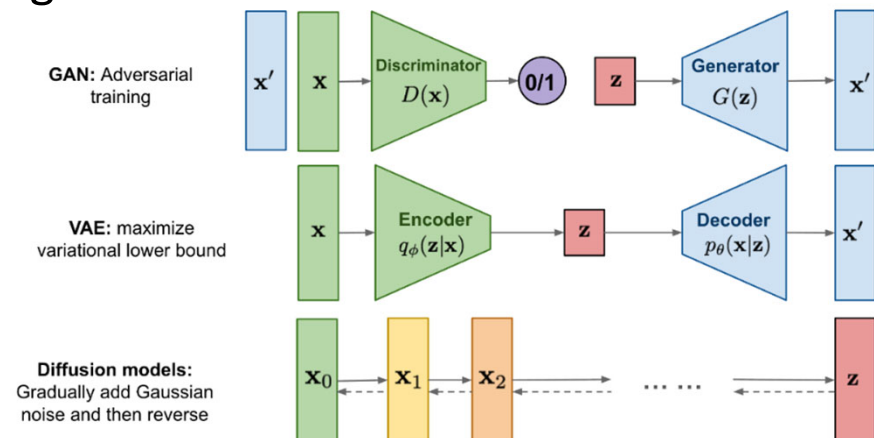
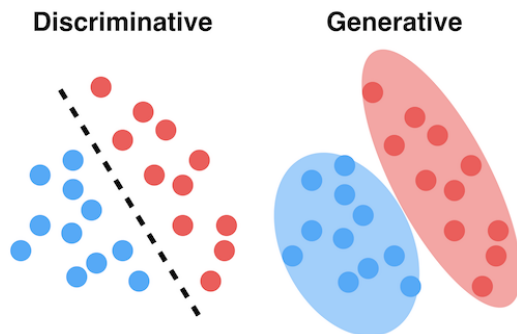
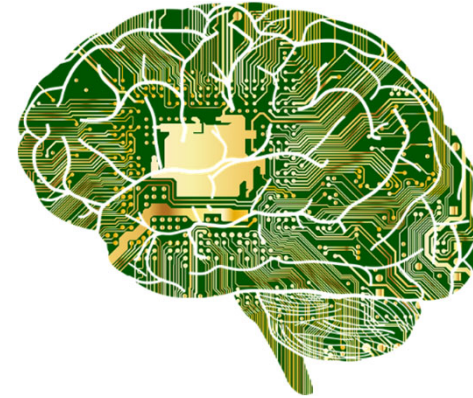
Yu-Chiang Frank Wang 王鈺強, Professor

Dept. Electrical Engineering, National Taiwan University

2025/09/24

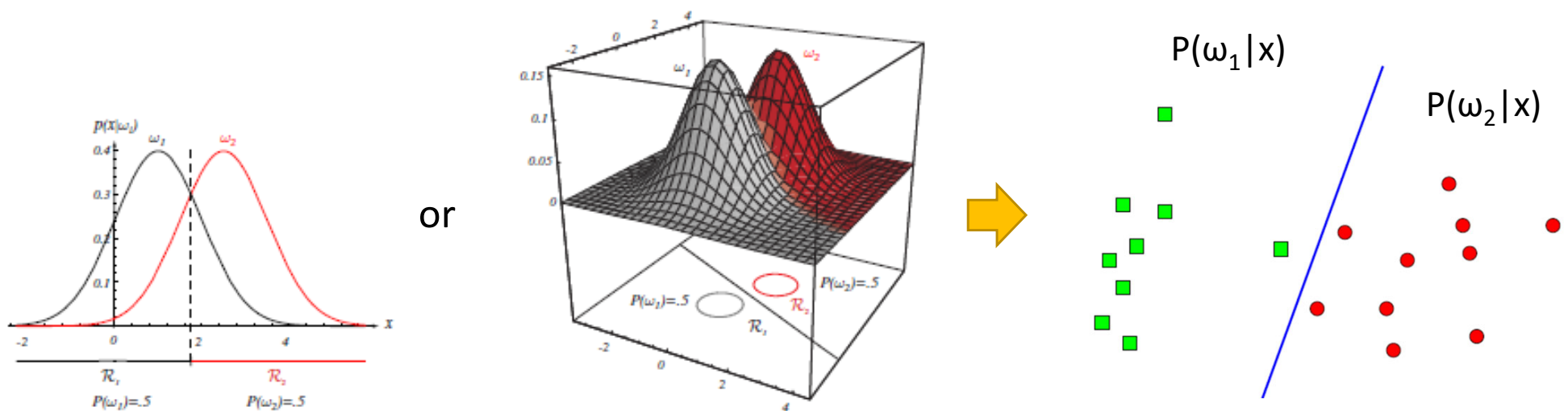
What's to Be Covered Today...

- Generative Models
 - Autoencoder
 - Variational Autoencoder
 - Diffusion Model
 - Generative Adversarial Network (next lecture)
- Updates on Final Projects
 - Company visits (capacity limitation)
 - Sponsored by **PicCollage**
 - Top teams w/ prizes & company sharing



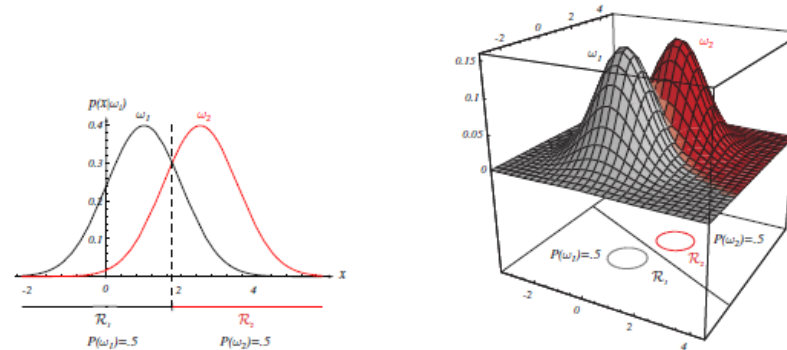
Discriminative vs. Generative Models

- Discriminative Models
 - Model posteriors $P(\omega|x)$ from likelihoods $P(x|\omega)$ where x is the input data, and ω indicates the class of interest
 - Example (posterior)



Discriminative vs. Generative Models (cont'd)

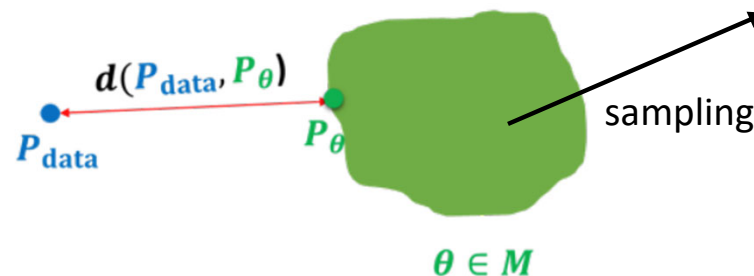
- Generative Models
 - Model likelihoods $P(x|\omega)$ with priors $P(\omega)$ (i.e., modeling $P(x|\omega) P(\omega)$) where x is the input data, and ω indicates the class of interest



- Example



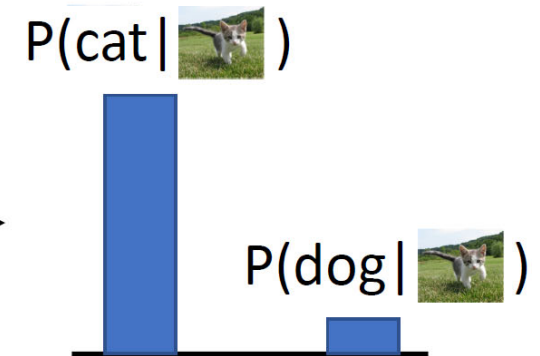
$$x_i \sim p_{data}$$



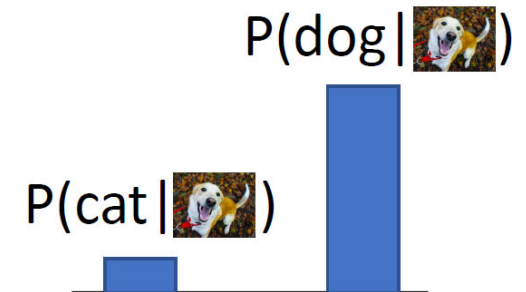
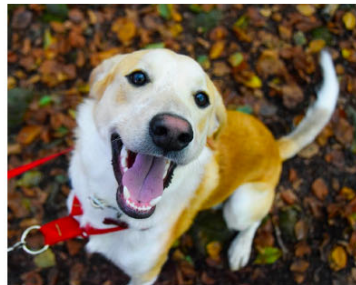
$$\hat{x}_i \sim p_{\theta}$$

Discriminative vs. Generative Models (cont'd)

Discriminative Model:
Learn a probability
distribution $p(y|x)$



Generative Model:
Learn a probability
distribution $p(x)$



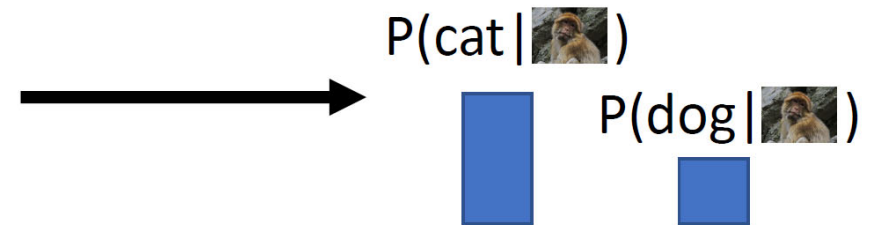
Conditional Generative Model: Learn $p(x|y)$

Discriminative model: the possible labels for each input "compete" for probability mass.
But no competition between **images**

Discriminative vs. Generative Models (cont'd)

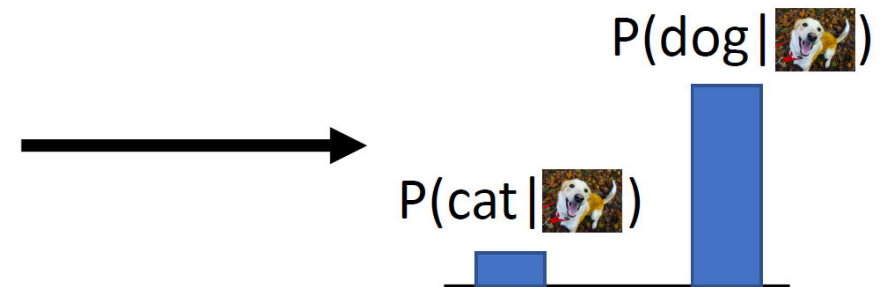
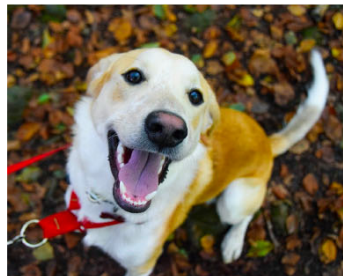
Discriminative Model:

Learn a probability distribution $p(y|x)$



Generative Model:

Learn a probability distribution $p(x)$



Conditional Generative Model: Learn $p(x|y)$

Discriminative model: No way for the model to handle unreasonable inputs; it must give label distributions for all images

Discriminative vs. Generative Models (cont'd)

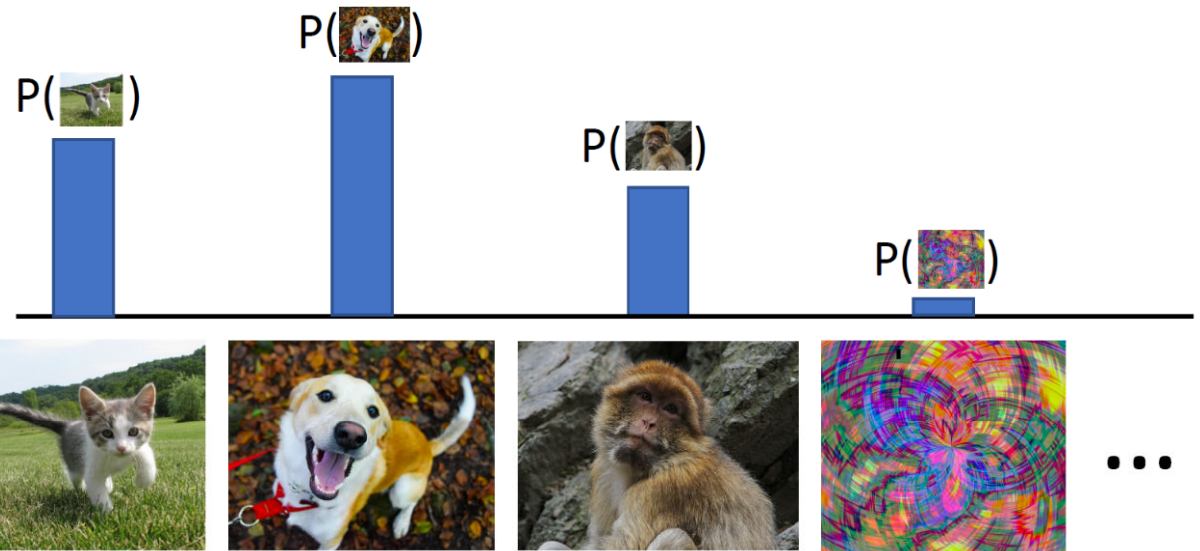
Discriminative Model:

Learn a probability distribution $p(y|x)$

Generative Model:

Learn a probability distribution $p(x)$

Conditional Generative Model: Learn $p(x|y)$



Generative model: All possible images compete with each other for probability mass

Model can “reject” unreasonable inputs by assigning them small values

Discriminative vs. Generative Models (cont'd)

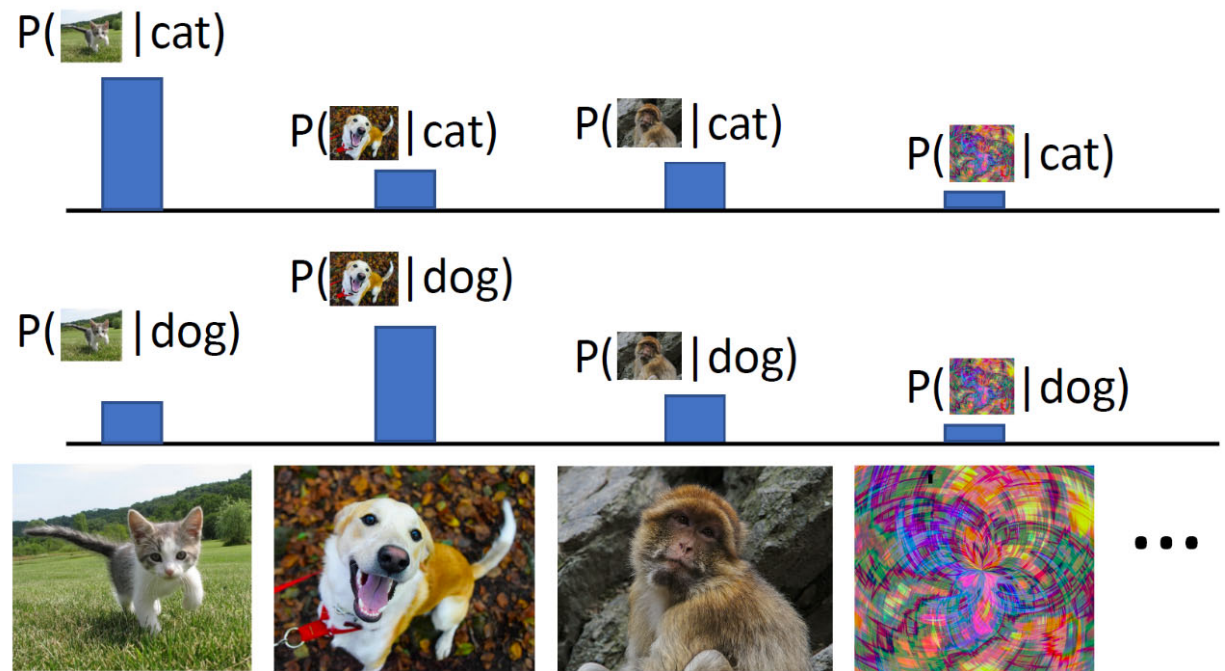
Discriminative Model:

Learn a probability distribution $p(y|x)$

Generative Model:

Learn a probability distribution $p(x)$

Conditional Generative Model: Learn $p(x|y)$



Conditional Generative Model: Each possible label induces a competition among all images

Discriminative vs. Generative Models (cont'd)

Discriminative Model:

Learn a probability distribution $p(y|x)$

Generative Model:

Learn a probability distribution $p(x)$

Conditional Generative Model: Learn $p(x|y)$

Recall **Bayes' Rule**:

$$\underbrace{P(x | y)}_{\text{Conditional Generative Model}} = \frac{\underbrace{P(y | x)}_{\text{Discriminative Model}} \underbrace{P(x)}_{\text{(Unconditional) Generative Model}}}{\underbrace{P(y)}_{\text{Prior over labels}}}$$

We can build a conditional generative model from other components!

Additional Remarks

- Discriminative Models
 - Goal: Learn a (posterior) probability distribution $p(y|x)$
 - Task: Assign labels to each instance x (e.g., classification, regression, etc.)
 - Supervised learning
- Generative Models
 - Goal: Learn a probability distribution $p(x)$
 - Task: Data representation, generation, detect outliers, etc.
 - (Mostly) unsupervised learning

What Have Been Done Using Deep Generative Models?

- 5+ years of progress on synthesizing face images



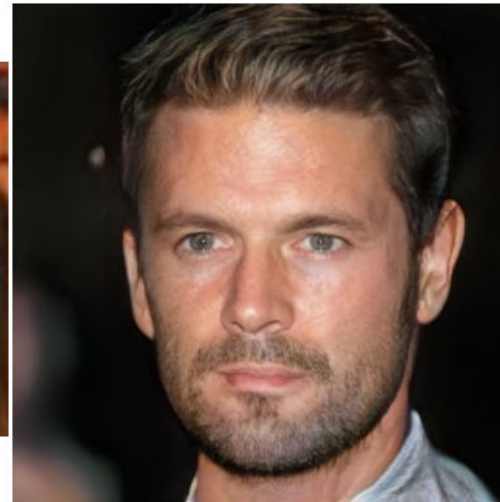
2014



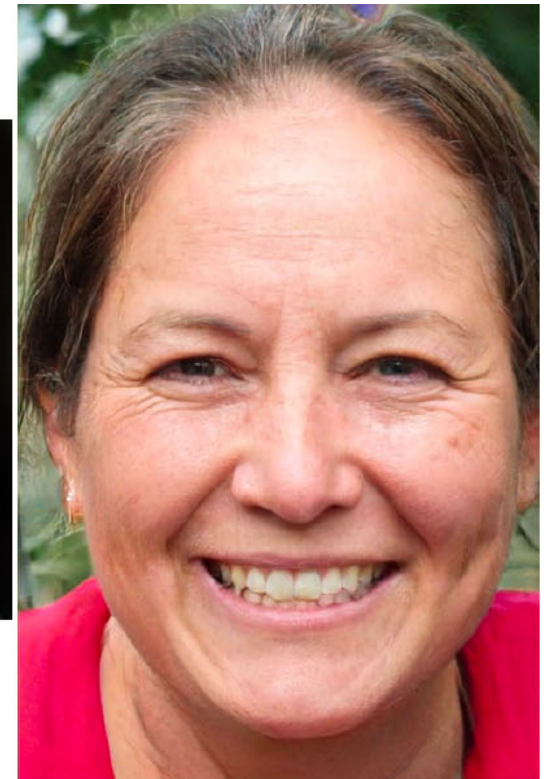
2015



2016



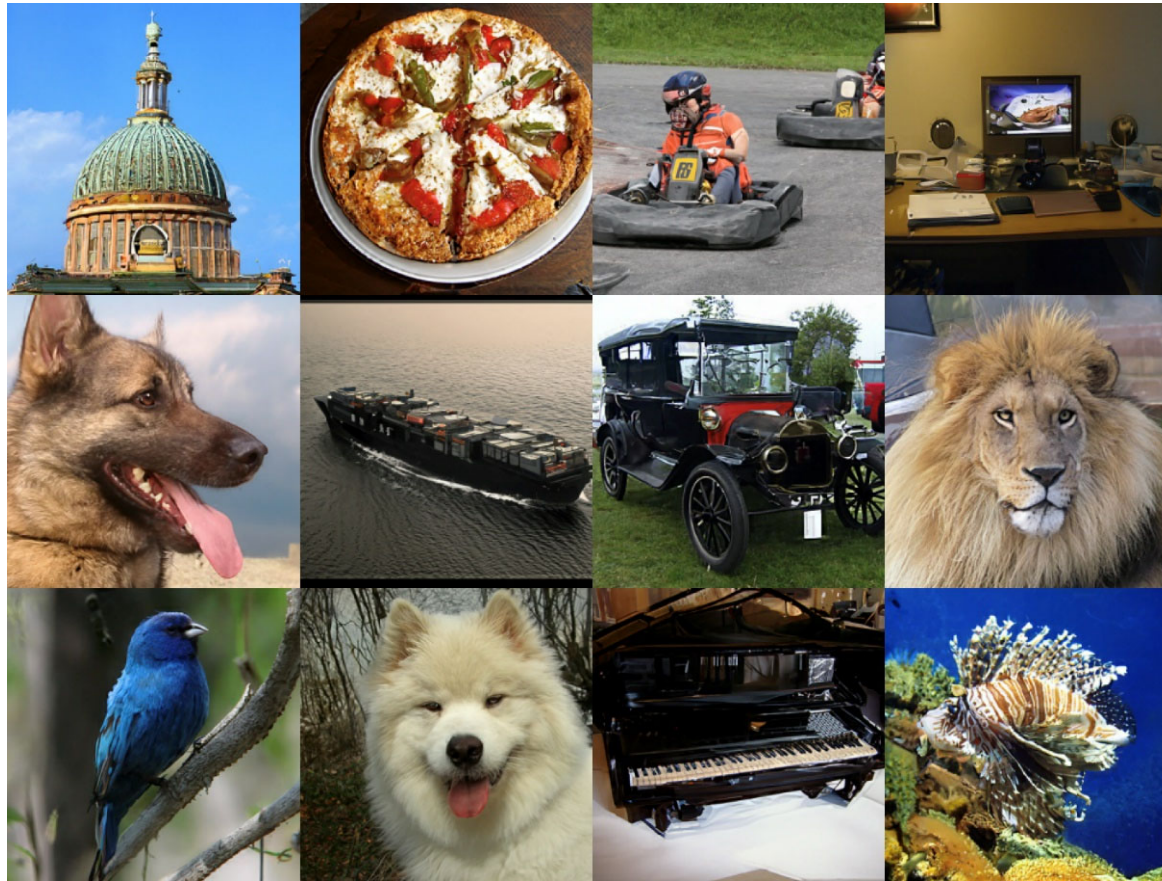
2017



2018

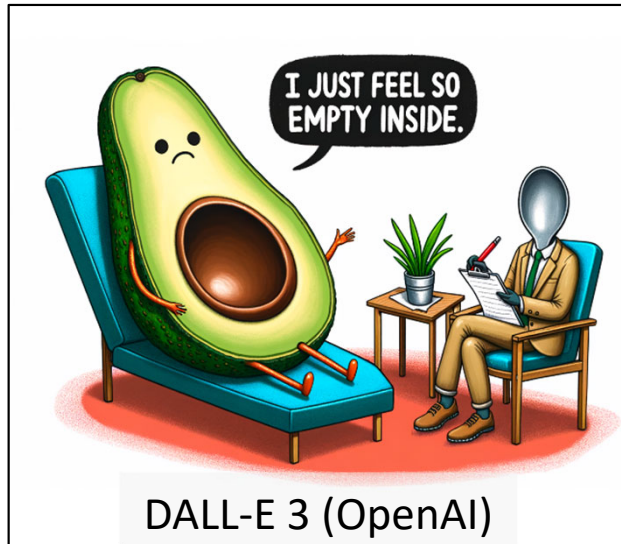
What Have Been Done Using Deep Generative Models?

- Progress on synthesizing images (ImageNet)



Super-Resolution via Repeated Refinements (SR3) by
Class Diffusion Models (Google, 2021)

What Have Been Done Using Deep Generative Models?



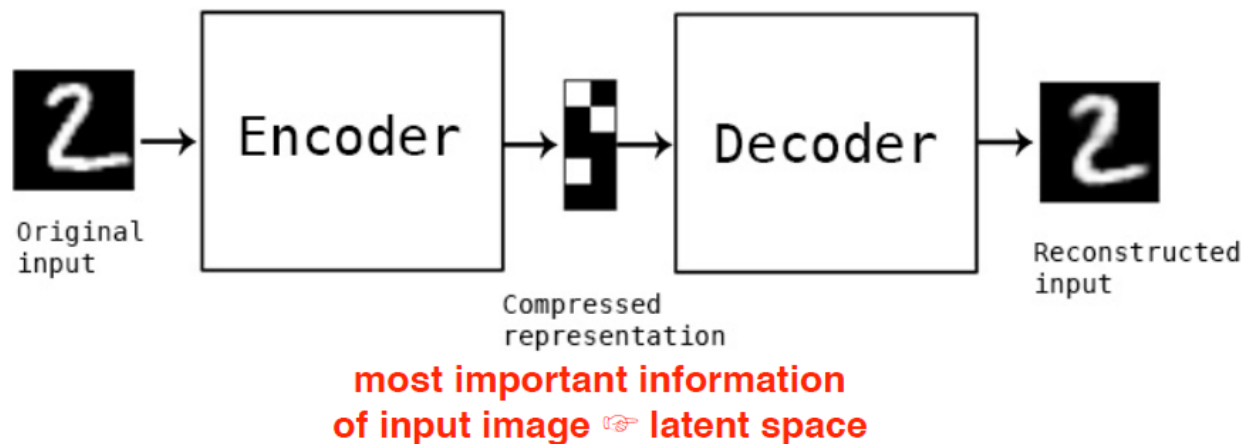
GPT-5 by OpenAI

GPT-5

How GenAI Works?

Let's Start from Autoencoder...

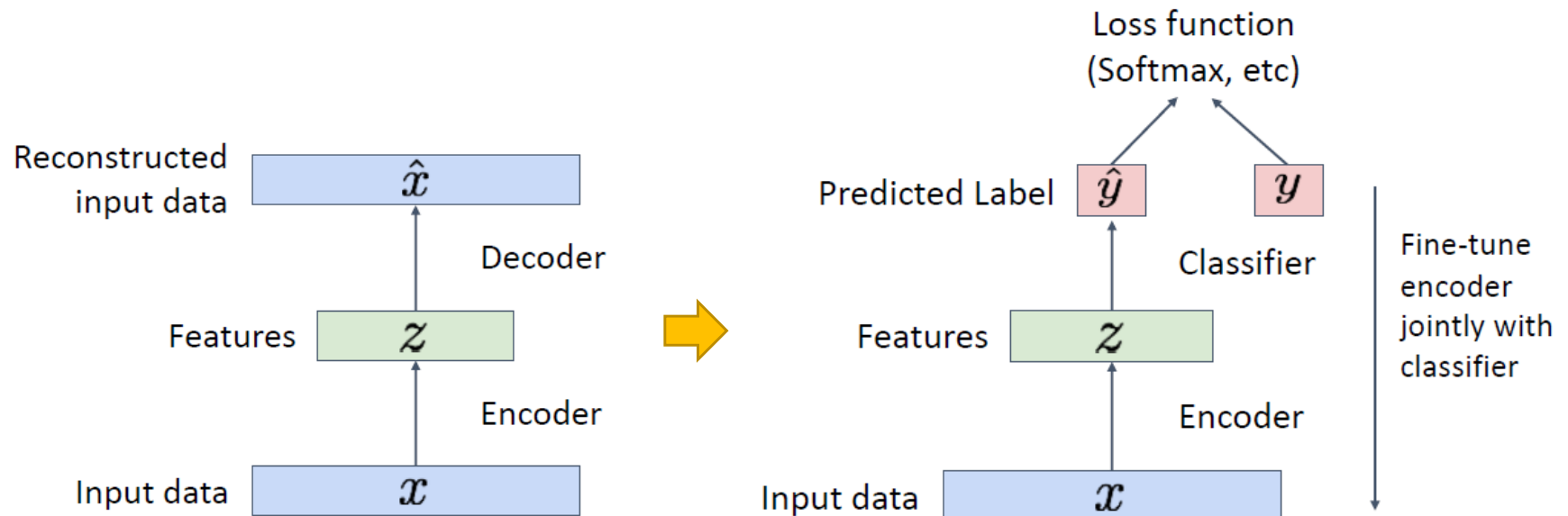
- Autoencoder (AE)
 - Autoencoding = encoding itself with recovery purposes
 - In other words, encode/decode data with reconstruction guarantees
 - **Latent variables/feature** as deep representation
 - Example objective/loss function at output:
 - L2 norm between input and output, i.e.,



- Since AE is trained in a unsupervised fashion, any **application** comes to your mind?

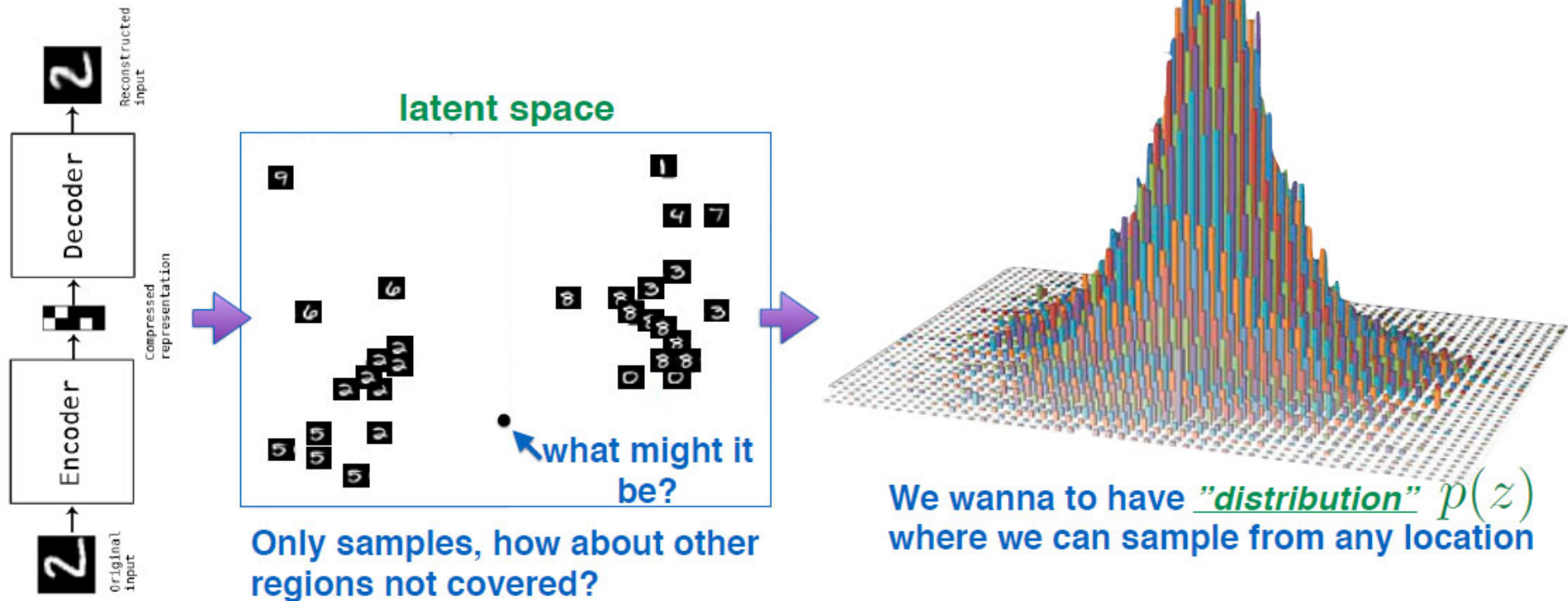
AE for Learning Latent Variables/Representations

- How AE benefits for downstream tasks?
 - Train AE with reconstruction guarantees
 - Keep encoder (and the derived features) for downstream tasks (e.g., classification)
 - Thus, a trained encoder can be applied to initialize a supervised model
 - This can be viewed as the **pre-training vs. finetuning stages** (discussed few weeks later)



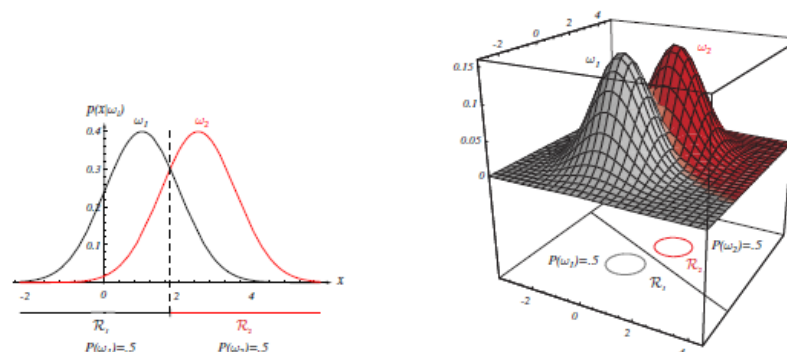
Take a Deep Look to Discover Latent Variables/Representations (cont'd)

- Limitation/concern of autoencoder?

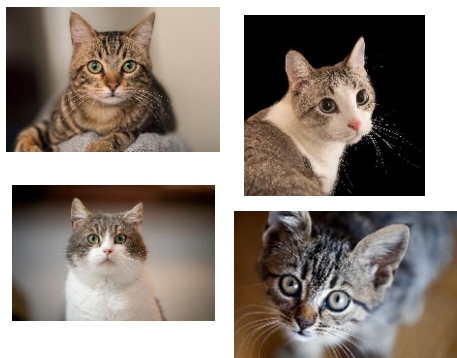


Recall: Generative Models

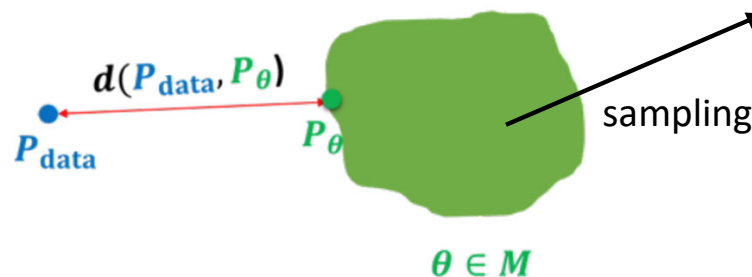
- Generative Models
 - Model likelihoods $P(x|\omega)$ with priors $P(\omega)$ (i.e., modeling $P(x|\omega) P(\omega)$) where x is the input data, and ω indicates the class of interest



- Take $P(x)$ for an example (i.e., unconditional generation)



$$x_i \sim p_{data}$$

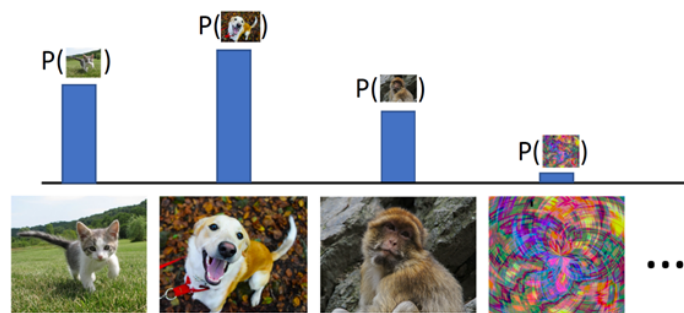


$$\hat{x}_i \sim p_{\theta}$$

Variational Autoencoder

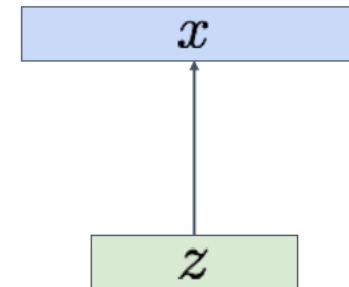
- Probabilistic Spin on AE

- Learn latent feature $\mathbf{z}$ from raw input data $\mathbf{x}_i \sim p_{data}$
- Sample from the latent space (via) to generate data
- For simplicity, assume simple prior $p(\mathbf{z})$ (e.g., Gaussian)

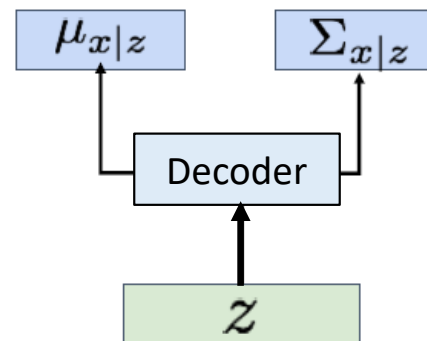
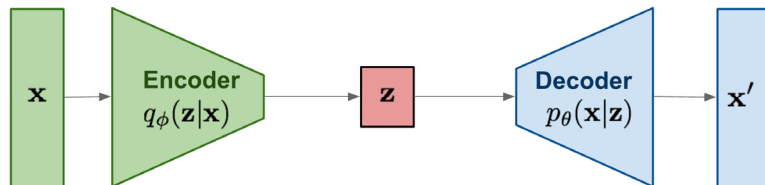


Sample from
conditional
 $p_{\theta}(\mathbf{x} | \mathbf{z}^{(i)})$

Sample $\mathbf{z}$
from prior
 $p_{\theta}(\mathbf{z})$



- Learn $p(\mathbf{x}|\mathbf{z})$ via a NN as a (probabilistic) **decoder**

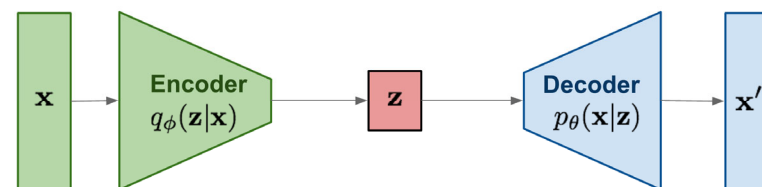


Decoder inputs $\mathbf{z}$, outputs mean $\mu_{\mathbf{x}|\mathbf{z}}$
and (diagonal) covariance $\Sigma_{\mathbf{x}|\mathbf{z}}$



Sample $\mathbf{x}$ from Gaussian with mean
 $\mu_{\mathbf{x}|\mathbf{z}}$ and (diagonal) covariance $\Sigma_{\mathbf{x}|\mathbf{z}}$

Variational Autoencoder (cont'd)



- Remarks

- Training objective: maximum likelihood of data $p(x)$
- Note that we can't possibly observe all latents z & need to marginalize it:

$$p_{\theta}(x) = \int p_{\theta}(x, z) dz = \int p_{\theta}(x|z) p_{\theta}(z) dz$$

- We can compute $p_{\theta}(x|z)$ with the decoder module, and we assume Gaussian prior for $p_{\theta}(z)$.

Still, can't integrate over all possible z ...

- What else can we do? Recall that we have Bayes' rule:

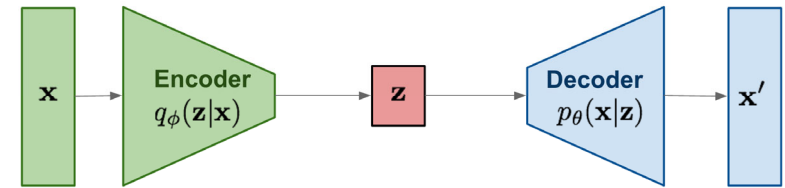
$$p_{\theta}(x) = \frac{p_{\theta}(x | z) p_{\theta}(z)}{p_{\theta}(z | x)}$$

We still need $p_{\theta}(z | x)$, which is not explicitly known. Instead, we train the encoder module to learn

$$q_{\phi}(z | x) \approx p_{\theta}(z | x)$$

Will see why in a minute...

Training Variational Autoencoder



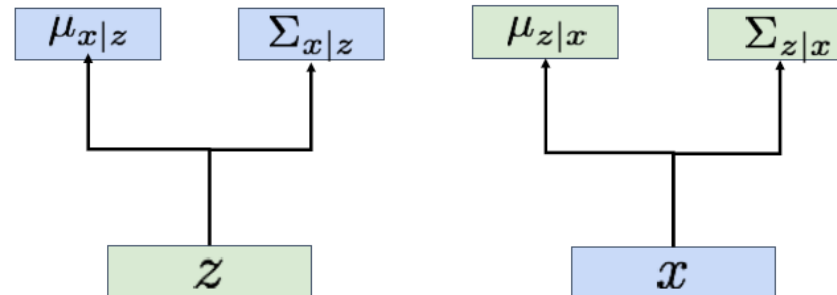
- Recall that, we aim to maximize

$$p_{\theta}(x) = \int p_{\theta}(x, z) dz = \int p_{\theta}(x|z)p_{\theta}(z) dz$$

we have...

Decoder network	inputs	Encoder network	inputs
latent code z , gives		data x , gives distribution	
distribution over data x		over latent codes z	

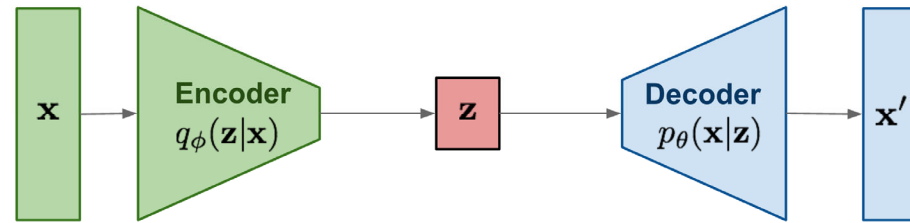
$$p_{\theta}(x | z) = N(\mu_{x|z}, \Sigma_{x|z}) \quad q_{\phi}(z | x) = N(\mu_{z|x}, \Sigma_{z|x})$$



- Via training, if we ensure $q_{\phi}(z | x) \approx p_{\theta}(z | x)$

then we have
$$p_{\theta}(x) = \frac{p_{\theta}(x | z)p_{\theta}(z)}{p_{\theta}(z | x)} \approx \frac{p_{\theta}(x | z)p(z)}{q_{\phi}(z | x)}$$

Training VAE



$$\log p_{\theta}(x) = \log \frac{p_{\theta}(x | z)p(z)}{p_{\theta}(z | x)} = \log \frac{p_{\theta}(x|z)p(z)q_{\phi}(z|x)}{p_{\theta}(z|x)q_{\phi}(z|x)}$$

$$= E_z [\log p_{\theta}(x|z)] - E_z \left[\log \frac{q_{\phi}(z|x)}{p(z)} \right] + E_z \left[\log \frac{q_{\phi}(z|x)}{p_{\theta}(z|x)} \right]$$

$$= E_{z \sim q_{\phi}(z|x)} [\log p_{\theta}(x|z)] - D_{KL} (q_{\phi}(z|x), p(z)) + D_{KL} (q_{\phi}(z|x), p_{\theta}(z|x))$$

Data reconstruction

KL divergence

between sample
distribution from the
encoder and the prior

KL divergence between

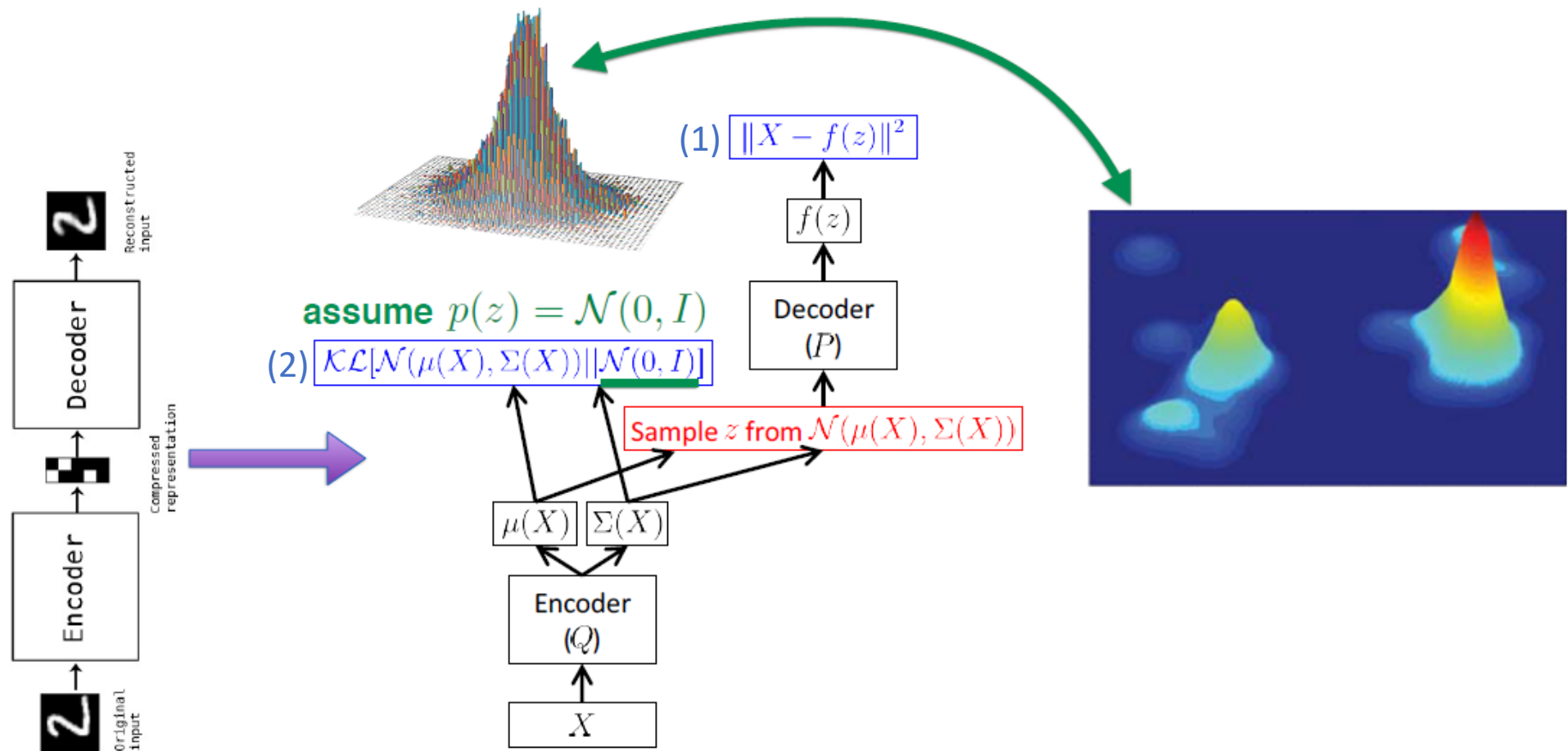
sample distribution
from the encoder and
the posterior of data

$$\Rightarrow \log p_{\theta}(x) \geq \underbrace{E_{z \sim q_{\phi}(z|x)} [\log p_{\theta}(x|z)]}_{(1)} - \underbrace{D_{KL} (q_{\phi}(z|x), p(z))}_{(2)}$$

which is the **variational lower bound**, aka **evidence lower bound (ELBO)**,
on the data likelihood $p_{\theta}(x)$

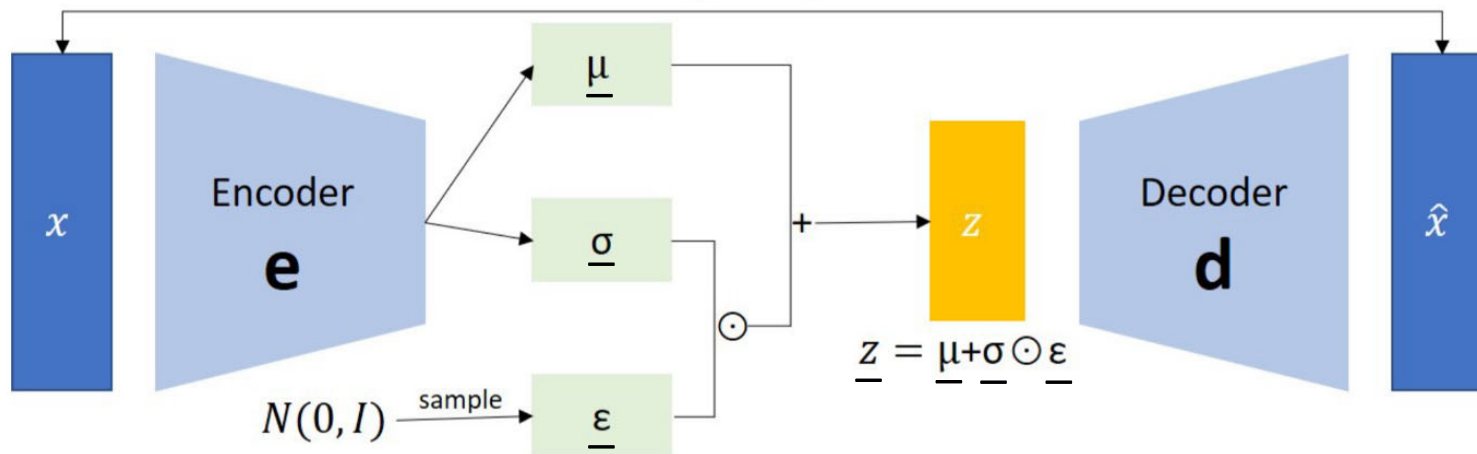
Overview of VAE Training Objectives

Now is a “*distribution*”, we can assume it to be a distribution easy to sample from, e.g. Gaussian

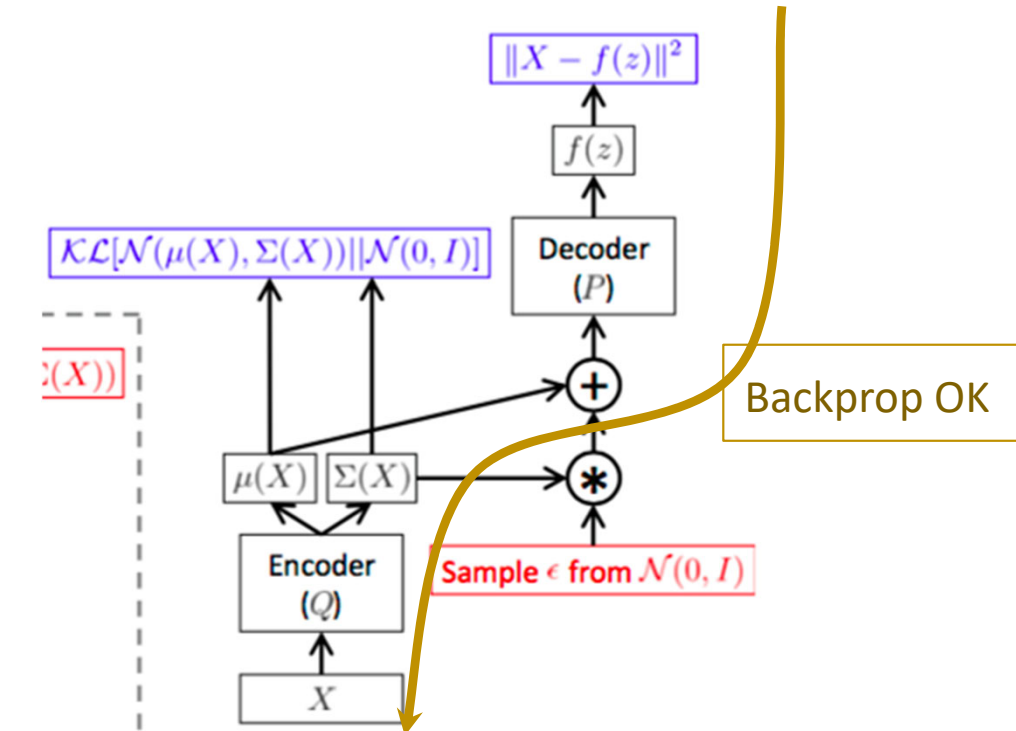


Reparameterization Trick for VAE

- Remarks
 - Given x , **sample** z from latent distribution (described by output parameters of encoder)
 - However, this creates a bottleneck since **backpropagation (BP) cannot flow through**
 - Alternatively, we apply $z = \mu + \sigma \odot \varepsilon$ (ε simply generated by **Normal distribution**).
 - This enables BP gradients in encoder through μ and σ , while maintaining stochasticity via ε (for generative synthesis purposes).



Implementation of VAE Training



Initialize parameters of encoder and decoder
Repeat:

Get mini-batch of X
 $\mathbf{mu_X}, \mathbf{var_X} = \text{encoder}(X)$

$\epsilon = \text{sampling from Normal}(0, I)$

$z = \mathbf{mu_X} + \epsilon * \mathbf{var_X}$

$X' = \text{decoder}(z)$

$\text{recon_loss} = \text{MSE}(X, X')$

$\text{latent_loss} =$

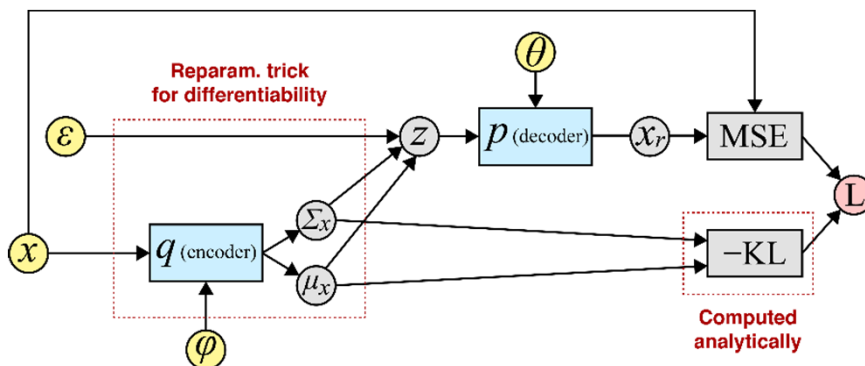
$\text{KLD}(\text{Normal}(\mathbf{mu_X}, \mathbf{var_X}) || \text{Normal}(0, I))$

$\text{all_loss} = \text{recon_loss} + \text{latent_loss}$

$\text{all_loss.backward}()$

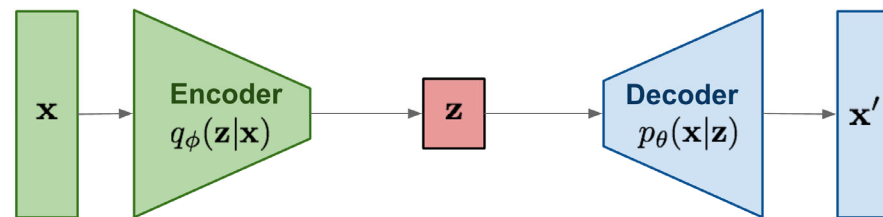
Until: parameters of **encoder** & **decoder** converge

Return parameters of **encoder** and **decoder**



First sample noise ϵ from $\text{Normal}(0, I)$,
 then reparameterize z by $\mu_x + \epsilon * \text{var}_x$,
 (equivalently sampled $\text{Normal}(\mu_x, \text{var}_x)$).
 The model is now differentiable!

Before We Move On...



$$\log p_{\theta}(x) = \log \frac{p_{\theta}(x | z)p(z)}{p_{\theta}(z | x)} = \log \frac{p_{\theta}(x|z) p(z) q_{\phi}(z|x)}{p_{\theta}(z|x) q_{\phi}(z|x)}$$

$$= E_z [\log p_{\theta}(x|z)] - E_z \left[\log \frac{q_{\phi}(z|x)}{p(z)} \right] + E_z \left[\log \frac{q_{\phi}(z|x)}{p_{\theta}(z|x)} \right]$$

$$= E_{z \sim q_{\phi}(z|x)} [\log p_{\theta}(x|z)] - D_{KL} \left(q_{\phi}(z|x), p(z) \right) + D_{KL} (q_{\phi}(z|x), p_{\theta}(z|x))$$

Data reconstruction

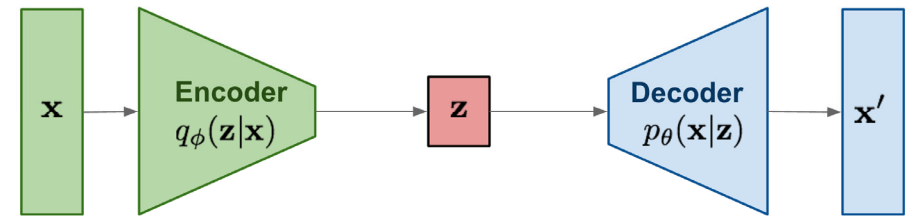
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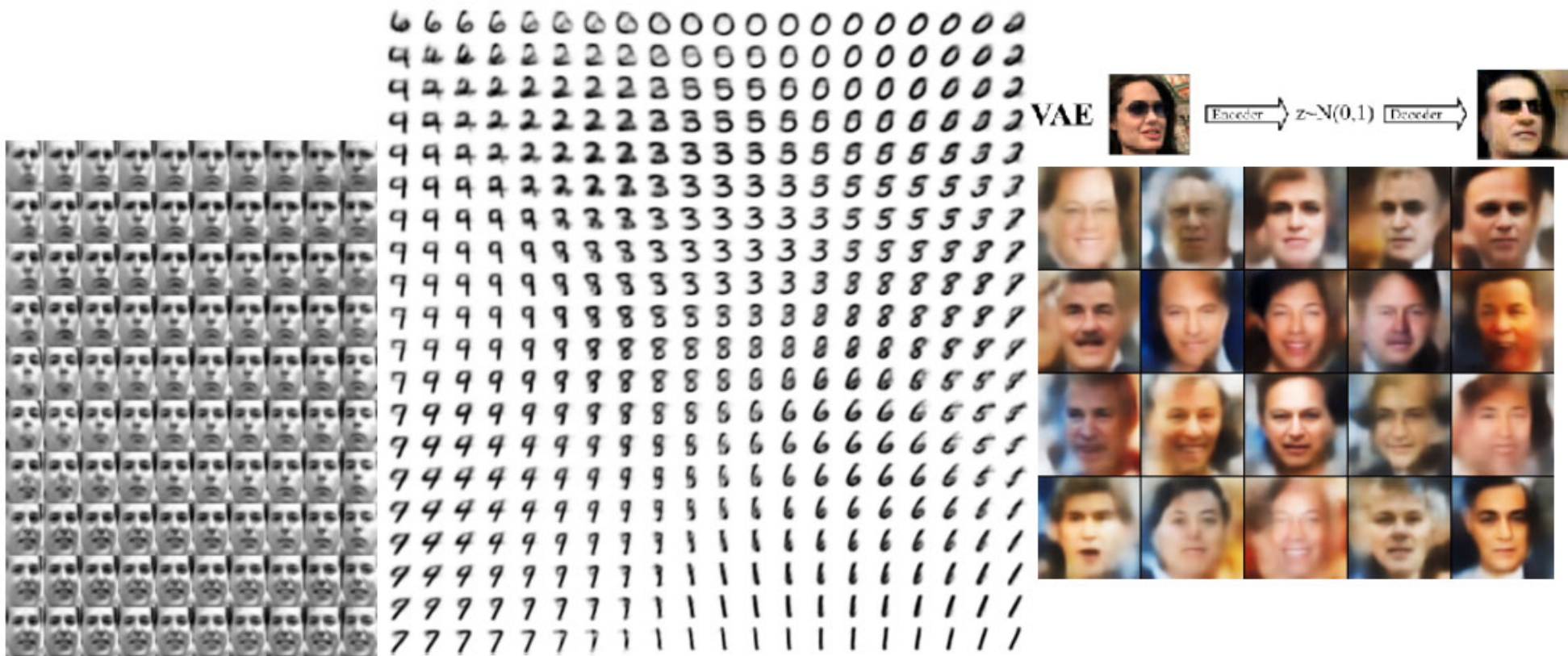
$$\Rightarrow \log p_{\theta}(x) \geq E_{z \sim q_{\phi}(z|x)} [\log p_{\theta}(x|z)] - D_{KL} \left(q_{\phi}(z|x), p(z) \right)$$

i.e., **variational lower bound**, aka **evidence lower bound (ELBO)**,
on the data likelihood $p_{\theta}(x)$

From Autoencoder to VAE (cont'd)



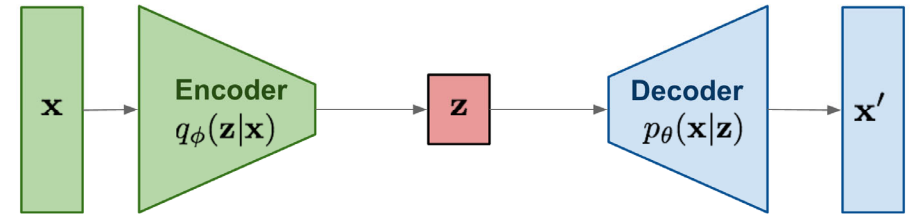
- Some empirical results



(a) Learned Frey Face manifold

(b) Learned MNIST manifold

From Autoencoder to VAE (cont'd)



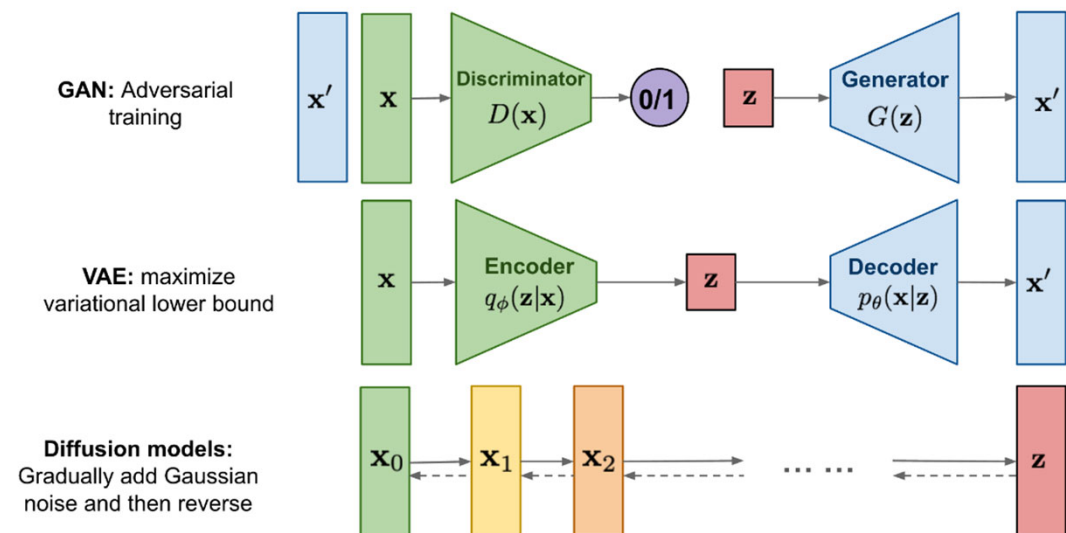
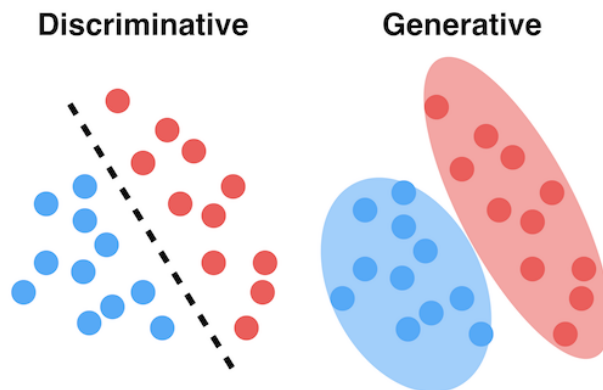
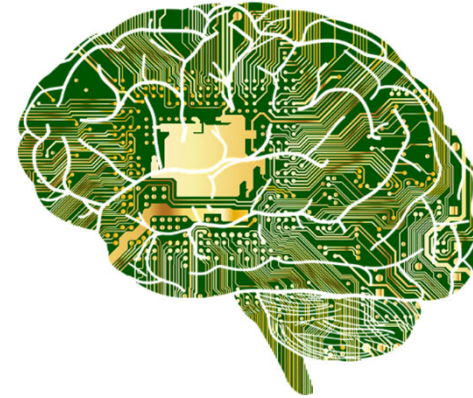
- Additional observations...

- $A' - A + B = B'$



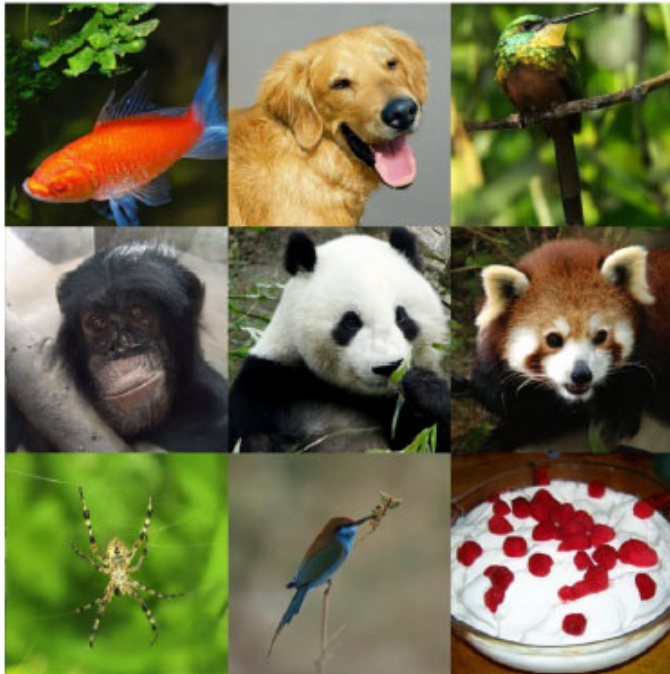
What's to Be Covered Today...

- Generative Models
 - Autoencoder
 - Variational Autoencoder
 - Diffusion Model
 - Denoising Diffusion Probabilistic Model (DDPM)
 - Latent Diffusion Model (LDM)
 - Denoising Diffusion Implicit Model (DDIM)



Denoising Diffusion Models

- Emerging as powerful **visual** generative models
 - Unconditional image synthesis
 - Conditional image synthesis



Diffusion Models Beat GANs on Image Synthesis,
Dhariwai & Nochoi, OpenAI, 2021



Cascaded Diffusion Models for High Fidelity Image
Generation, Ho et al., Google, 2021

Denoising Diffusion Models

- Emerging as powerful **visual** generative models
 - Unconditional image synthesis
 - Conditional image synthesis

DALL·E 2

"a teddy bear on a skateboard in times square"



Diffusion Models Beat GANs on Image Synthesis,
Dhariwai & Nochol, OpenAI, 2021

Imagen

A group of teddy bears in suit in a corporate office celebrating the birthday of their friend. There is a pizza cake on the desk.

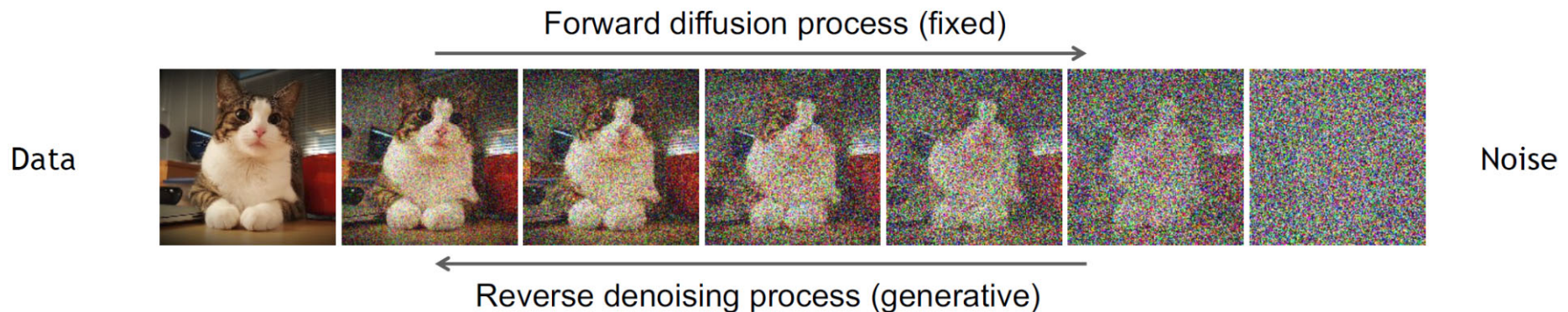
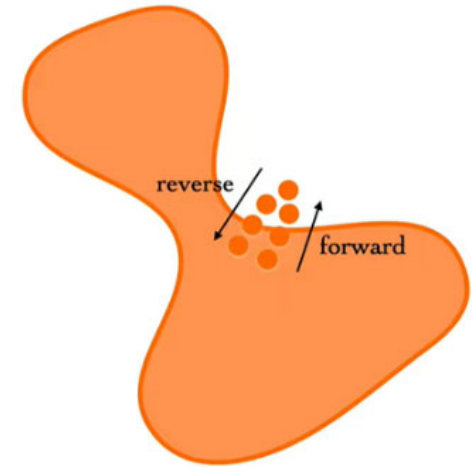


Cascaded Diffusion Models for High Fidelity Image
Generation, Ho et al., Google, 2021

Denoising Diffusion Probabilistic Models (DDPM)

Learning to generate by denoising

- 2 processes required for training:
 - **Forward diffusion process**
 - gradually add noise to input
 - **Reverse diffusion process**
 - learns to generate/restore data by denoising
 - typically implemented via a **conditional U-net**

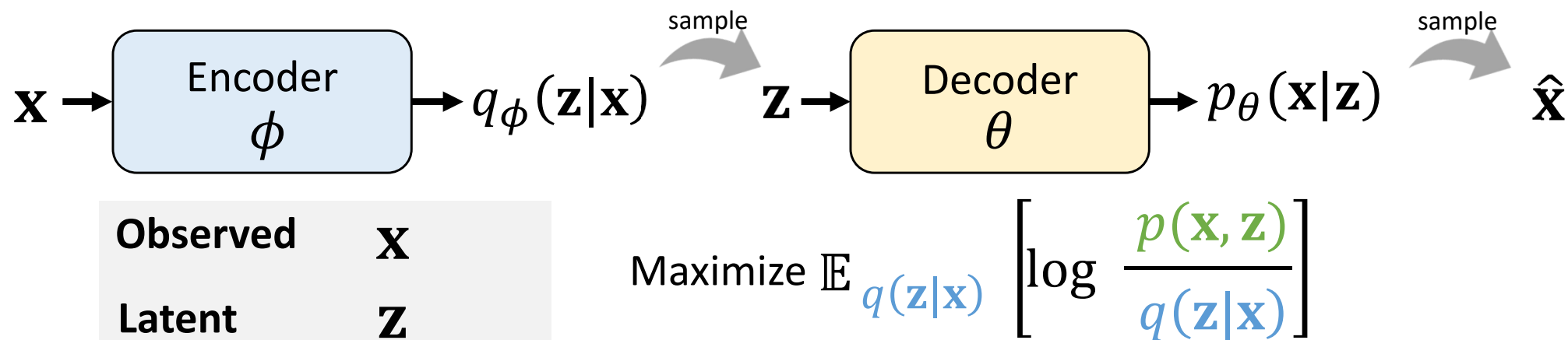


Ho et al., Denoising Diffusion Probabilistic Models, NeurIPS 2020

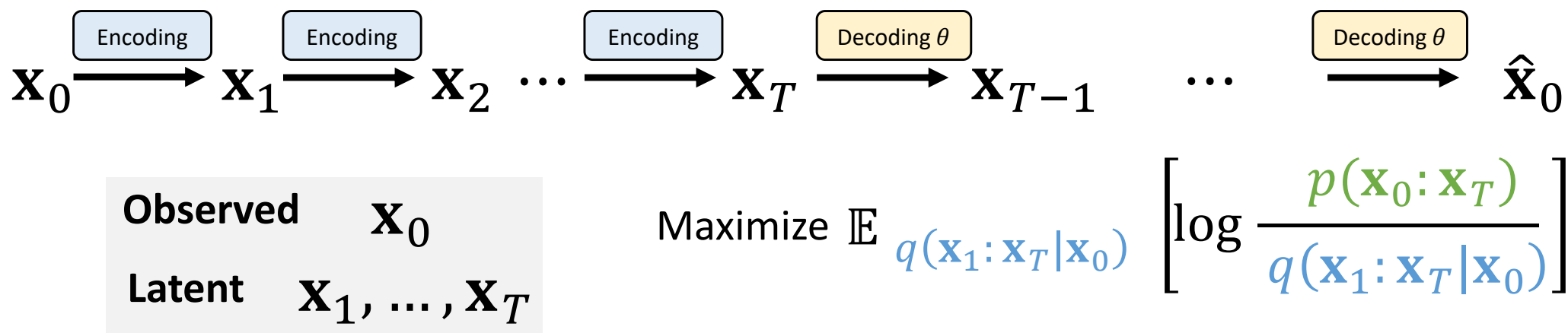
Song et al., Score-Based Generative Modeling through Stochastic Differential Equations, ICLR 2021

VAE vs. DDPM

Variational Autoencoder



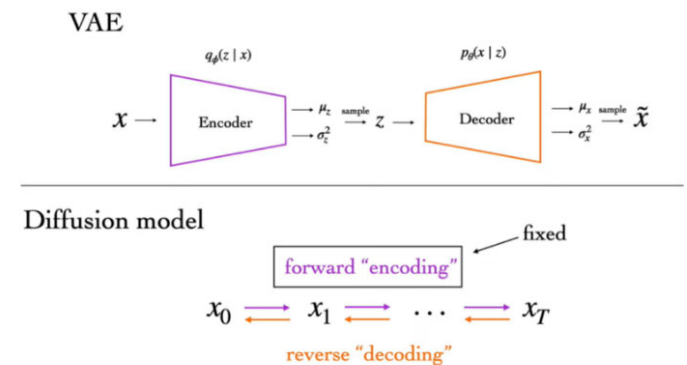
Diffusion model



Learning of Diffusion Models

- $\log p_\theta(x) \geq \text{variational lower bound}$

- Variational bound for optimizing VAE models



Variational Autoencoder $\log p_\theta(x) \geq E_{z \sim q_\phi(z|x)} [\log p_\theta(x|z)] - D_{KL}(q_\phi(z|x), p(z))$

vs. **Diffusion model** $\mathbb{E}_{q(\mathbf{x}_0)} [-\log p_\theta(\mathbf{x}_0)] \leq \mathbb{E}_{q(\mathbf{x}_0)q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[-\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \right] =: L$

- According to Ho *et al.* [NeurIPS'20], it is shown that

$$L = \mathbb{E}_q \left[\underbrace{D_{KL}(q(\mathbf{x}_T|\mathbf{x}_0) || p(\mathbf{x}_T))}_{L_T} + \sum_{t>1} \underbrace{D_{KL}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) || p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{L_{t-1}} - \underbrace{\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)}_{L_0} \right]$$

Prior matching
Denoising matching
Reconstruction

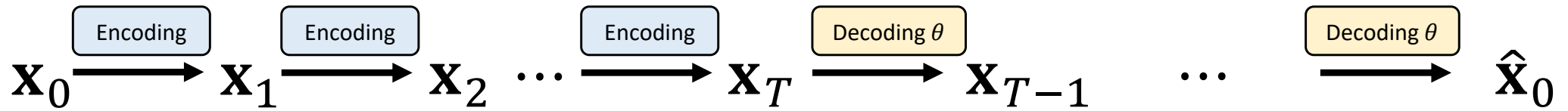


$$\mathcal{N}(x_{t-1}; \tilde{\mu}_t(x_t, x_0), \tilde{\beta}_t I)$$

$$\tilde{\mu}_t(x_t, x_0) = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t} x_0 \quad \text{fixed}$$

$$= \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \varepsilon \right)$$

Let's Take a Look at the Encoding Part...



Observed $\mathbf{x}_0$
Latent $\mathbf{x}_1, \dots, \mathbf{x}_T$

Maximize $\mathbb{E}_{q(\mathbf{x}_1:\mathbf{x}_T|\mathbf{x}_0)} \left[\log \frac{p(\mathbf{x}_0:\mathbf{x}_T)}{q(\mathbf{x}_1:\mathbf{x}_T|\mathbf{x}_0)} \right]$

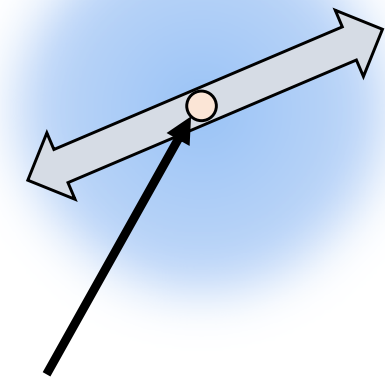
$$q(\mathbf{x}_1:\mathbf{x}_T|\mathbf{x}_0) = \prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1})$$

Encoding



$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{\alpha_t}\mathbf{x}_{t-1}, (1 - \alpha_t)\mathbf{I})$$

Encoding



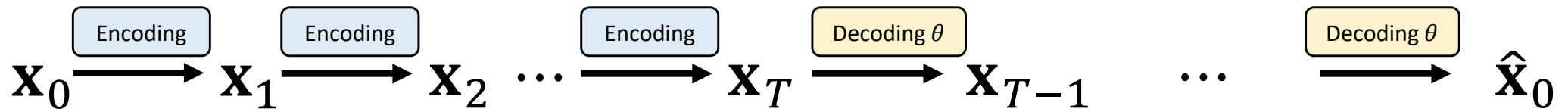
Mean

Variance

Coeff < 1...why?

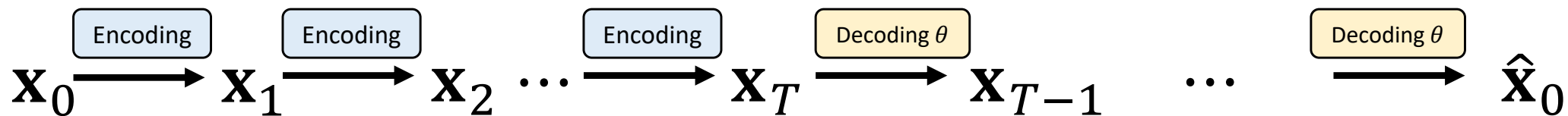
Will get back to this encoding part later. 35

Now Let's Focus on the Training Objective



$$\text{Maximize } \mathbb{E}_{q(\mathbf{x}_1:\mathbf{x}_T|\mathbf{x}_0)} \left[\log \frac{p(\mathbf{x}_0:\mathbf{x}_T)}{q(\mathbf{x}_1:\mathbf{x}_T|\mathbf{x}_0)} \right]$$

- ① $-\text{D}_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0)||p(\mathbf{x}_T))$ **Prior matching**
- ② $\mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)}[\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)]$ **Reconstruction**
- ③ $-\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)}[\text{D}_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)||p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))]$ **Denoising matching**

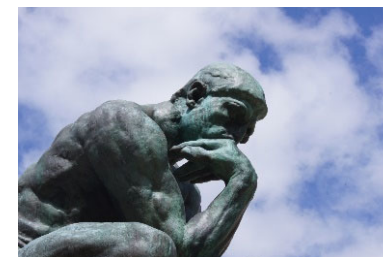


$$\mathbb{E}_{q(\mathbf{x}_1:\mathbf{x}_T|\mathbf{x}_0)} \left[\log \frac{p(\mathbf{x}_0:\mathbf{x}_T)}{q(\mathbf{x}_1:\mathbf{x}_T|\mathbf{x}_0)} \right]$$

① $-\text{D}_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0)||p(\mathbf{x}_T))$ Prior matching ✗

② $\mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} [\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)]$ Reconstruction ✗

③ $-\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [\text{D}_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)||p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))]$ Denoising matching





$$\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) || p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))]$$

$$q(\mathbf{x}_t|\mathbf{x}_0) = ?$$

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{\alpha_t}\mathbf{x}_{t-1}, (1 - \alpha_t)\mathbf{I})$$

Encoding

Mean

Variance

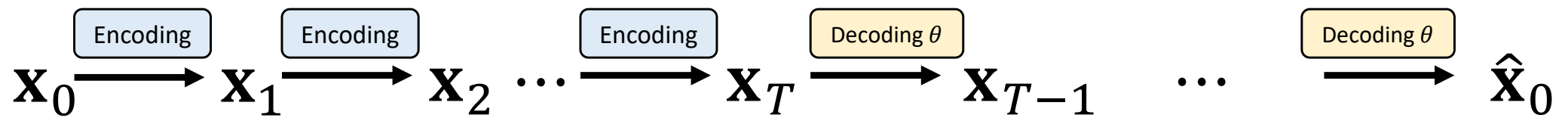
Reparameterization trick

$$\mathbf{x} \sim \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \sigma^2)$$

$$\mathbf{x} = \boldsymbol{\mu} + \sigma\boldsymbol{\epsilon}, \text{ where } \boldsymbol{\epsilon} \sim \mathcal{N}(\boldsymbol{\epsilon}; 0, \mathbf{I})$$

$$\mathcal{N}(\boldsymbol{\epsilon}; 0, \mathbf{I})$$

$$\mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \sigma^2)$$



$$\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [\text{D}_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) || p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))]$$

$$q(\mathbf{x}_t|\mathbf{x}_0) = ?$$

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \underbrace{\sqrt{\alpha_t}\mathbf{x}_{t-1}}_{\text{Mean}}, \underbrace{(1 - \alpha_t)\mathbf{I}}_{\text{Variance}})$$

Encoding

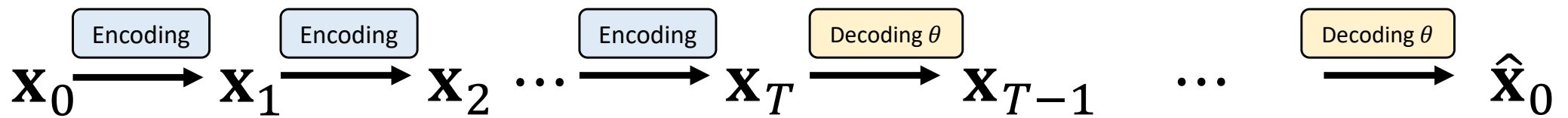
$$\mathbf{x}_1 = \sqrt{\alpha_1}\mathbf{x}_0 + \sqrt{1 - \alpha_1}\boldsymbol{\epsilon}_0$$

$$\alpha_t = 1 - \beta_t, \bar{\alpha}_t = \prod_{i=1}^t \alpha_i$$

$$\boldsymbol{\epsilon}_0 \sim \mathcal{N}(\boldsymbol{\epsilon}; 0, \mathbf{I})$$

$$\mathbf{x}_2 = \sqrt{\alpha_2} \mathbf{x}_1 + \sqrt{1 - \alpha_2}\boldsymbol{\epsilon}_1$$

$$\boldsymbol{\epsilon}_1 \sim \mathcal{N}(\boldsymbol{\epsilon}; 0, \mathbf{I})$$



$$\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) || p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))]$$

$$q(\mathbf{x}_t|\mathbf{x}_0) = ?$$

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \underbrace{\sqrt{\alpha_t}\mathbf{x}_{t-1}}_{\text{Mean}}, \underbrace{(1 - \alpha_t)\mathbf{I}}_{\text{Variance}})$$

Encoding

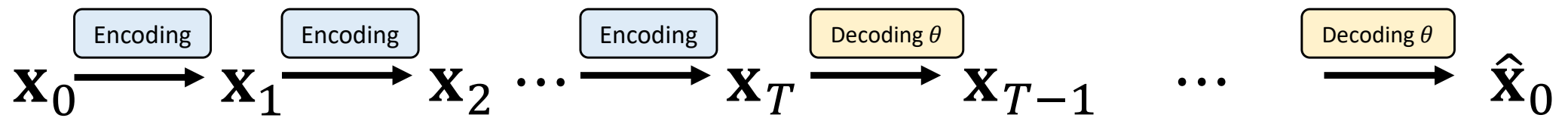
$$\mathbf{x}_1 = \sqrt{\alpha_1}\mathbf{x}_0 + \sqrt{1 - \alpha_1}\boldsymbol{\epsilon}_0$$

$$\boldsymbol{\epsilon}_0 \sim \mathcal{N}(\boldsymbol{\epsilon}; 0, \mathbf{I})$$

$$\mathbf{x}_2 = \sqrt{\alpha_2} (\sqrt{\alpha_1}\mathbf{x}_0 + \sqrt{1 - \alpha_1}\boldsymbol{\epsilon}_0) + \sqrt{1 - \alpha_2}\boldsymbol{\epsilon}_1$$

$$\boldsymbol{\epsilon}_1 \sim \mathcal{N}(\boldsymbol{\epsilon}; 0, \mathbf{I})$$

$$= \sqrt{\alpha_1\alpha_2}\mathbf{x}_0 + \sqrt{\alpha_2(1 - \alpha_1)}\boldsymbol{\epsilon}_0 + \sqrt{1 - \alpha_2}\boldsymbol{\epsilon}_1$$



$$\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [\text{D}_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) || p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))]$$

$$q(\mathbf{x}_t|\mathbf{x}_0) = ?$$

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \underbrace{\sqrt{\alpha_t}\mathbf{x}_{t-1}}_{\text{Mean}}, \underbrace{(1 - \alpha_t)\mathbf{I}}_{\text{Variance}})$$

Encoding

$$\mathbf{x}_1 = \sqrt{\alpha_1}\mathbf{x}_0 + \sqrt{1 - \alpha_1}\boldsymbol{\epsilon}_0$$

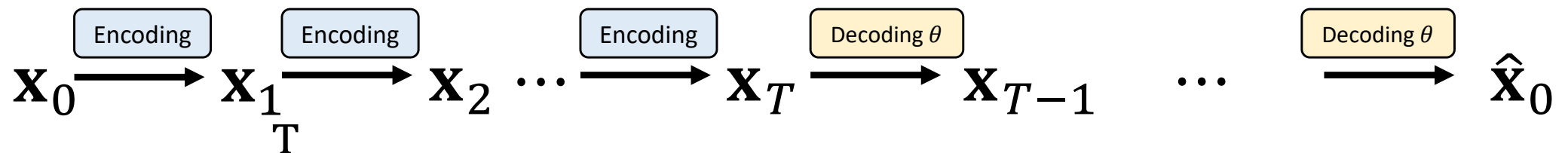
$$\boldsymbol{\epsilon}_0 \sim \mathcal{N}(\boldsymbol{\epsilon}; 0, \mathbf{I})$$

$$\mathbf{x}_2 = \sqrt{\alpha_2} (\sqrt{\alpha_1}\mathbf{x}_0 + \sqrt{1 - \alpha_1}\boldsymbol{\epsilon}_0) + \sqrt{1 - \alpha_2}\boldsymbol{\epsilon}_1$$

$$\boldsymbol{\epsilon}_1 \sim \mathcal{N}(\boldsymbol{\epsilon}; 0, \mathbf{I})$$

$$\vdots = \sqrt{\alpha_1\alpha_2}\mathbf{x}_0 + \sqrt{1 - \alpha_1\alpha_2}\boldsymbol{\epsilon}$$

$$\boldsymbol{\epsilon} \sim \mathcal{N}(\boldsymbol{\epsilon}; 0, \mathbf{I})$$

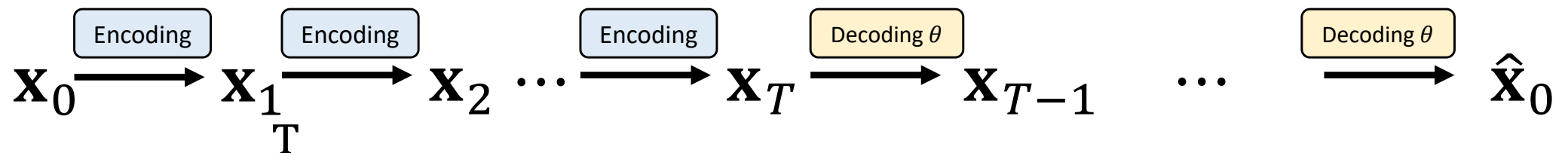


$$\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [\text{D}_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) || p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))]$$

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \underbrace{\sqrt{\bar{\alpha}_t}\mathbf{x}_0}_{\text{Mean}}, \underbrace{(1 - \bar{\alpha}_t)\mathbf{I}}_{\text{Variance}})$$

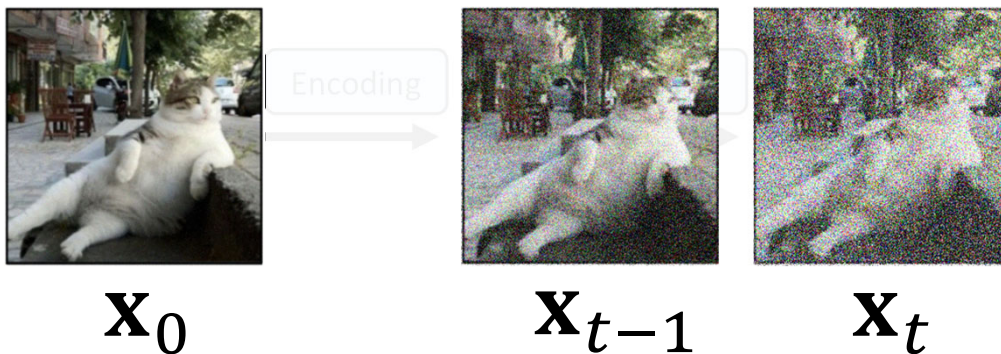
$$\Rightarrow \boxed{\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t}\boldsymbol{\epsilon}} \quad \bar{\alpha}_t = \prod_{i=1}^t \alpha_i \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\boldsymbol{\epsilon}; 0, \mathbf{I})$$

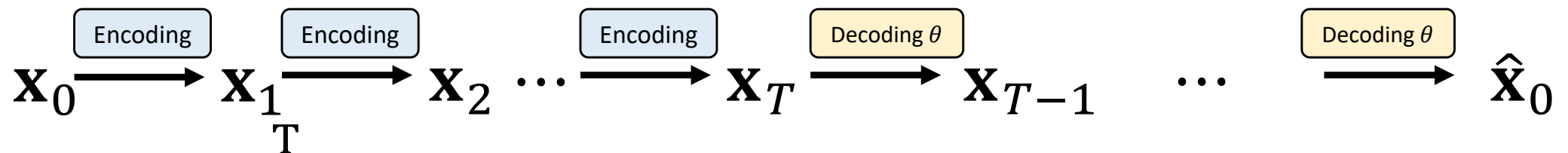
What does this equation imply?



$$\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [\text{D}_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) || p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))]$$

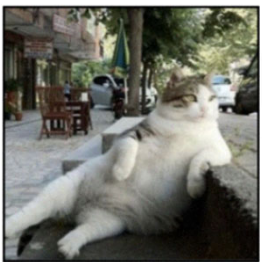
$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$$





$$\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [\text{D}_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) || p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))]$$

$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$



$\mathbf{x}_0$



$\mathbf{x}_{t-1}$



$\mathbf{x}_t$

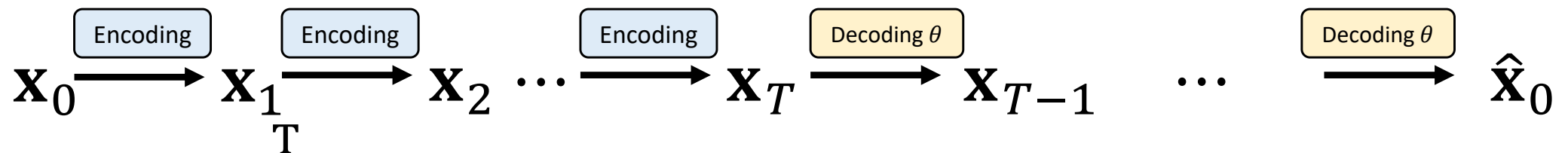
$p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)$



$\mathbf{x}_{t-1}$



$\mathbf{x}_t$

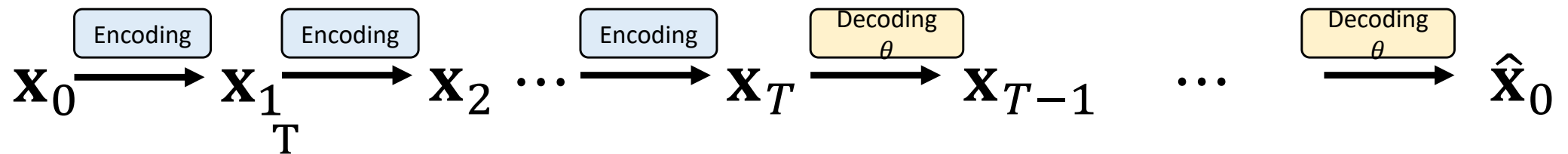


$$\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [\text{D}_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) || p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))]$$

$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$

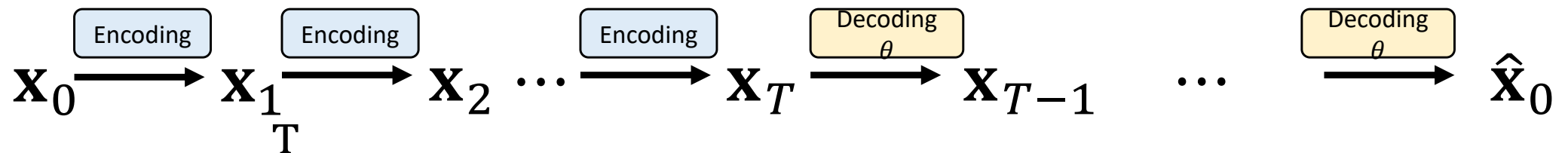


$p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)$



$$\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [\text{D}_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) || p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))]$$

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1})q(\mathbf{x}_{t-1}|\mathbf{x}_0)q(\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)q(\mathbf{x}_0)}$$



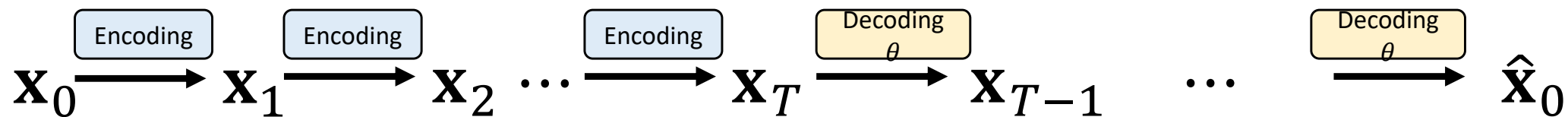
$$\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [\text{D}_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) || p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))]$$

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1})q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)}$$

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{\alpha_t}\mathbf{x}_{t-1}, (1 - \alpha_t)\mathbf{I}) \quad \text{😊}$$

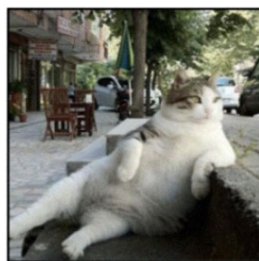
$$q(\mathbf{x}_{t-1}|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}; \sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0, (1 - \bar{\alpha}_{t-1})\mathbf{I}) \quad \text{😊}$$

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1 - \bar{\alpha}_t)\mathbf{I}) \quad \text{😊}$$



$$\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [\text{D}_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) || p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))]$$

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}; \mu_q(\mathbf{x}_t, \mathbf{x}_0), \Sigma_q(t))$$



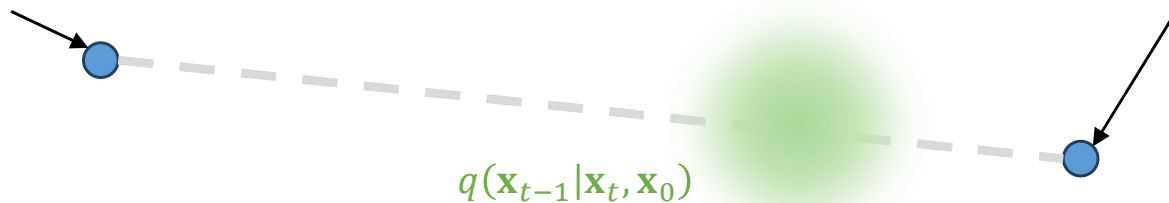
$\mathbf{x}_0$

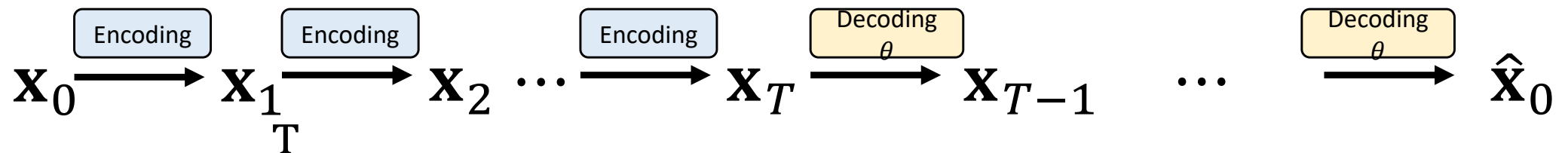


$\mathbf{x}_{t-1}$



$\mathbf{x}_t$



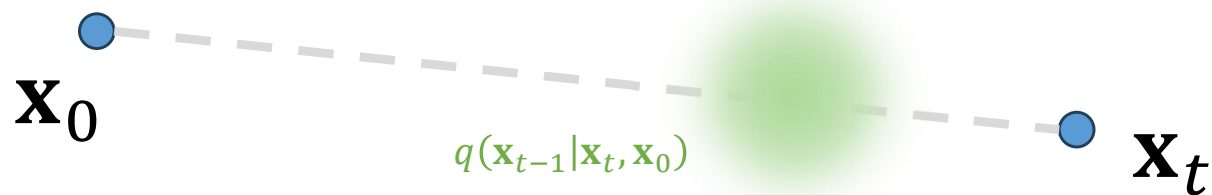


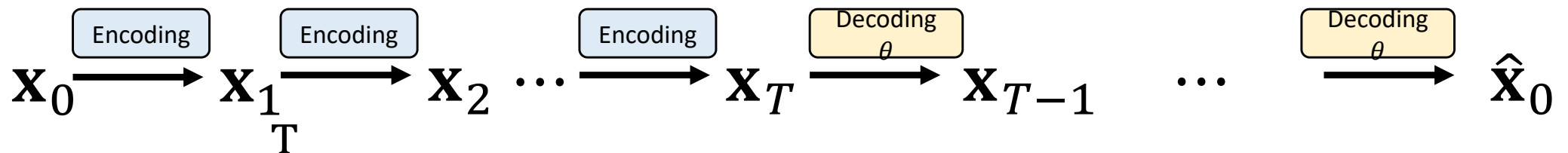
$$\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [\text{D}_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) || p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))]$$

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}; \mu_q(\mathbf{x}_t, \mathbf{x}_0), \Sigma_q(t))$$

$$\mu_q(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)}{1 - \bar{\alpha}_t} \mathbf{x}_0$$

Mean





$$\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [\text{D}_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) || p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))]$$

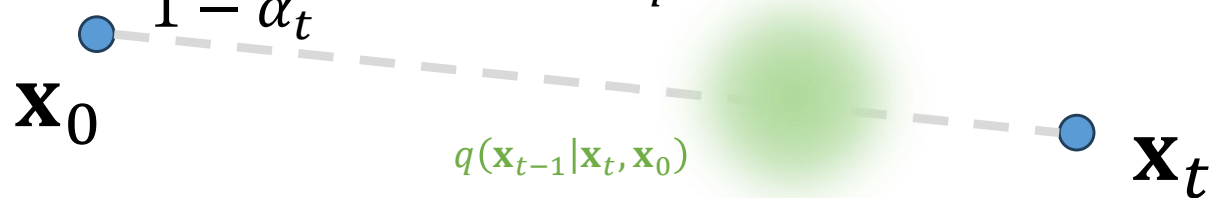
$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}; \mu_q(\mathbf{x}_t, \mathbf{x}_0), \Sigma_q(t))$$

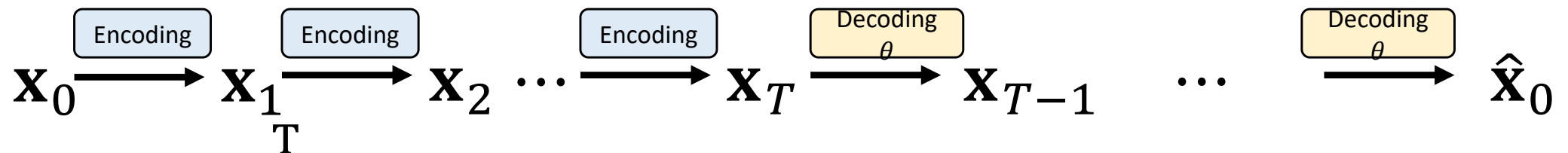
$$\mu_q(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)}{1 - \bar{\alpha}_t} \mathbf{x}_0$$

Mean

$$\Sigma_q(t) = \frac{(1 - \alpha_t)(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{I} = \sigma_q^2(t) \mathbf{I}$$

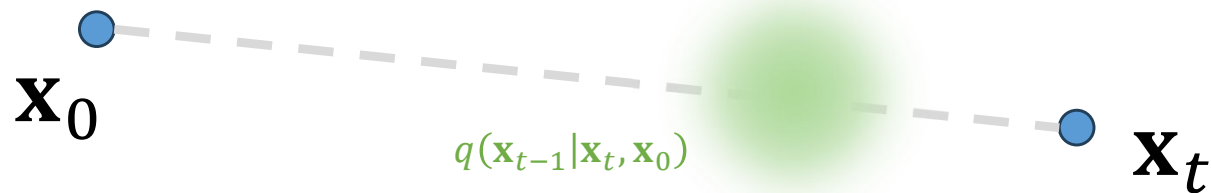
Variance

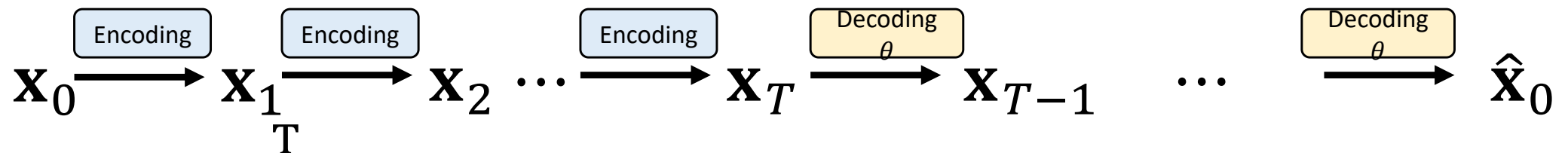




$$\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [\text{D}_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) || p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))]$$

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}; \mu_q(\mathbf{x}_t, \mathbf{x}_0), \Sigma_q(t))$$

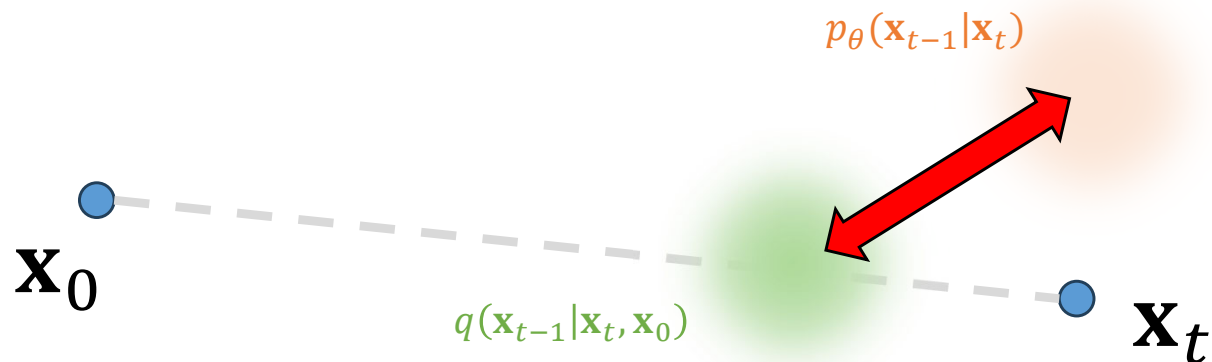


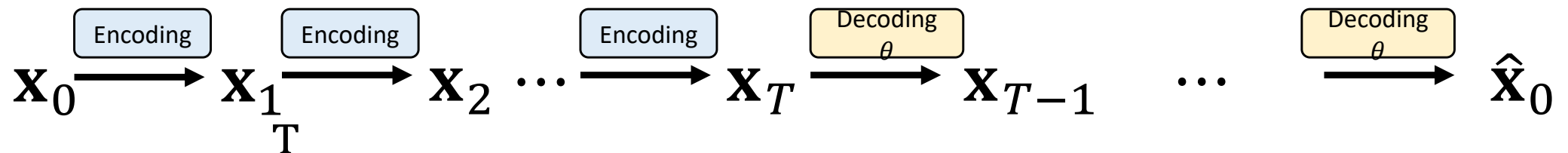


$$\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [\text{D}_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) || p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))]$$

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}; \mu_q(\mathbf{x}_t, \mathbf{x}_0), \Sigma_q(t))$$

$$p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \mu_{\theta}(\mathbf{x}_t, t), \Sigma_q(t))$$

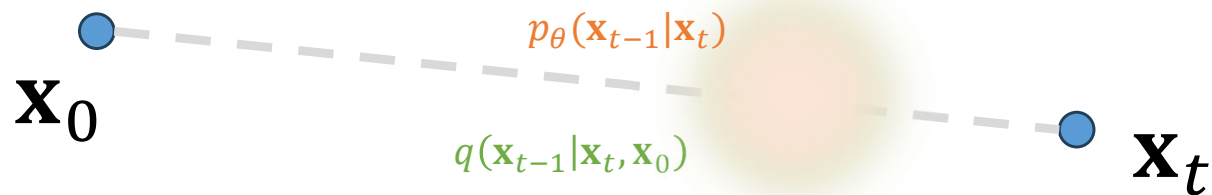


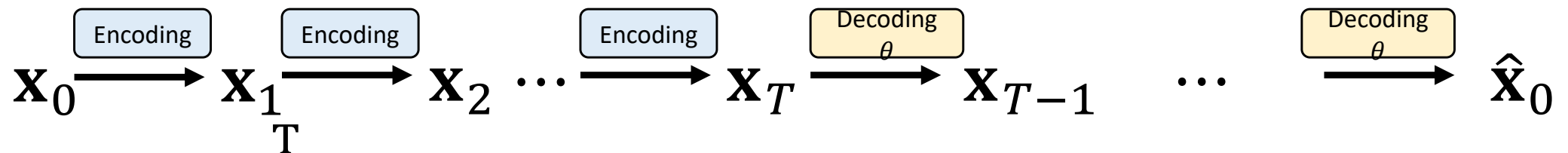


$$\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [\text{D}_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) || p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))]$$

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}; \mu_q(\mathbf{x}_t, \mathbf{x}_0), \Sigma_q(t))$$

$$p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \mu_{\theta}(\mathbf{x}_t, t), \Sigma_q(t))$$

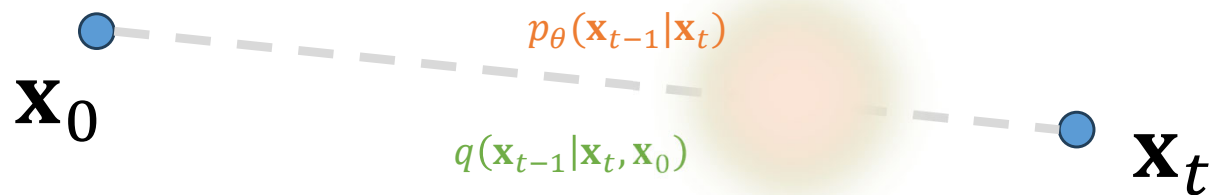


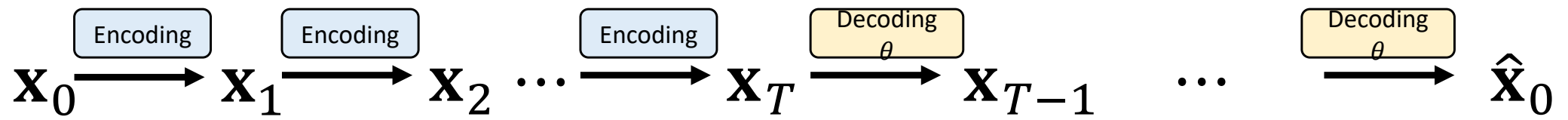


$$\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [\text{D}_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) || p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))]$$

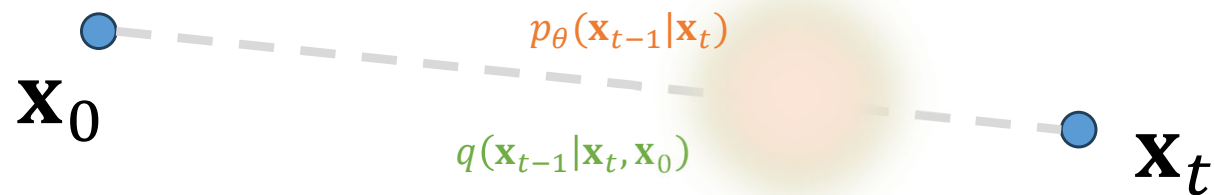
$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_q(\mathbf{x}_t, \mathbf{x}_0), \Sigma_q(t))$$

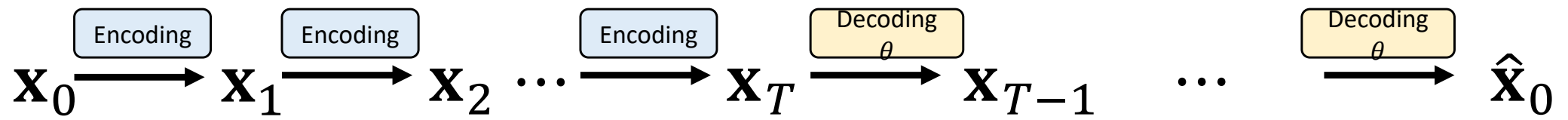
$$p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_{\theta}(\mathbf{x}_t, t), \Sigma_q(t))$$



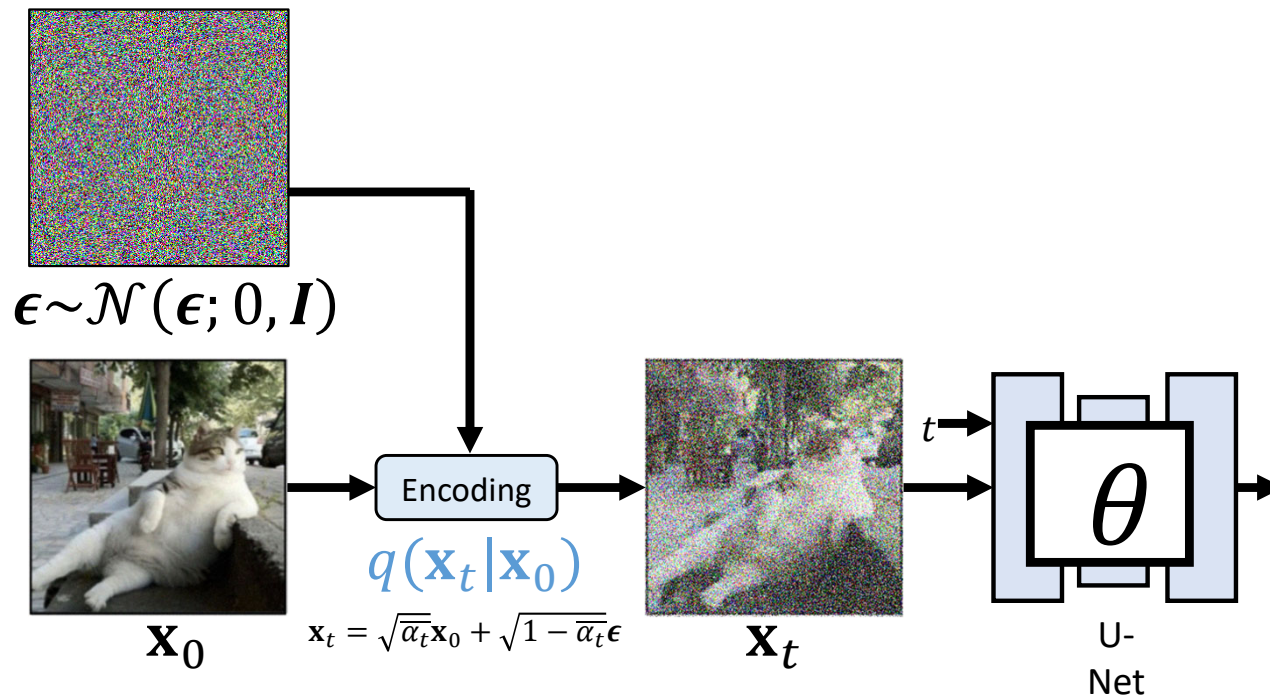


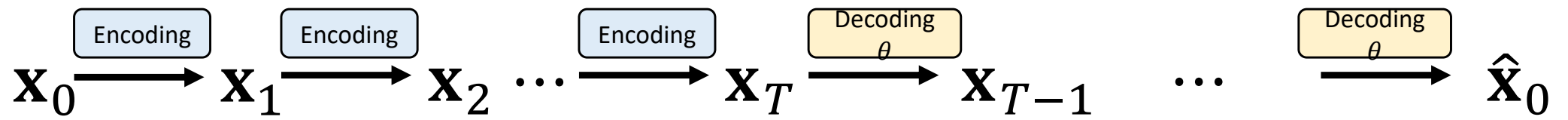
$$\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[\frac{1}{2\sigma_q^2(t)} \left[\|\mu_{\theta}(\mathbf{x}_t, t) - \mu_q(\mathbf{x}_t, \mathbf{x}_0)\|_2^2 \right] \right]$$



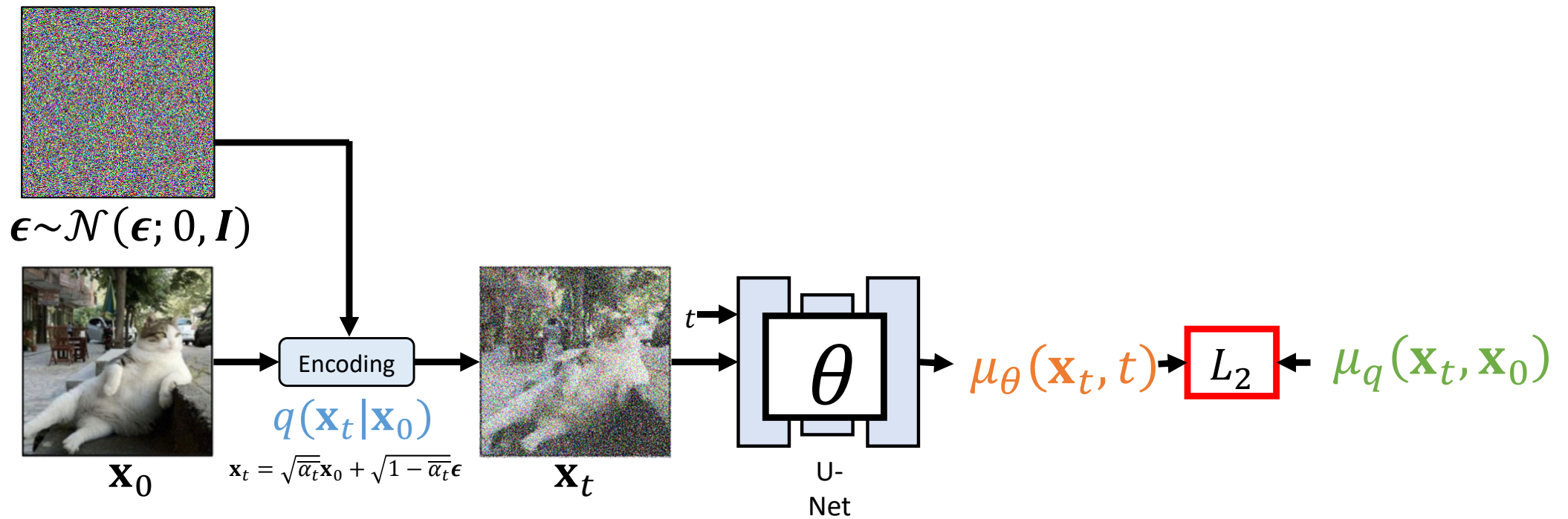


$$\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[\frac{1}{2\sigma_q^2(t)} \left[\|\mu_{\theta}(\mathbf{x}_t, t) - \mu_q(\mathbf{x}_t, \mathbf{x}_0)\|_2^2 \right] \right]$$

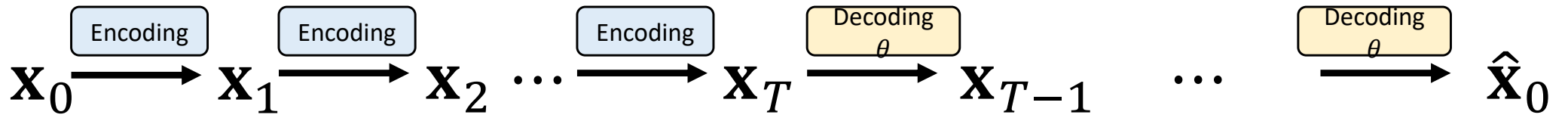




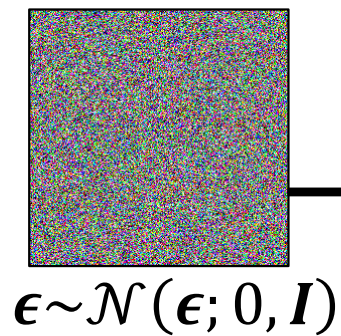
$$\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[\frac{1}{2\sigma_q^2(t)} \left[\|\mu_{\theta}(\mathbf{x}_t, t) - \mu_q(\mathbf{x}_t, \mathbf{x}_0)\|_2^2 \right] \right]$$



Observation #1

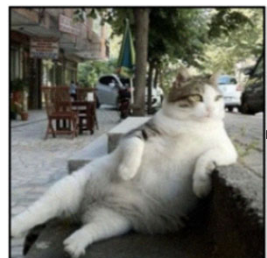


$$\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[\frac{1}{2\sigma_q^2(t)} \left[\|\mu_\theta(\mathbf{x}_t, t) - \mu_q(\mathbf{x}_t, \mathbf{x}_0)\|_2^2 \right] \right]$$



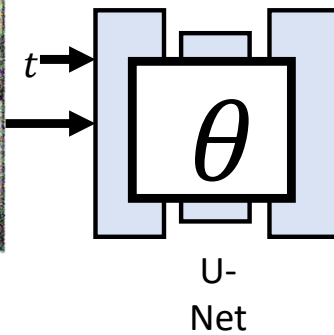
$$\mu_q(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\alpha_{t-1}}(1 - \alpha_t)}{1 - \bar{\alpha}_t} \mathbf{x}_0$$

$$\mu_\theta(\mathbf{x}_t, t) = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\alpha_{t-1}}(1 - \alpha_t)}{1 - \bar{\alpha}_t} \hat{\mathbf{x}}_\theta(\mathbf{x}_t, t)$$



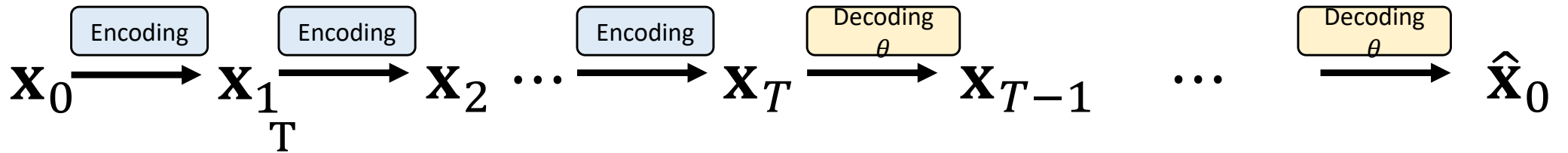
Encoding
 $q(\mathbf{x}_t|\mathbf{x}_0)$

$$\mathbf{x}_t = \sqrt{\alpha_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$$

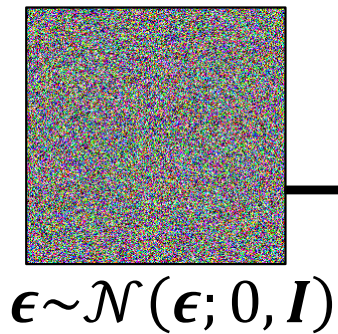


$$\mu_\theta(\mathbf{x}_t, t) \rightarrow L_2 \leftarrow \mu_q(\mathbf{x}_t, \mathbf{x}_0)$$

Observation #1

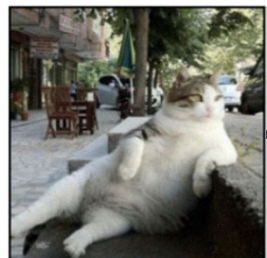


$$\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [w(t) [|| \hat{\mathbf{x}}_{\theta}(\mathbf{x}_t, t) - \mathbf{x}_0 ||_2^2]]$$



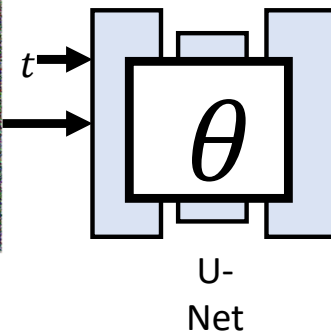
$$\mu_q(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\alpha_{t-1}}(1 - \alpha_t)}{1 - \bar{\alpha}_t} \mathbf{x}_0$$

$$\mu_{\theta}(\mathbf{x}_t, t) = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\alpha_{t-1}}(1 - \alpha_t)}{1 - \bar{\alpha}_t} \hat{\mathbf{x}}_{\theta}(\mathbf{x}_t, t)$$



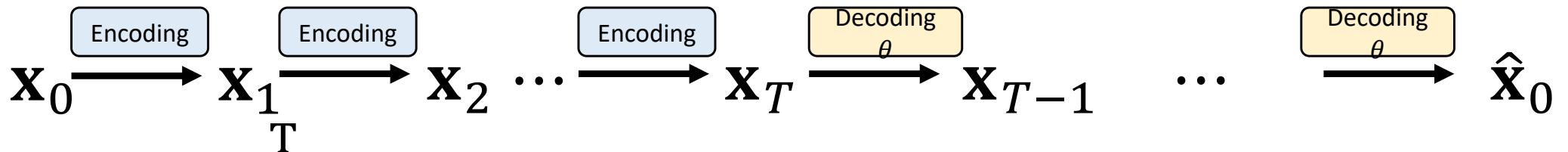
Encoding
 $q(\mathbf{x}_t|\mathbf{x}_0)$

$$\mathbf{x}_t = \sqrt{\alpha_t} \mathbf{x}_0 + \sqrt{1 - \alpha_t} \epsilon$$

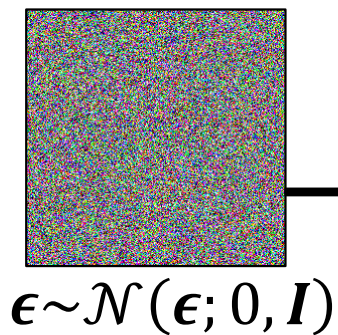


$$\mu_{\theta}(\mathbf{x}_t, t) \rightarrow L_2 \leftarrow \mu_q(\mathbf{x}_t, \mathbf{x}_0)$$

Observation #1

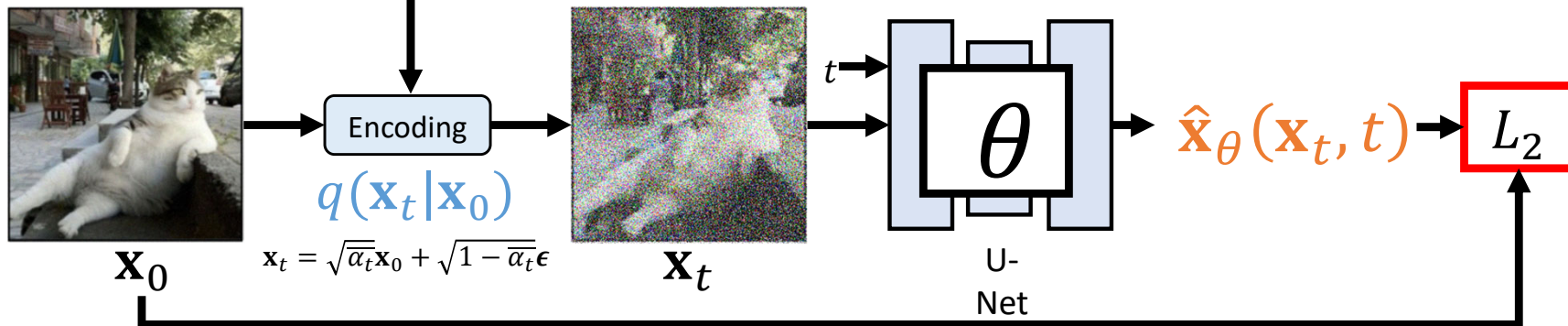


$$\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [w(t) [|| \hat{\mathbf{x}}_{\theta}(\mathbf{x}_t, t) - \mathbf{x}_0 ||_2^2]]$$

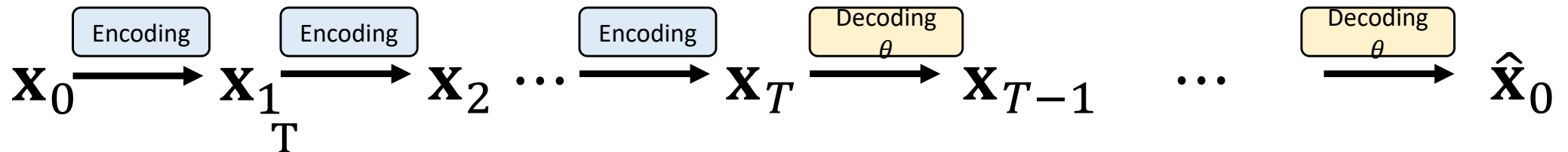


$$\mu_q(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\alpha_{t-1}}(1 - \alpha_t)}{1 - \bar{\alpha}_t} \mathbf{x}_0$$

$$\mu_{\theta}(\mathbf{x}_t, t) = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\alpha_{t-1}}(1 - \alpha_t)}{1 - \bar{\alpha}_t} \hat{\mathbf{x}}_{\theta}(\mathbf{x}_t, t)$$

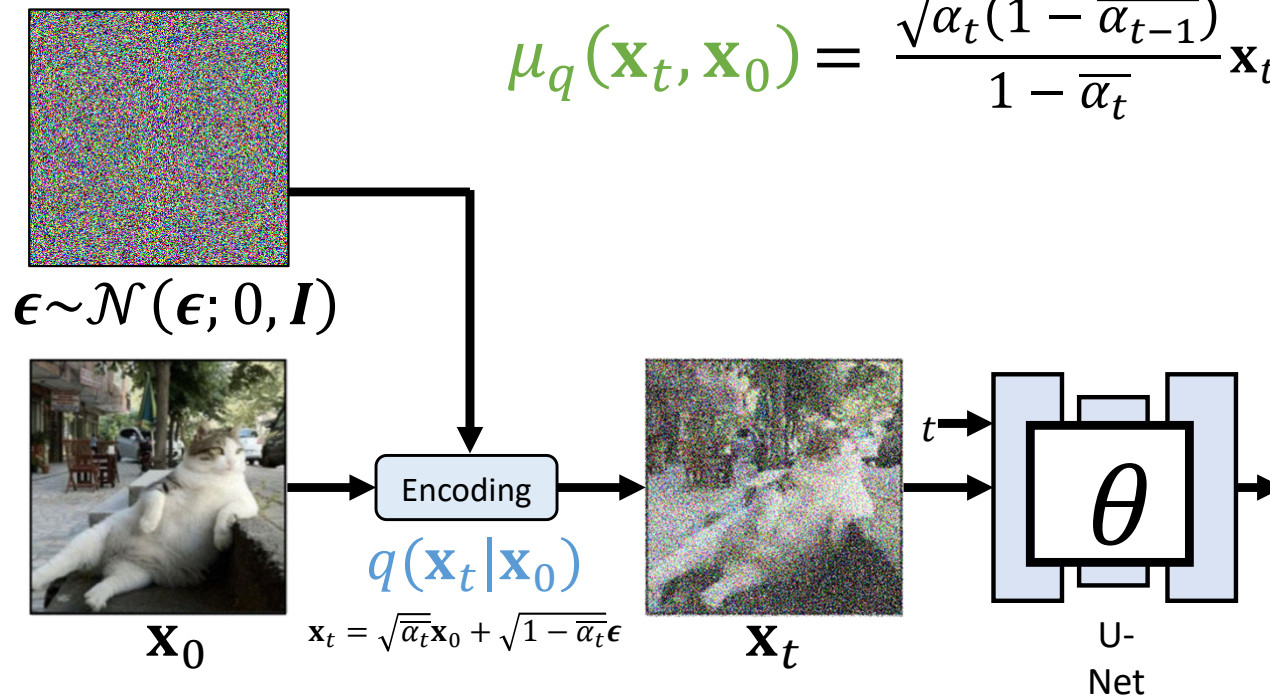


Observation #1

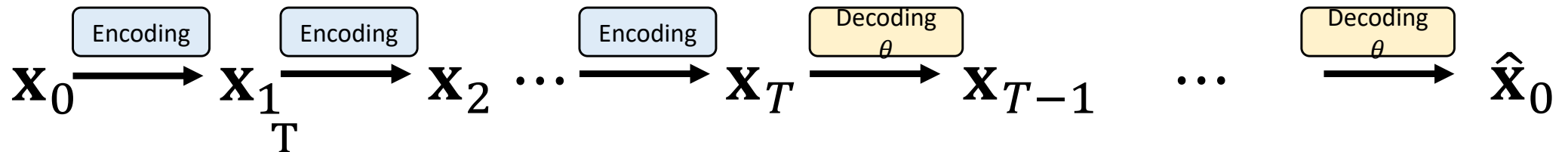


$$\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[w(t) \left[\|\mu_\theta(\mathbf{x}_t, t) - \mu_q(\mathbf{x}_t, \mathbf{x}_0)\|_2^2 \right] \right]$$

$$\mu_q(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\alpha_{t-1}}(1 - \alpha_t)}{1 - \bar{\alpha}_t} \mathbf{x}_0$$



Observation #2



$$\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[w(t) \left[\|\mu_\theta(\mathbf{x}_t, t) - \mu_q(\mathbf{x}_t, \mathbf{x}_0)\|_2^2 \right] \right]$$

$$\mu_q(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)}{1 - \bar{\alpha}_t} \mathbf{x}_0$$



$$\epsilon \sim \mathcal{N}(\epsilon; 0, I)$$



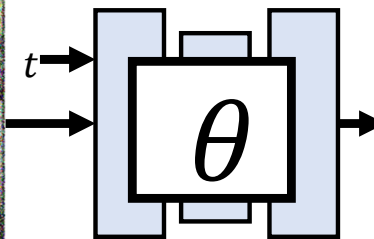
$\mathbf{x}_0$

Encoding
 $q(\mathbf{x}_t|\mathbf{x}_0)$

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$$



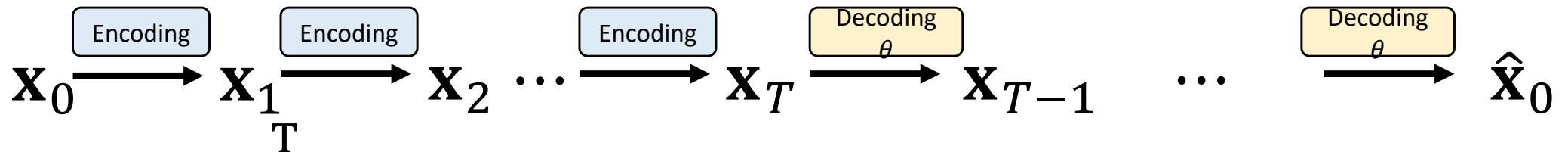
$\mathbf{x}_t$



U-
Net

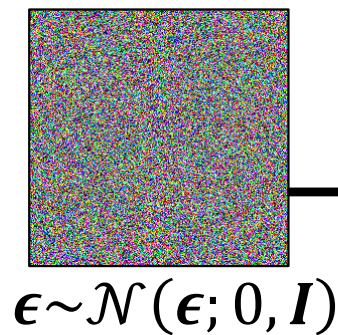
$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$$

Observation #2

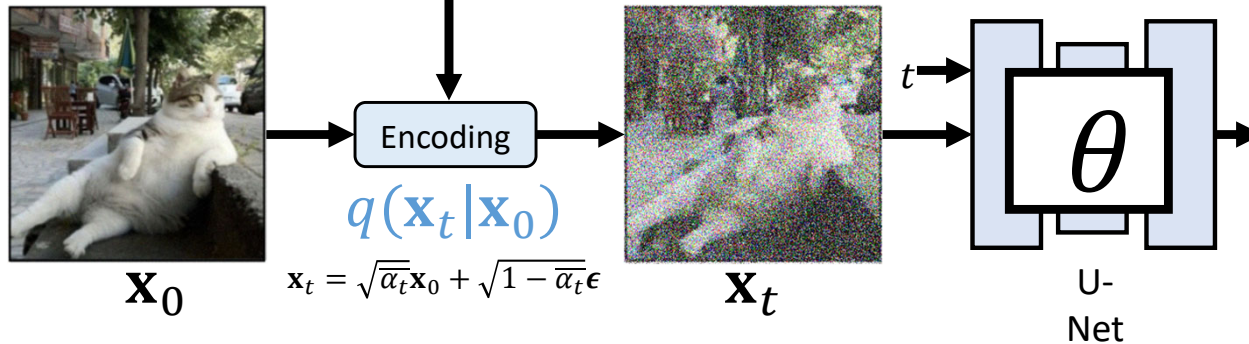


$$\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[w(t) \left[\|\mu_\theta(\mathbf{x}_t, t) - \mu_q(\mathbf{x}_t, \mathbf{x}_0)\|_2^2 \right] \right]$$

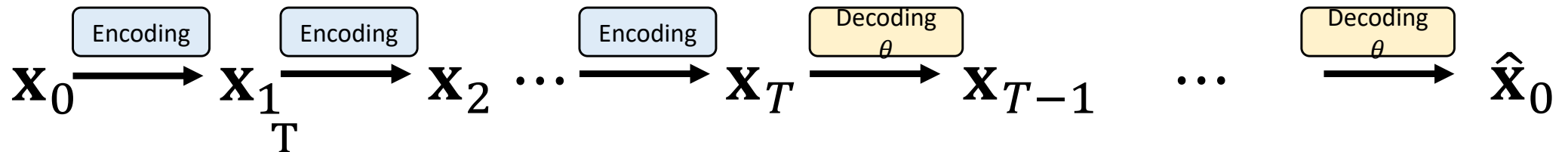
$$\mu_q(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\alpha_{t-1}}(1 - \alpha_t)}{1 - \bar{\alpha}_t} \mathbf{x}_0$$



$$\sqrt{\alpha_t} \mathbf{x}_0 = \mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \epsilon$$



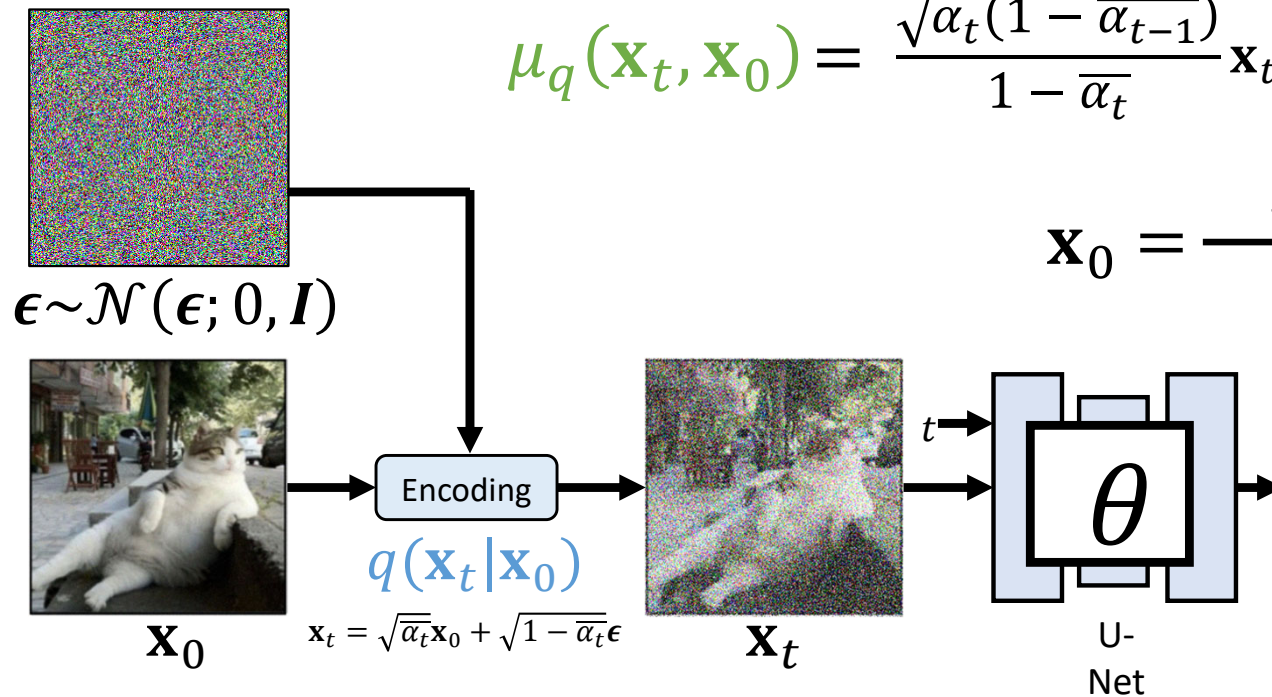
Observation #2



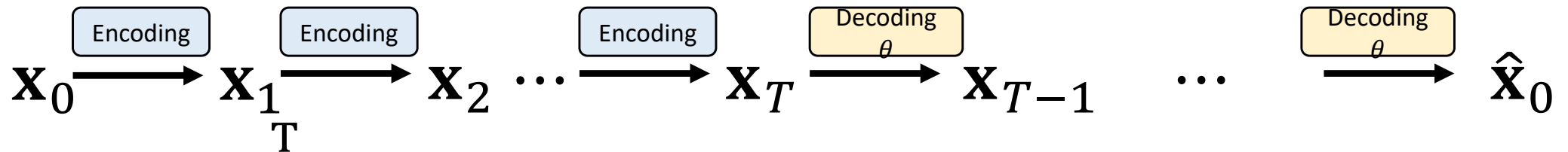
$$\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[w(t) \left[\|\mu_\theta(\mathbf{x}_t, t) - \mu_q(\mathbf{x}_t, \mathbf{x}_0)\|_2^2 \right] \right]$$

$$\mu_q(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)}{1 - \bar{\alpha}_t} \mathbf{x}_0$$

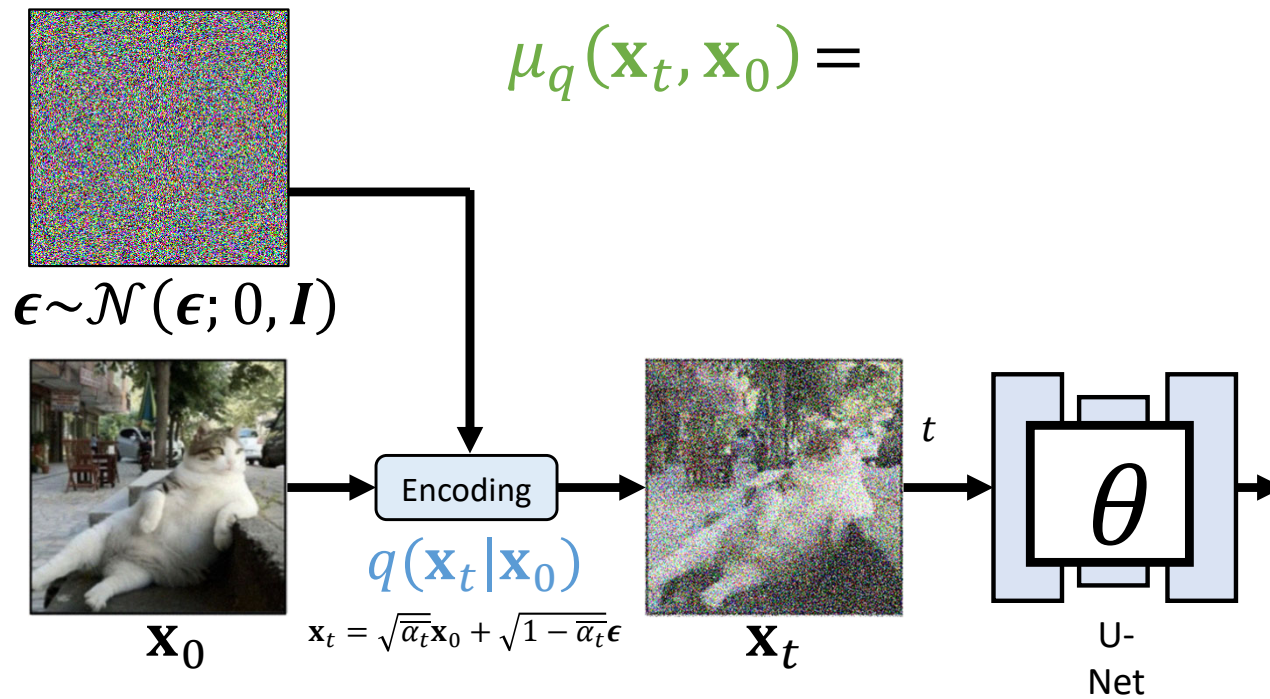
$$\mathbf{x}_0 = \frac{\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}}{\sqrt{\bar{\alpha}_t}}$$



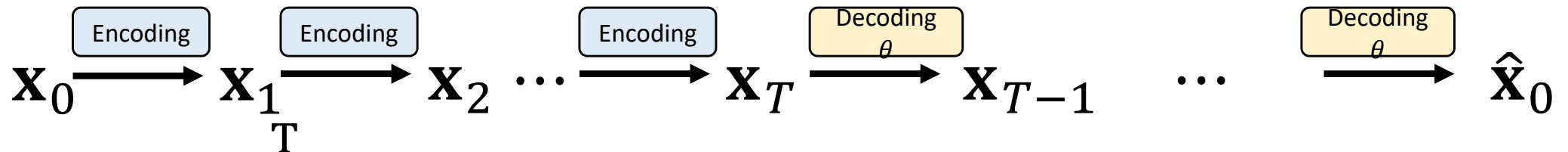
Observation #2



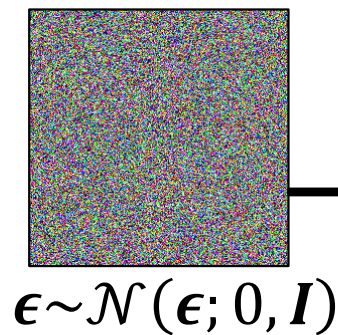
$$\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[w(t) \left[\|\mu_{\theta}(\mathbf{x}_t, t) - \mu_q(\mathbf{x}_t, \mathbf{x}_0)\|_2^2 \right] \right]$$



Observation #2

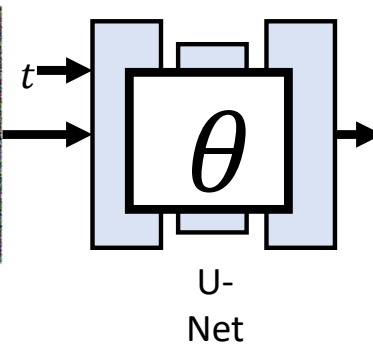
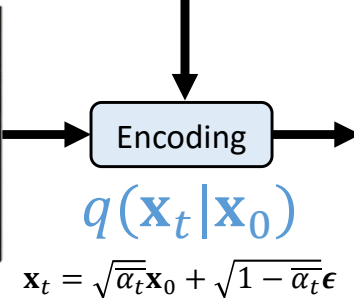
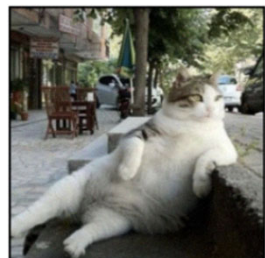


$$\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[w(t) \left[\|\mu_{\theta}(\mathbf{x}_t, t) - \mu_q(\mathbf{x}_t, \mathbf{x}_0)\|_2^2 \right] \right]$$

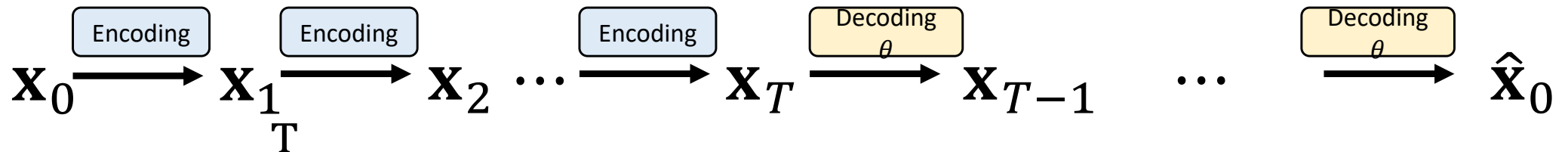


$$\mu_q(\mathbf{x}_t, \mathbf{x}_0) = \frac{1}{\sqrt{\alpha_t}} \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{(1 - \bar{\alpha}_t)\alpha_t}} \epsilon$$

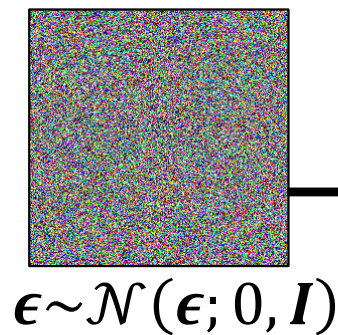
$$\mu_{\theta}(\mathbf{x}_t, t) = \frac{1}{\sqrt{\alpha_t}} \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{(1 - \bar{\alpha}_t)\alpha_t}} \hat{\epsilon}_{\theta}(\mathbf{x}_t, t)$$



Observation #2

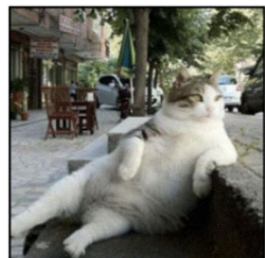


$$\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[w(t) \left[\|\mu_{\theta}(\mathbf{x}_t, t) - \mu_q(\mathbf{x}_t, \mathbf{x}_0)\|_2^2 \right] \right]$$



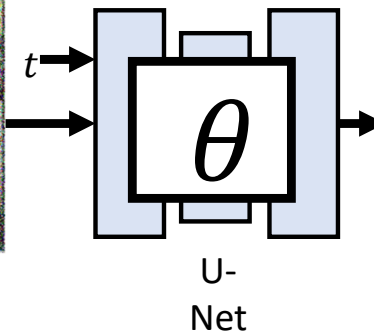
$$\mu_q(\mathbf{x}_t, \mathbf{x}_0) = \frac{1}{\sqrt{\alpha_t}} \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{(1 - \bar{\alpha}_t)\alpha_t}} \epsilon$$

$$\mu_{\theta}(\mathbf{x}_t, t) = \frac{1}{\sqrt{\alpha_t}} \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{(1 - \bar{\alpha}_t)\alpha_t}} \hat{\epsilon}_{\theta}(\mathbf{x}_t, t)$$

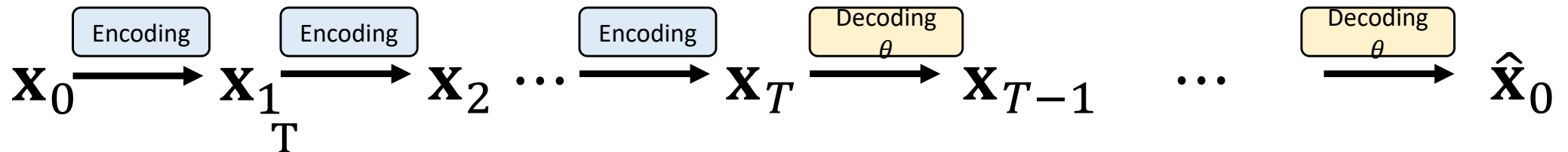


Encoding
 $q(\mathbf{x}_t|\mathbf{x}_0)$

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$$



Observation #2



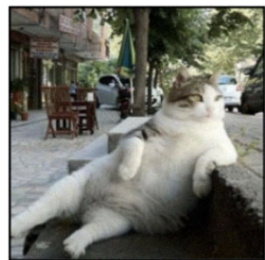
$$\sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[w(t) \left[\|\hat{\epsilon}_{\theta}(\mathbf{x}_t, t) - \epsilon\|_2^2 \right] \right]$$



$$\epsilon \sim \mathcal{N}(\epsilon; 0, I)$$

$$\mu_q(\mathbf{x}_t, \mathbf{x}_0) = \frac{1}{\sqrt{\alpha_t}} \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{(1 - \bar{\alpha}_t)\alpha_t}} \epsilon$$

$$\mu_{\theta}(\mathbf{x}_t, t) = \frac{1}{\sqrt{\alpha_t}} \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{(1 - \bar{\alpha}_t)\alpha_t}} \hat{\epsilon}_{\theta}(\mathbf{x}_t, t)$$



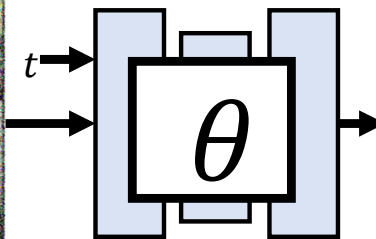
$\mathbf{x}_0$

Encoding
 $q(\mathbf{x}_t|\mathbf{x}_0)$

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$$

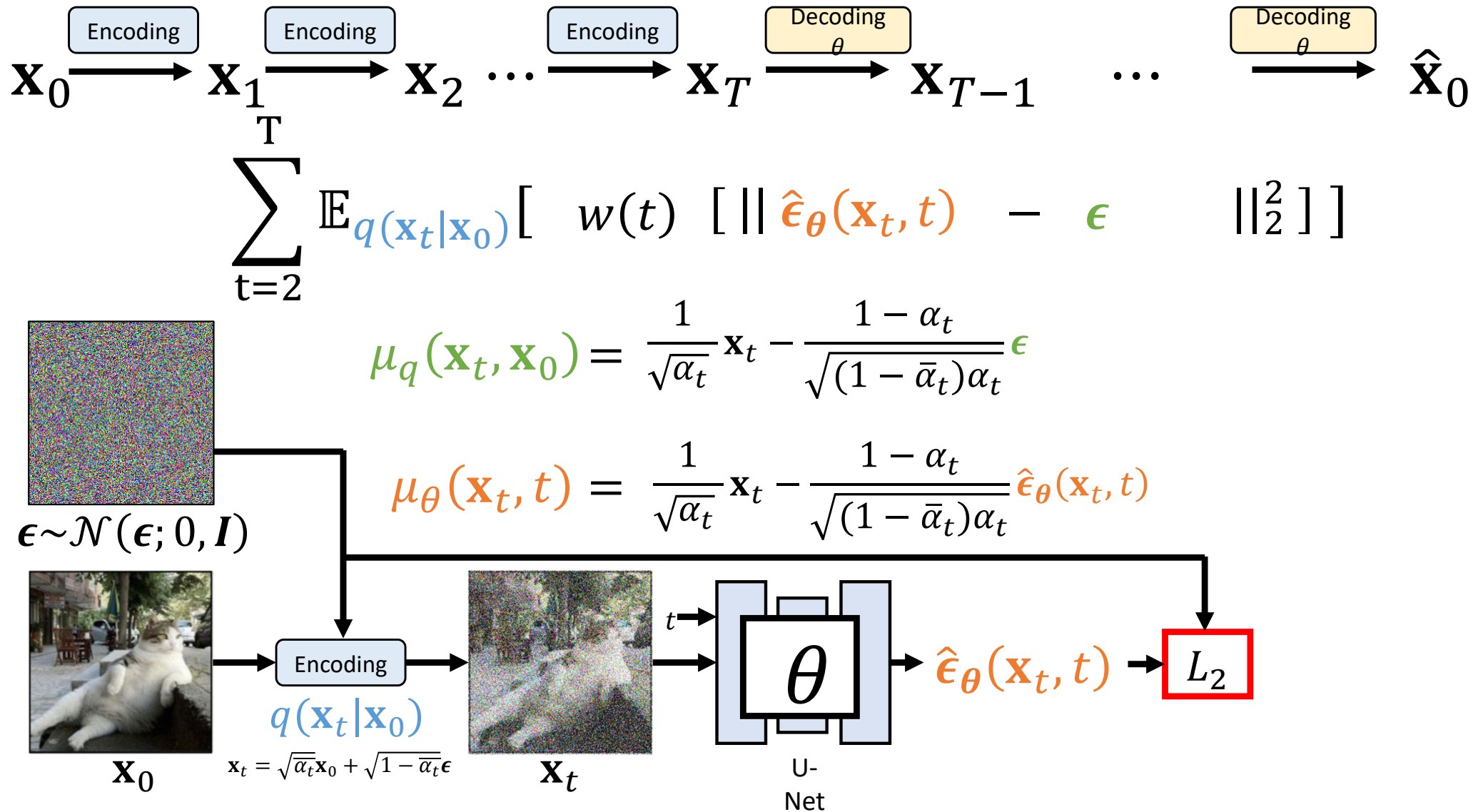


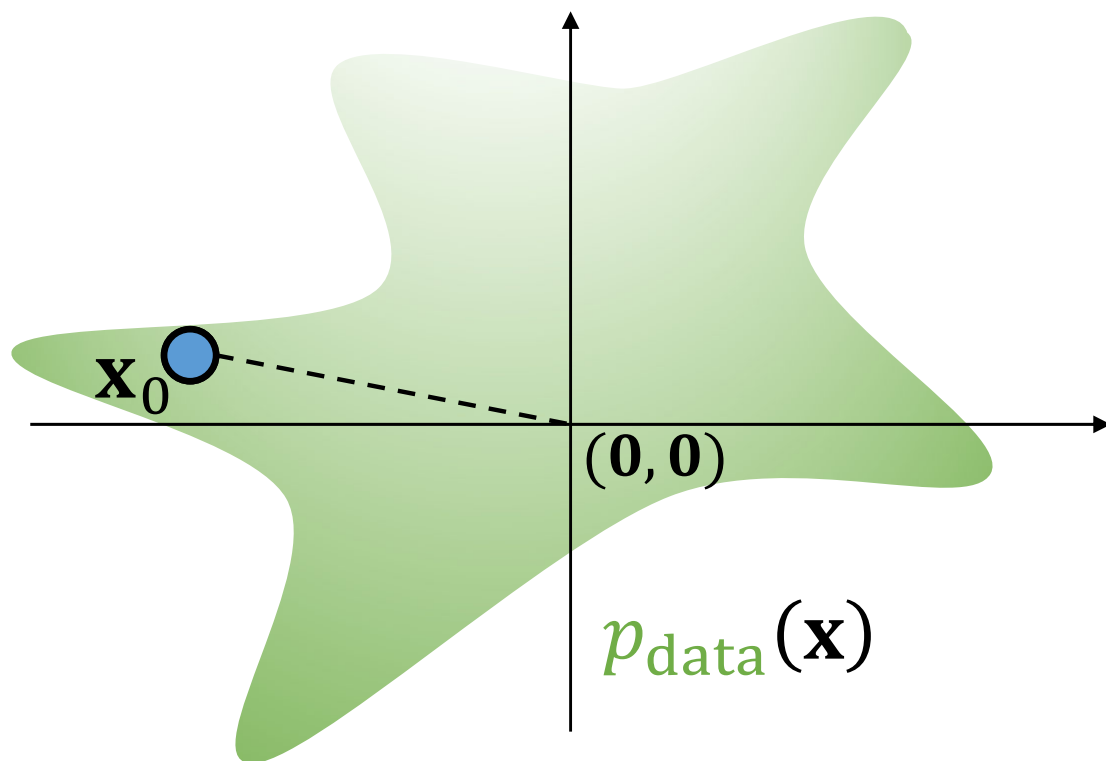
$\mathbf{x}_t$

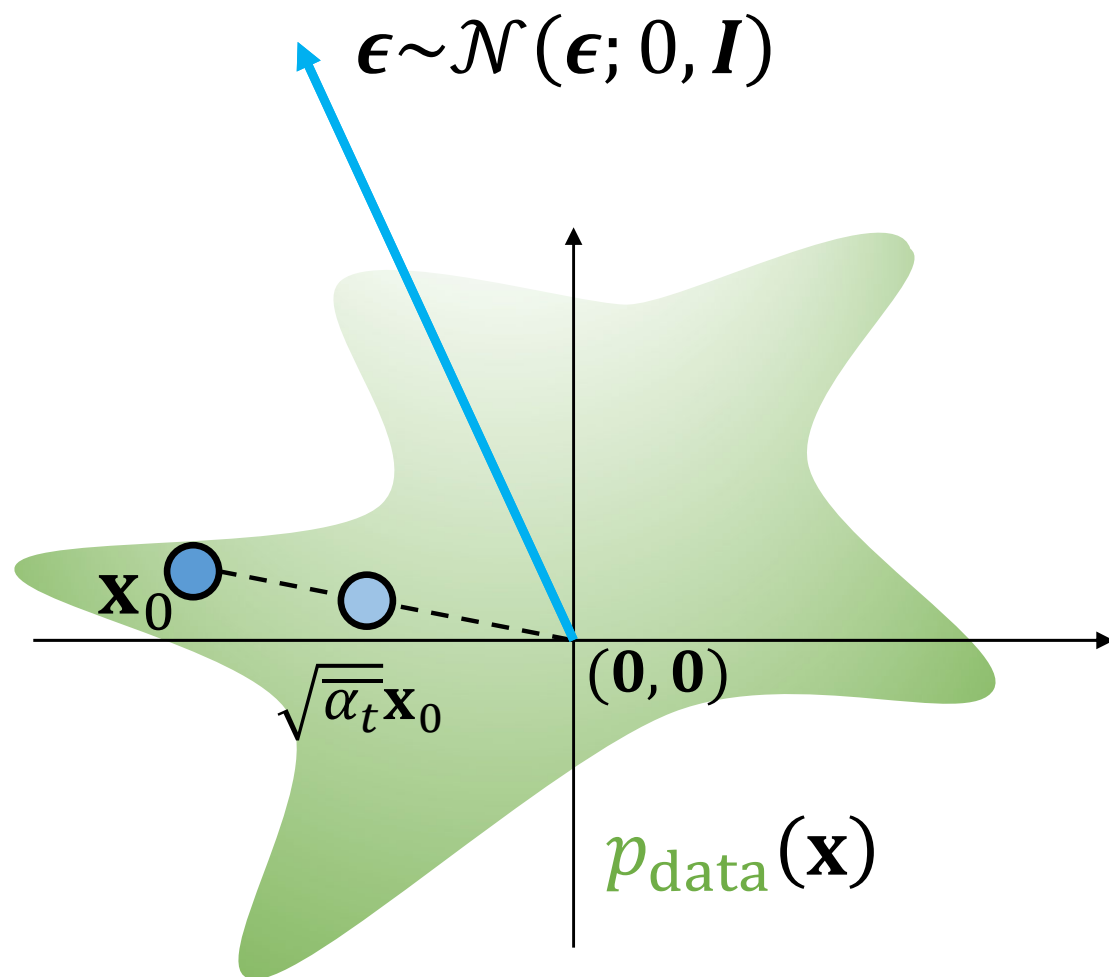


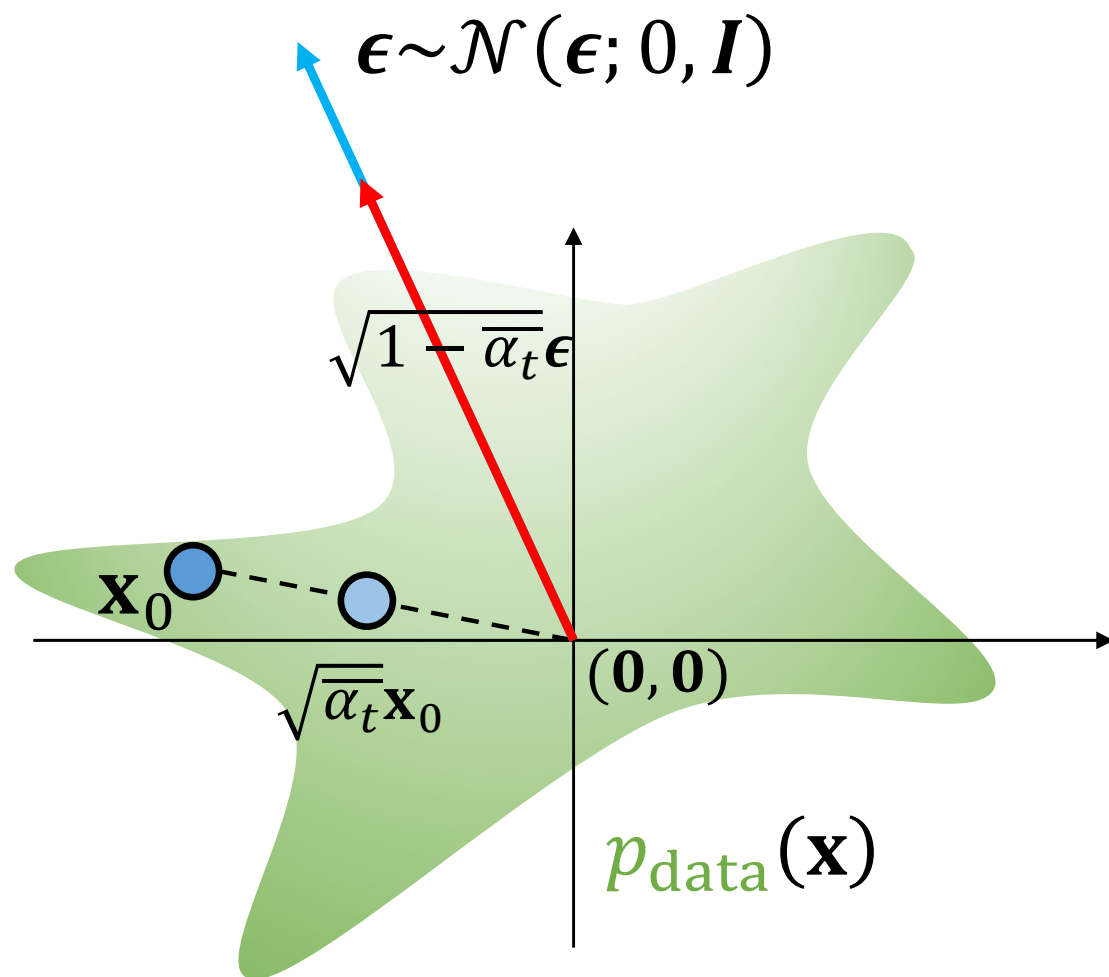
U-Net

Observation #2

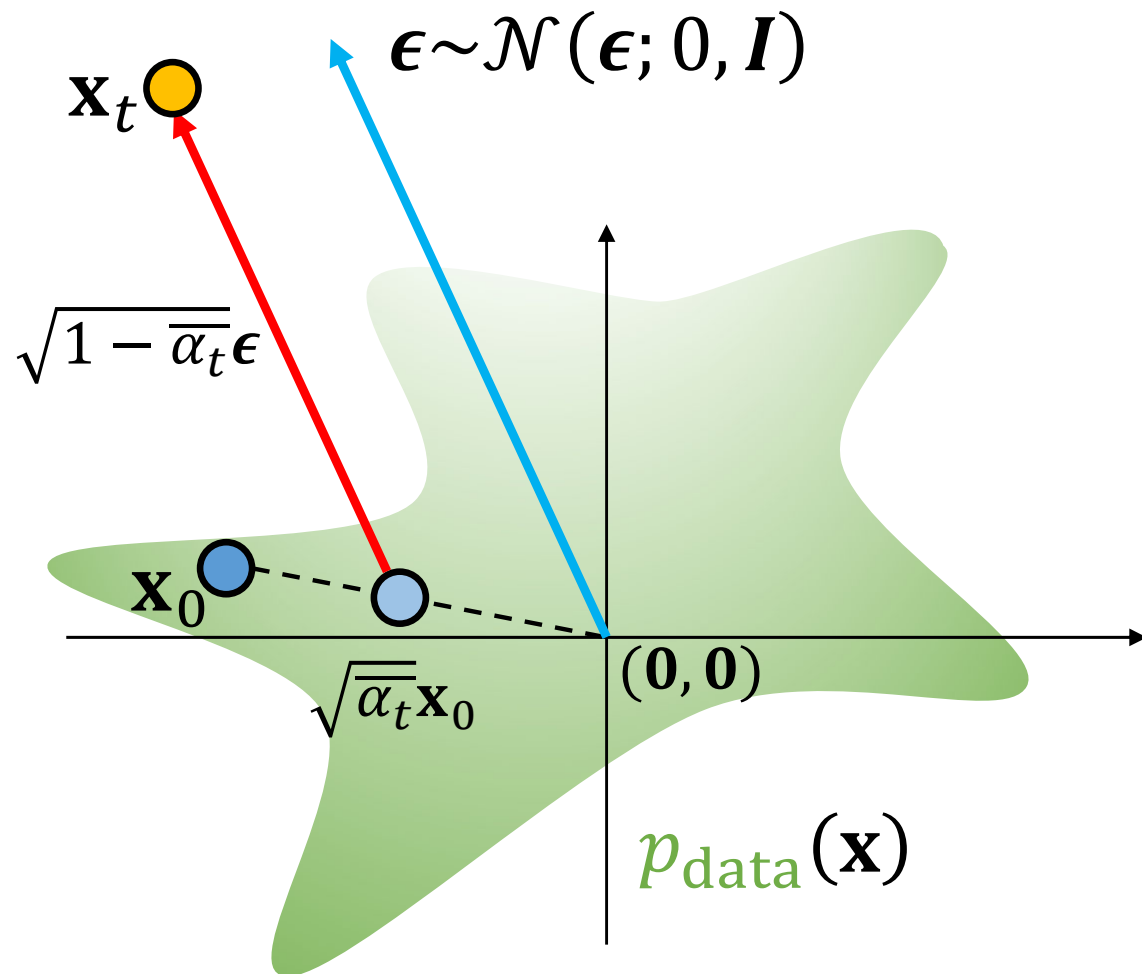




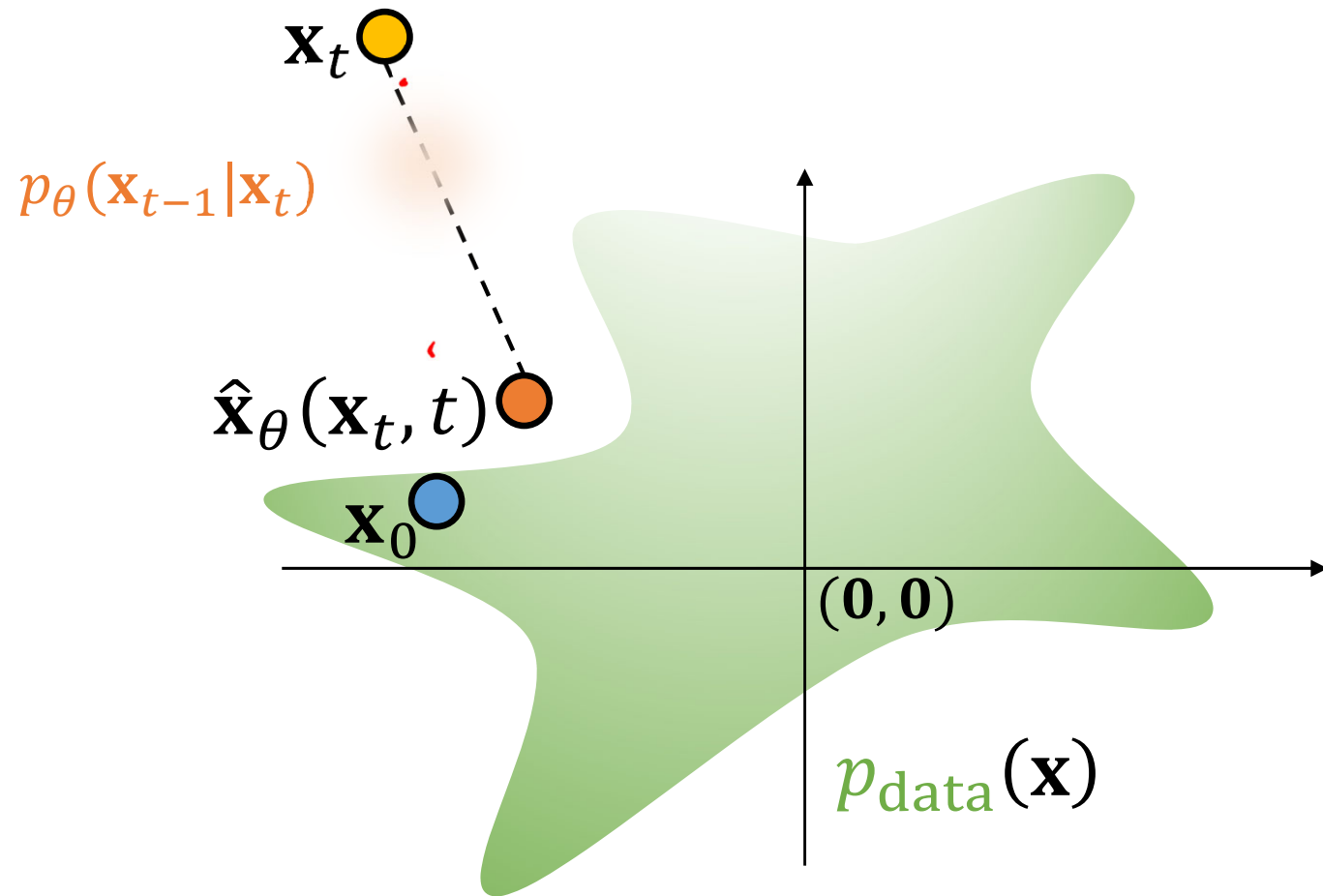




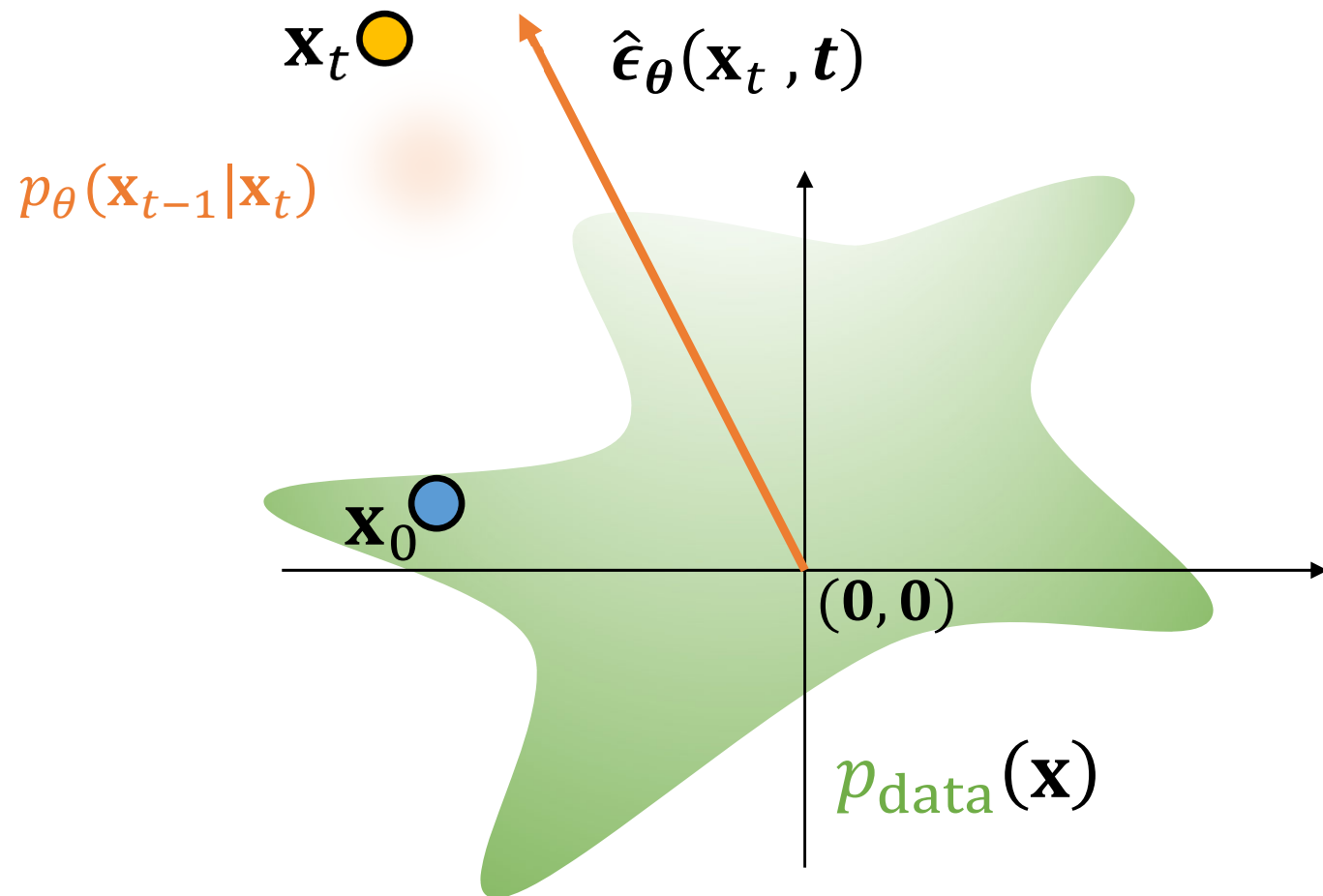
$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}$$



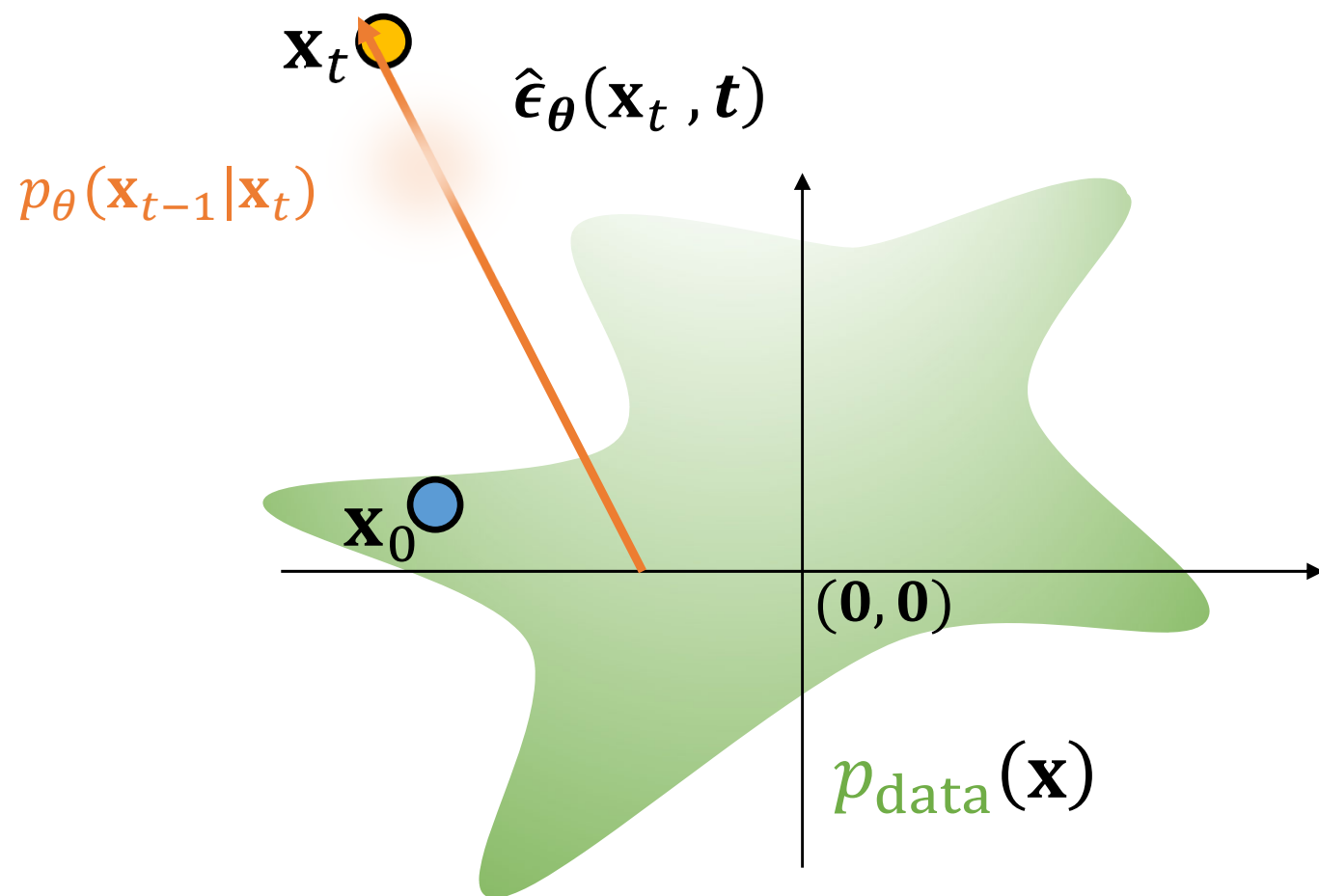
$$\mu_{\theta}(\mathbf{x}_t, t) = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)}{1 - \bar{\alpha}_t} \hat{\mathbf{x}}_{\theta}(\mathbf{x}_t, t)$$



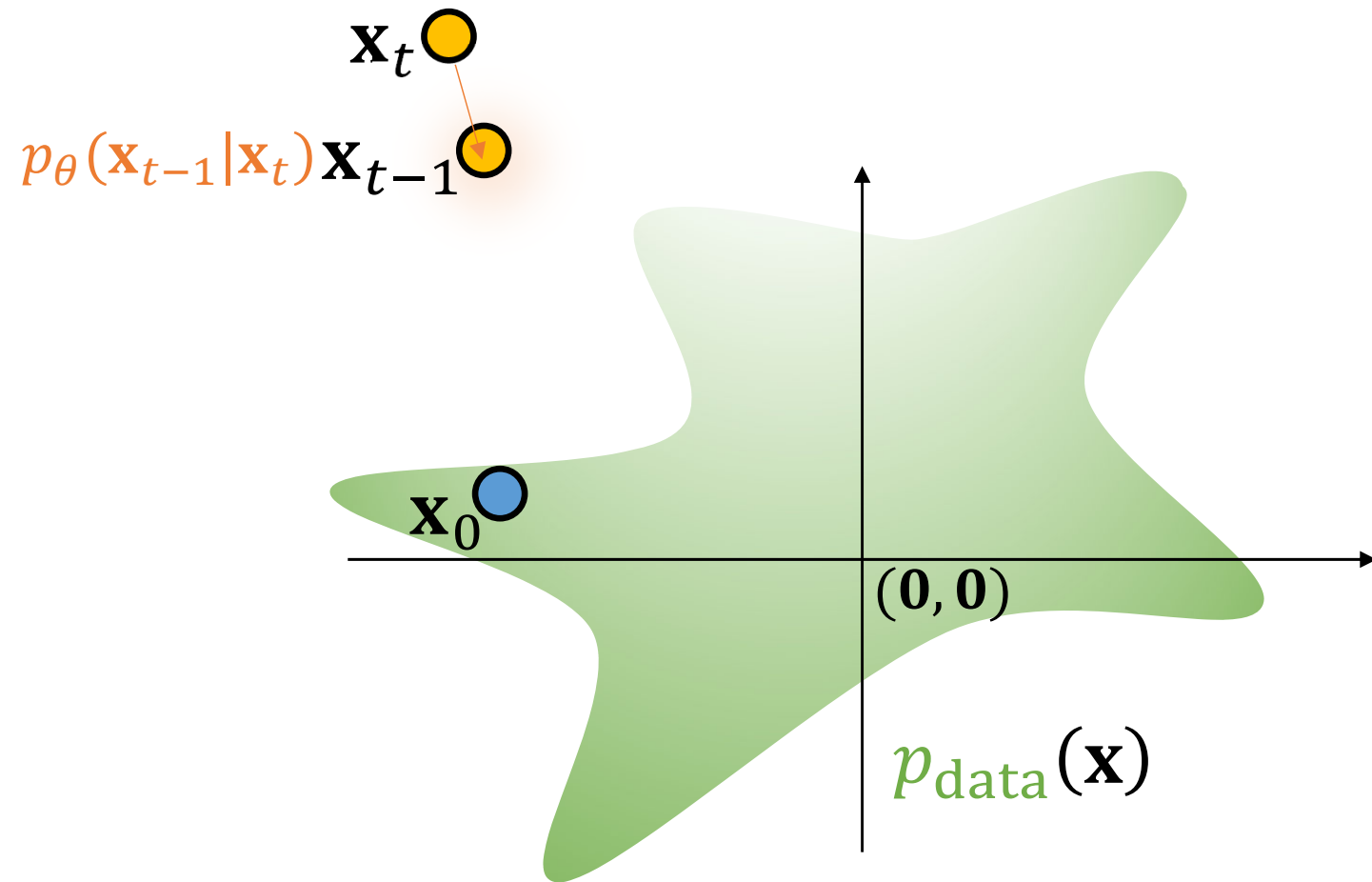
$$\mu_{\theta}(\mathbf{x}_t, t) = \frac{1}{\sqrt{\alpha_t}} \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{(1 - \bar{\alpha}_t)\alpha_t}} \hat{\epsilon}_{\theta}(\mathbf{x}_t, t)$$



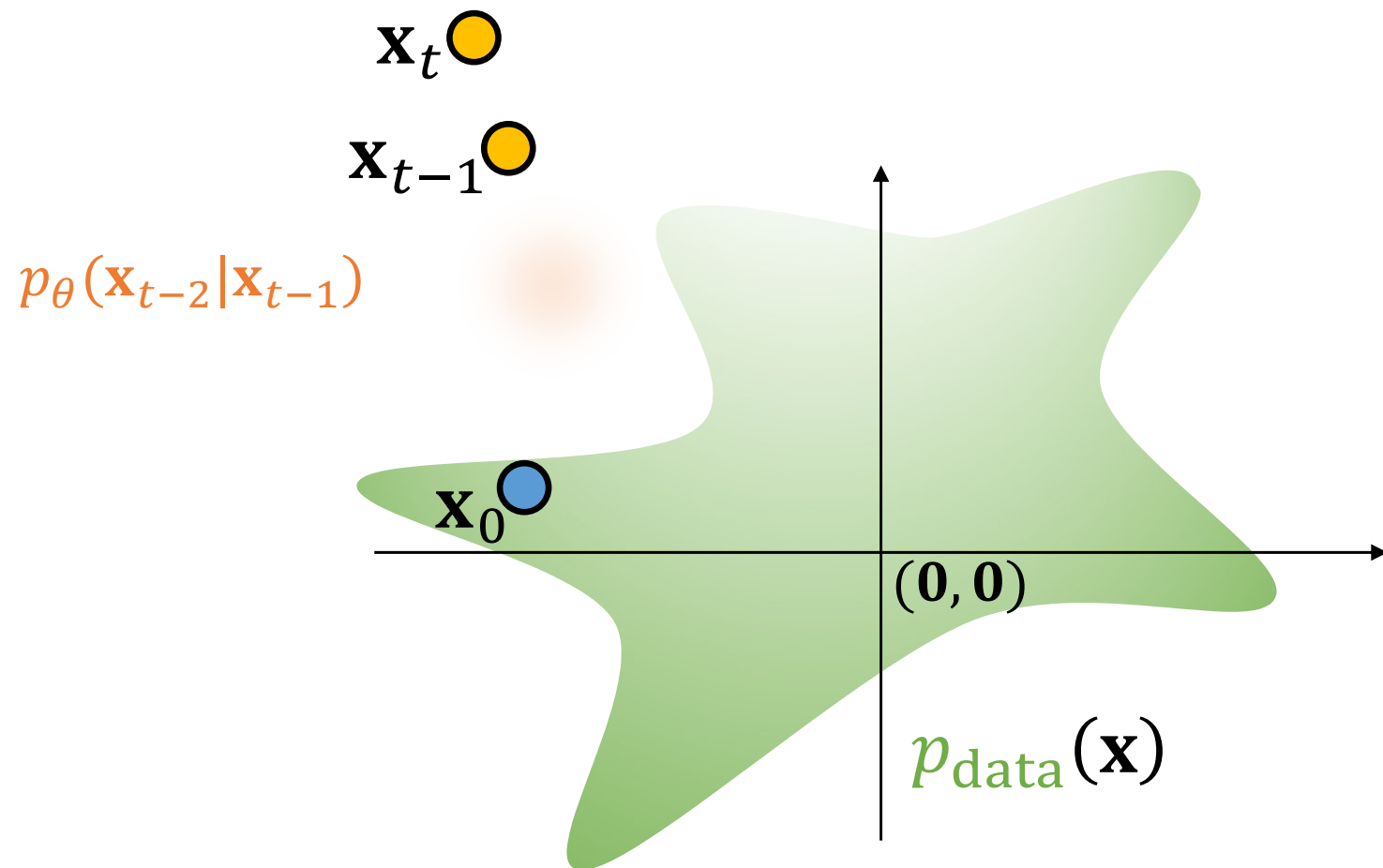
$$\mu_{\theta}(\mathbf{x}_t, t) = \frac{1}{\sqrt{\alpha_t}} \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{(1 - \bar{\alpha}_t)\alpha_t}} \hat{\epsilon}_{\theta}(\mathbf{x}_t, t)$$



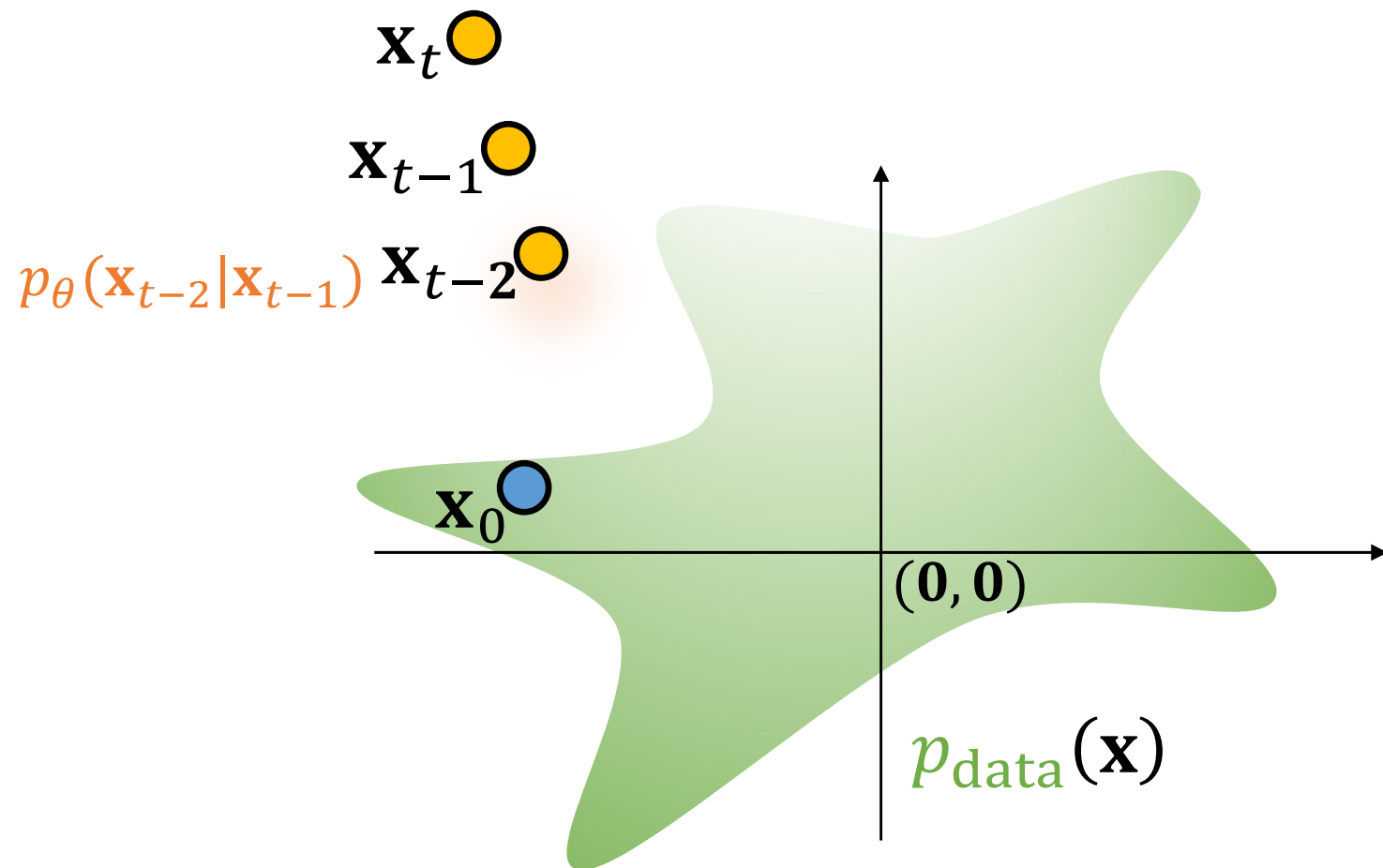
$$\mu_{\theta}(\mathbf{x}_t, t) = \frac{1}{\sqrt{\alpha_t}} \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{(1 - \bar{\alpha}_t)\alpha_t}} \hat{\epsilon}_{\theta}(\mathbf{x}_t, t)$$



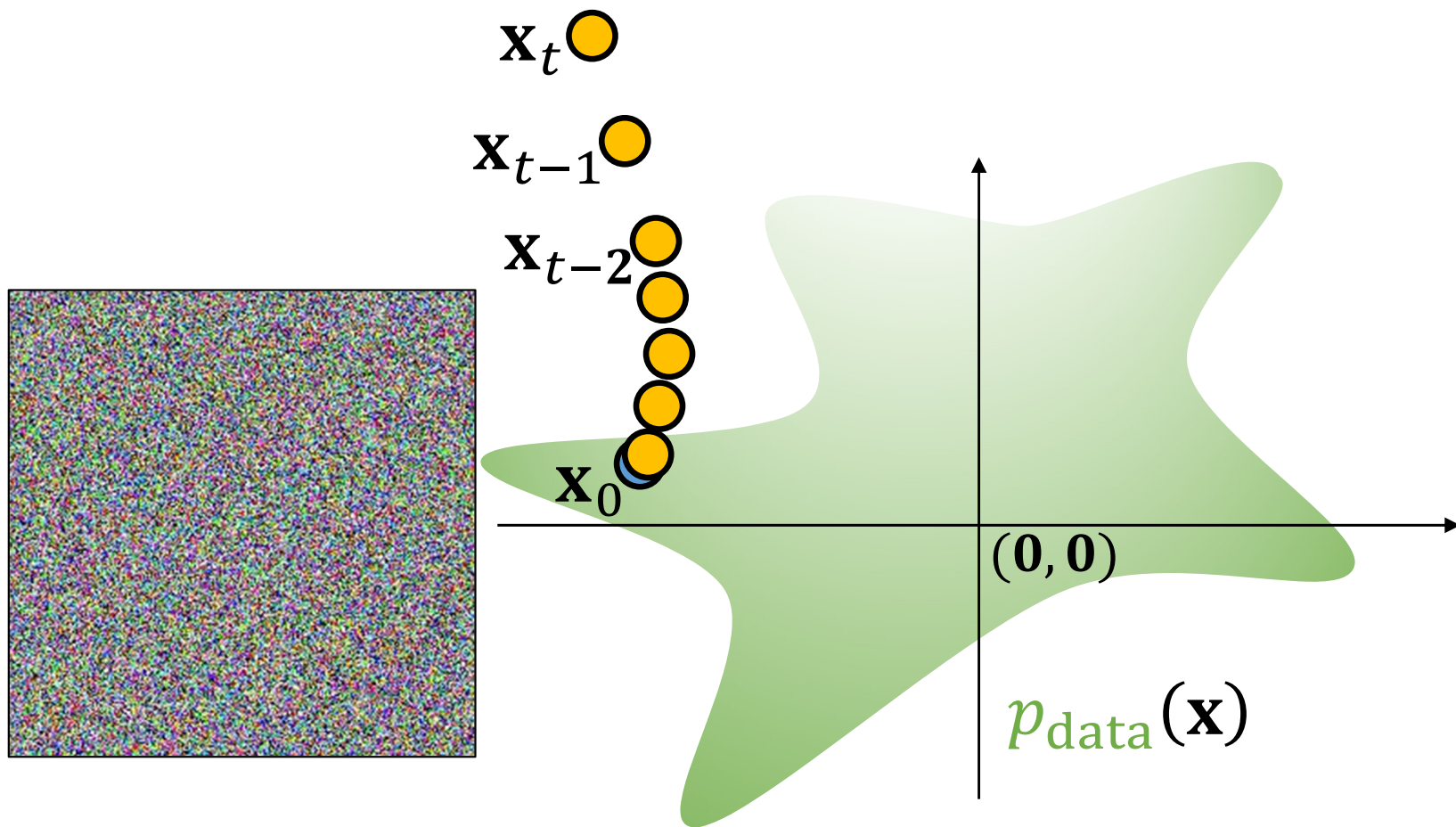
$$\mu_{\theta}(\mathbf{x}_t, t) = \frac{1}{\sqrt{\alpha_t}} \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{(1 - \bar{\alpha}_t)\alpha_t}} \hat{\epsilon}_{\theta}(\mathbf{x}_t, t)$$



$$\mu_{\theta}(\mathbf{x}_t, t) = \frac{1}{\sqrt{\alpha_t}} \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{(1 - \bar{\alpha}_t)\alpha_t}} \hat{\epsilon}_{\theta}(\mathbf{x}_t, t)$$



$$\mu_{\theta}(\mathbf{x}_t, t) =$$



Training vs. Inference

- Summary

Algorithm 1 Training

```

1: repeat
2:    $\mathbf{x}_0 \sim q(\mathbf{x}_0)$ 
3:    $t \sim \text{Uniform}(\{1, \dots, T\})$ 
4:    $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
5:   Take gradient descent step on
      $\nabla_{\theta} \|\epsilon - \epsilon_{\theta}(\sqrt{\alpha_t} \mathbf{x}_0 + \sqrt{1 - \alpha_t} \epsilon, t)\|^2$ 
6: until converged
    
```

$\mathbf{x}_t$

Algorithm 2 Sampling

```

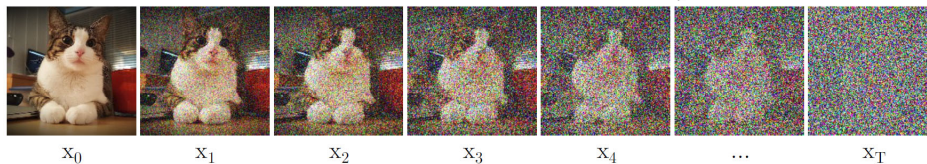
1:  $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
2: for  $t = T, \dots, 1$  do
3:    $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
4:    $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left( \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \alpha_t}} \epsilon_{\theta}(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$ 
5: end for
6: return  $\mathbf{x}_0$ 
    
```

allowing
generation
diversity

$$\mathbf{x}_t \xrightarrow{\text{model}} \epsilon_{\theta}(\mathbf{x}_t, t) \xrightarrow{P(\mathbf{x}_t|\mathbf{x}_0) \rightarrow P(\mathbf{x}_0|\mathbf{x}_t, \epsilon_{\theta})} \hat{\mathbf{x}}_0(\mathbf{x}_t, \epsilon_{\theta}) \longrightarrow \mu(\mathbf{x}_t, \hat{\mathbf{x}}_0), \beta_t \xrightarrow{P(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)} \hat{\mathbf{x}}_{t-1}$$

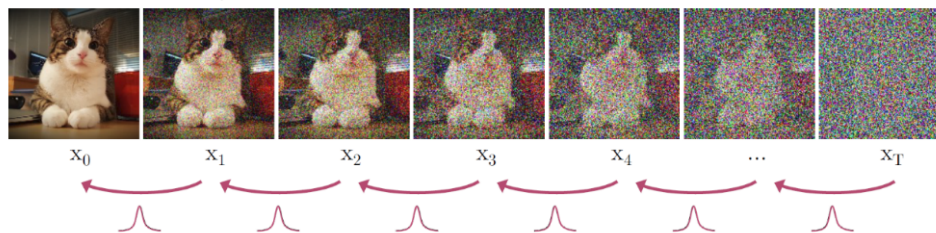
Forward diffusion process (fixed)

Data

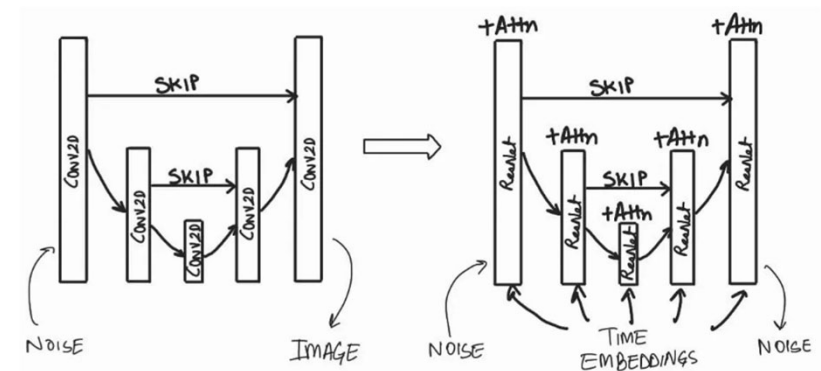


Reverse denoising process (generative)

Data



Noise



<https://medium.com/@vedantjumble/class-conditioned-diffusion-models-using-keras-and-tensorflow-9997fa6d958c>

Training vs. Inference

- Summary

Algorithm 1 Training

```

1: repeat
2:    $\mathbf{x}_0 \sim q(\mathbf{x}_0)$ 
3:    $t \sim \text{Uniform}(\{1, \dots, T\})$ 
4:    $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
5:   Take gradient descent step on
        $\nabla_{\theta} \|\epsilon - \epsilon_{\theta}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t)\|^2$ 
6: until converged
    
```

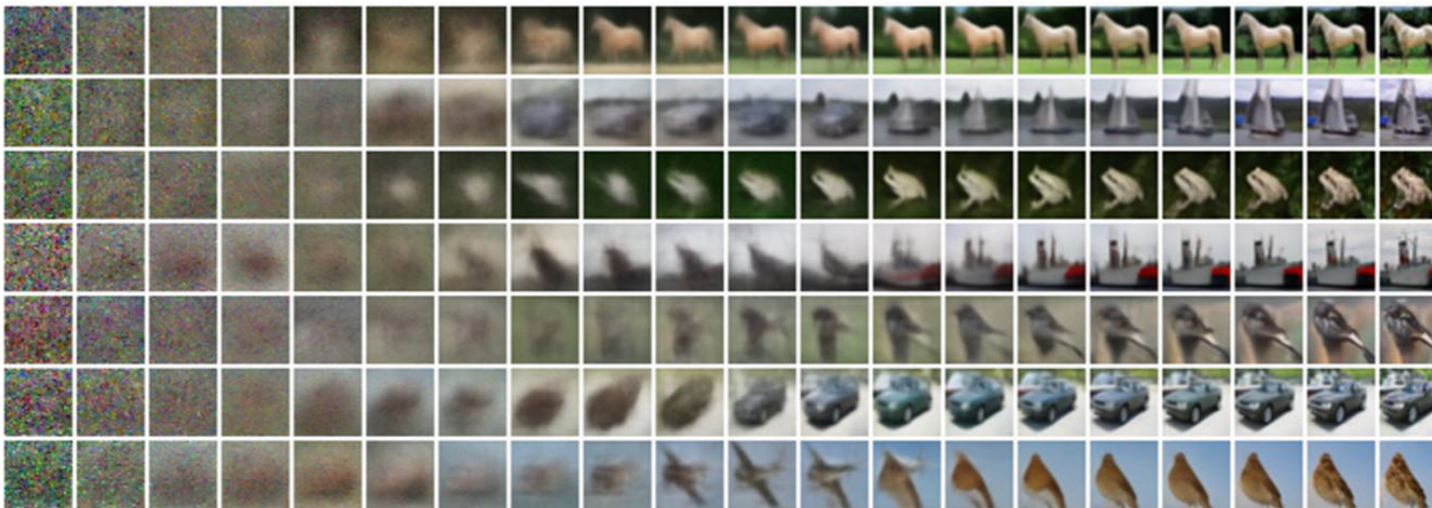
$\mathbf{x}_t$

Algorithm 2 Sampling

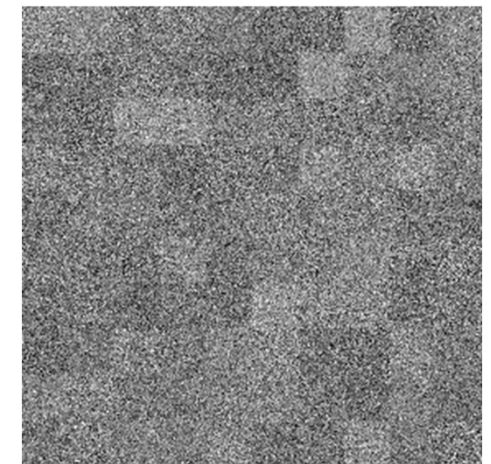
```

1:  $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
2: for  $t = T, \dots, 1$  do
3:    $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
4:    $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left( \mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon_{\theta}(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$ 
5: end for
6: return  $\mathbf{x}_0$ 
    
```

allowing
generation
diversity



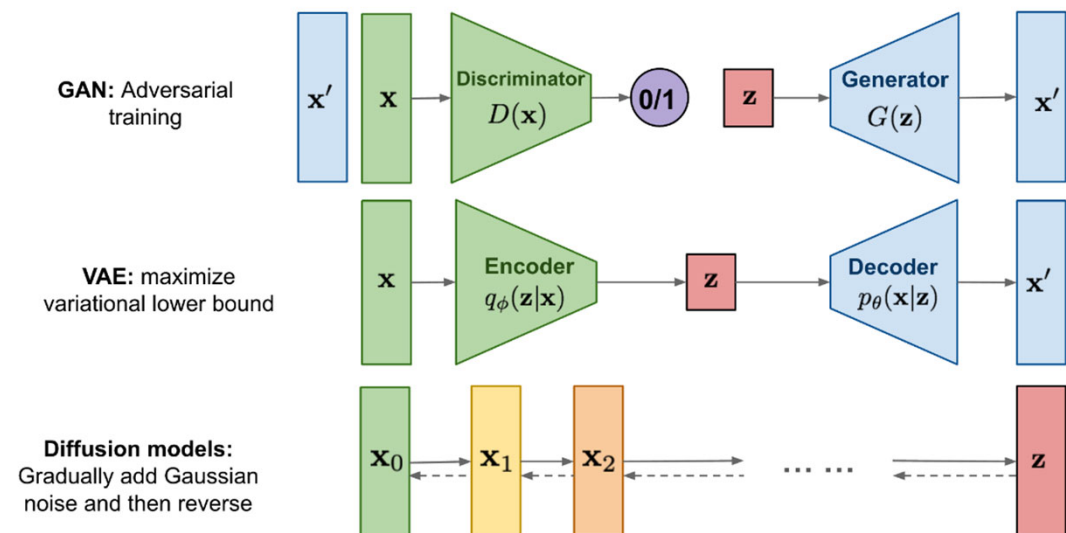
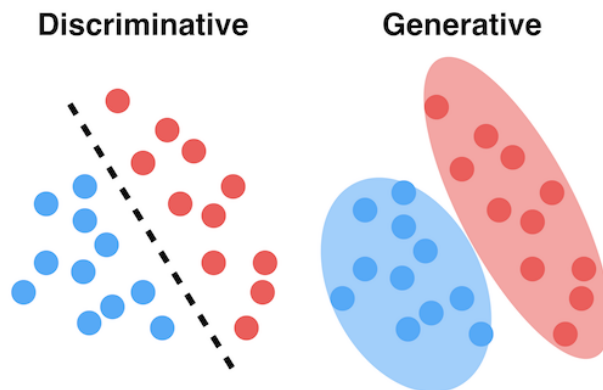
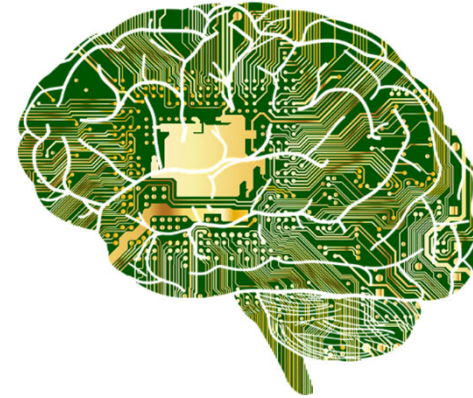
More steps

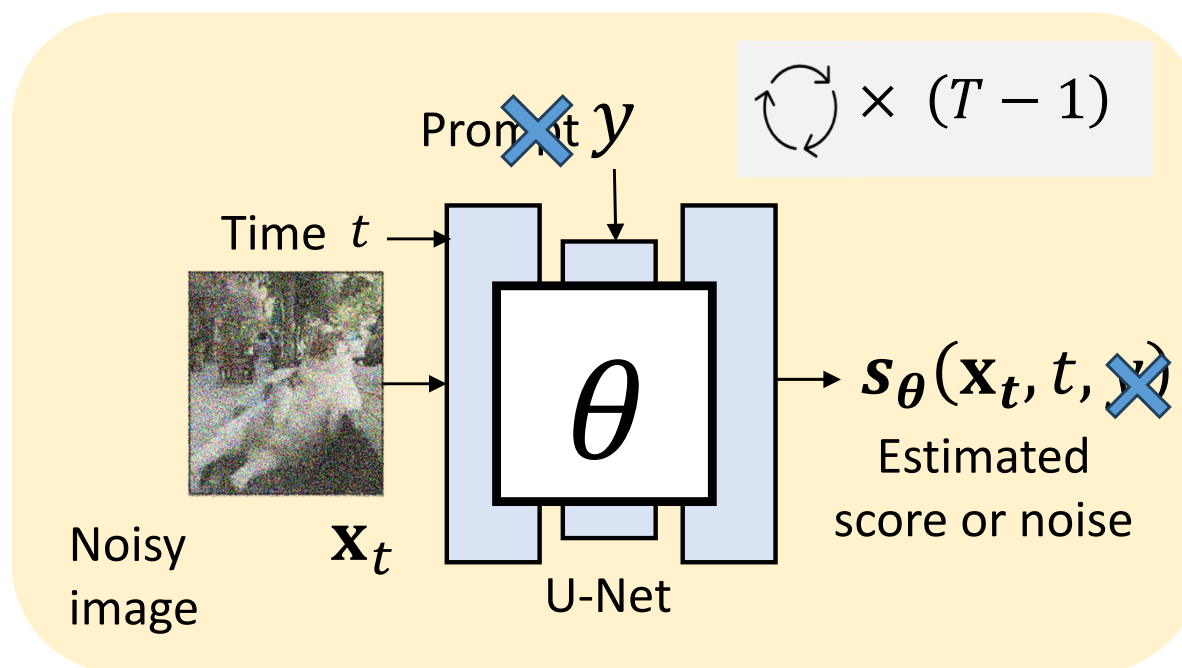
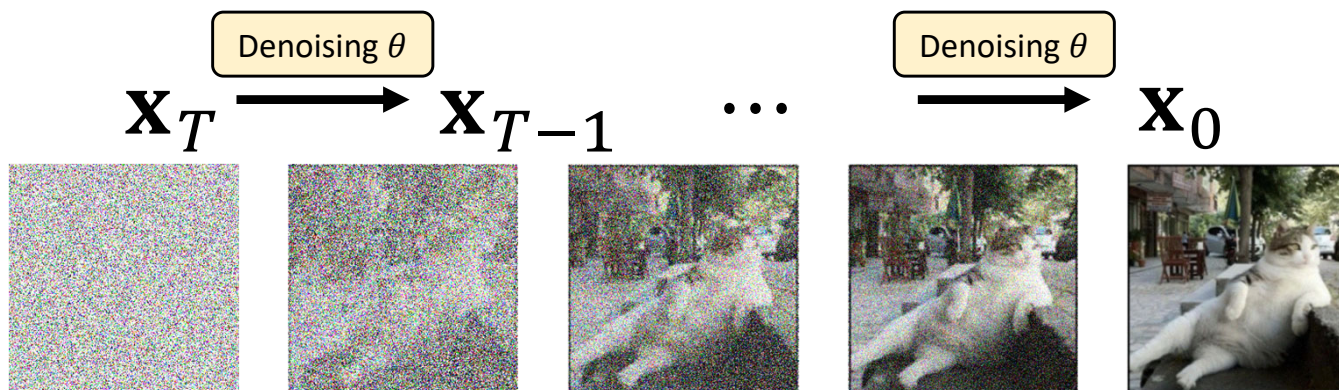


MNIST handwritten
image data

What's to Be Covered Today...

- Generative Models
 - Autoencoder
 - Variational Autoencoder
 - Diffusion Model
 - Denoising Diffusion Probabilistic Model (DDPM)
 - Latent Diffusion Model
 - Denoising Diffusion Implicit Model (DDIM)

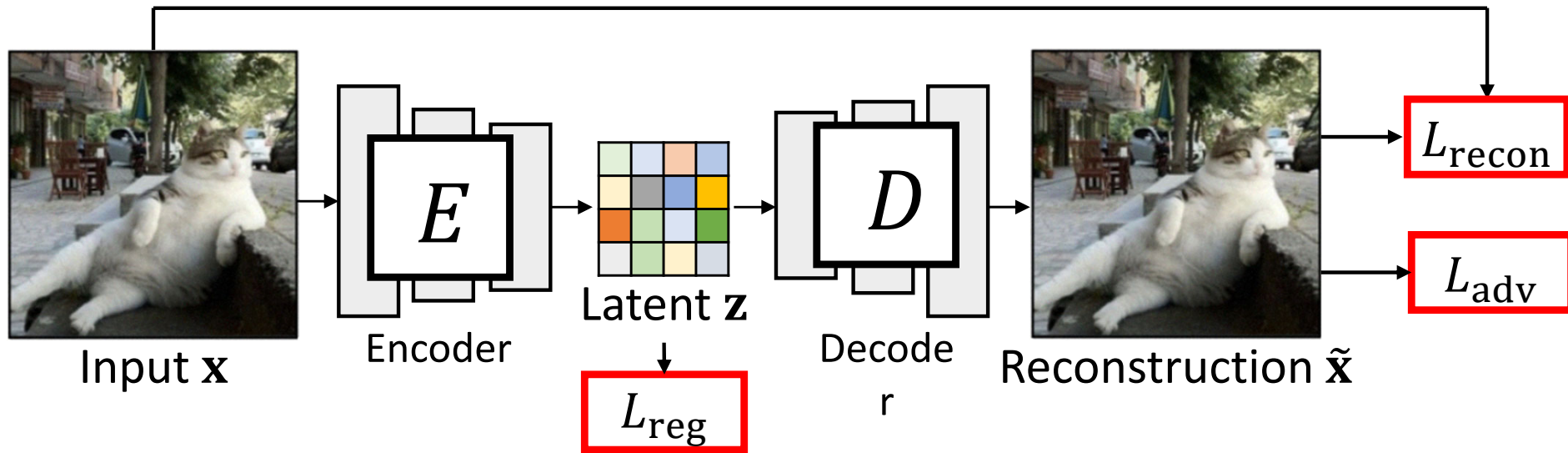




Any concern?

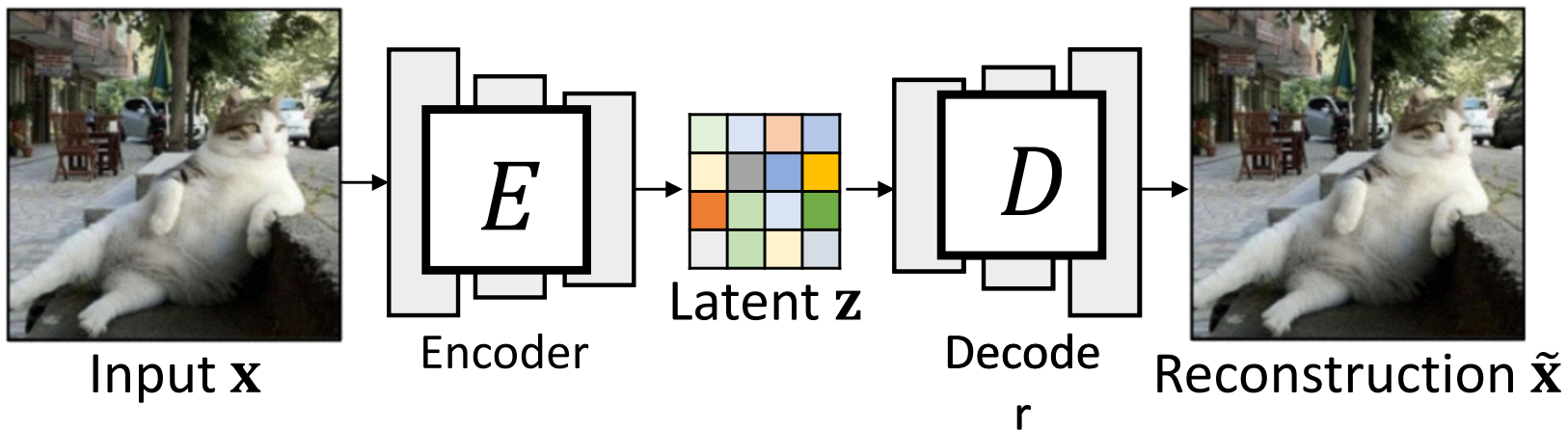


Latent Diffusion Models



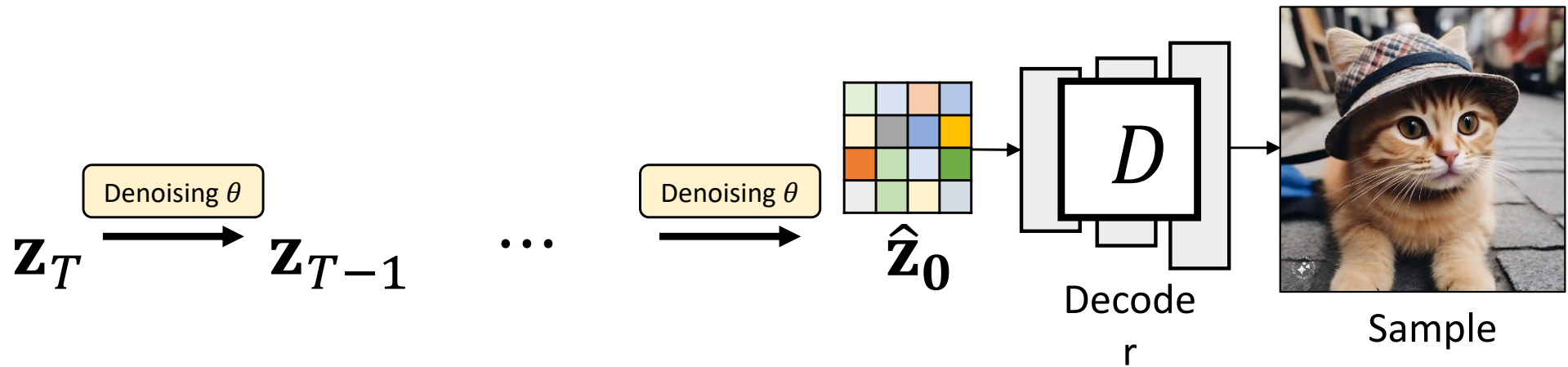
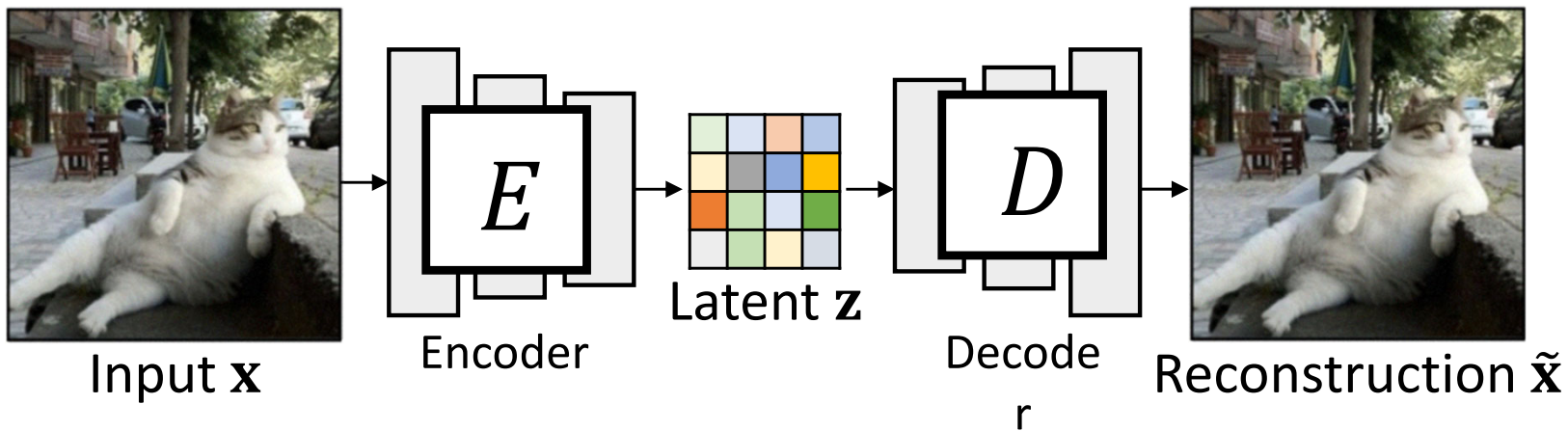


Latent Diffusion Models



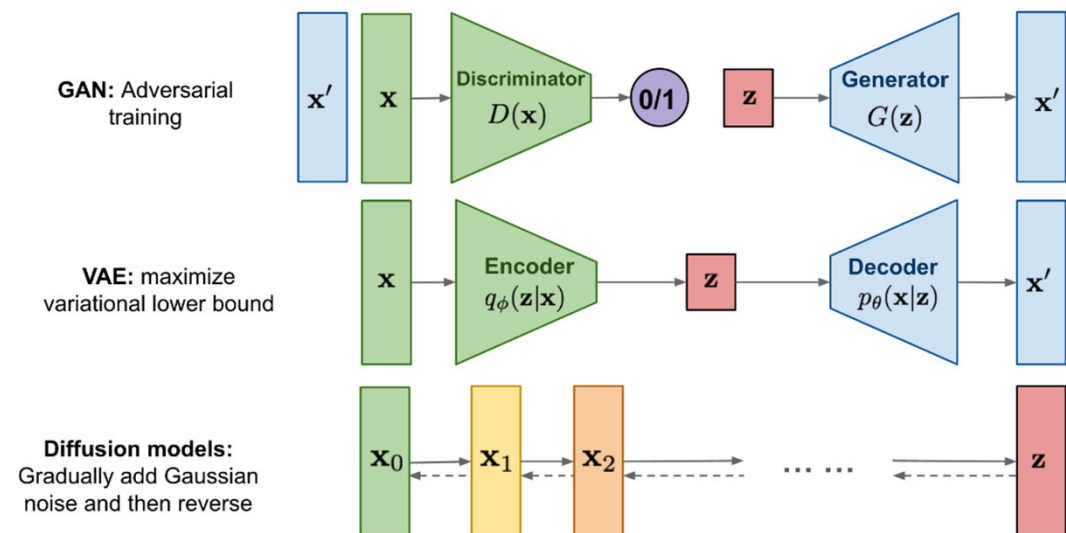
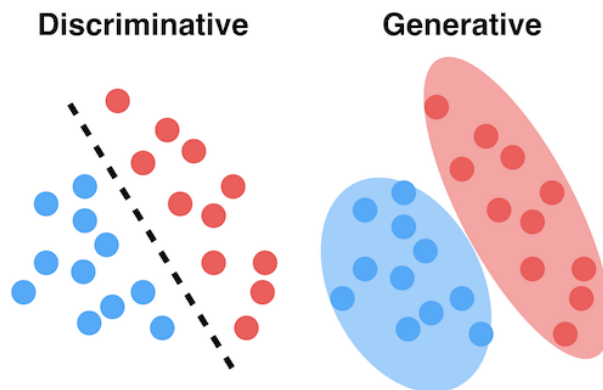
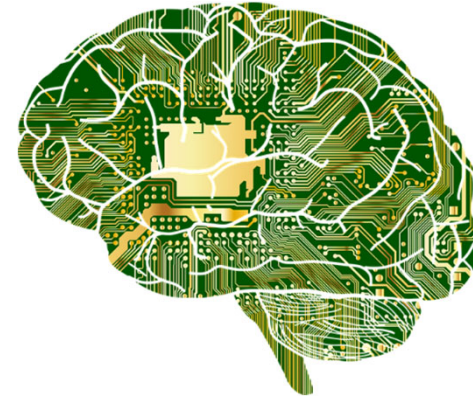


Latent Diffusion Models



What's to Be Covered Today...

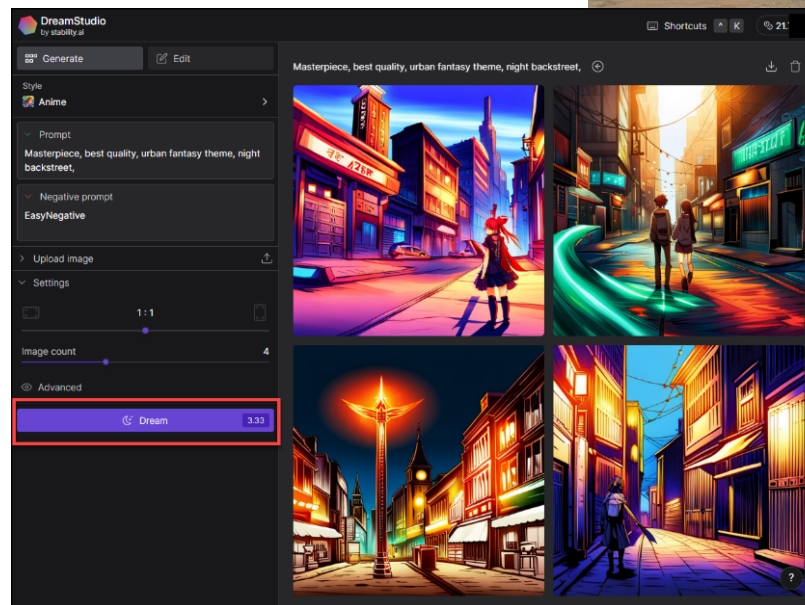
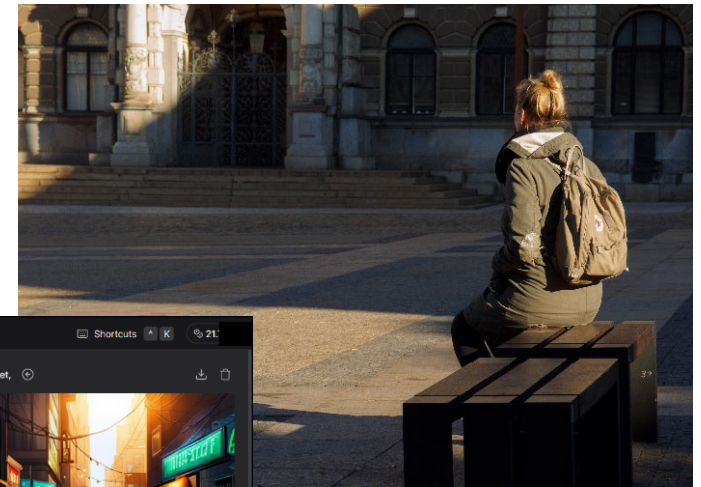
- Generative Models
 - Autoencoder
 - Variational Autoencoder
 - Diffusion Model
 - Denoising Diffusion Probabilistic Model (DDPM)
 - Latent Diffusion Model
 - Denoising Diffusion Implicit Model (DDIM)



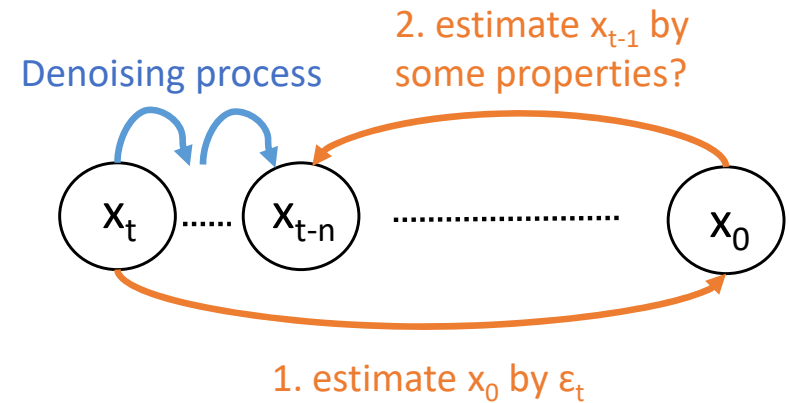
From DDPM to DDIM:

Denoising Diffusion Implicit Models (Song *et al.*, ICLR'21)

- Song et al. 2020:
"It takes around 20 hours to sample 50k images of size 32×32 pixels from a DDPM, but less than a minute to do so from a GAN on an Nvidia 2080 Ti GPU."
- Motivation/Goals
 - Accelerating sampling process
 - A deterministic process
 - Inversion
 - Manipulation
 - Interpolation
 - ...
- Applications
 - Stable Diffusion



From DDPM to DDIM: Denoising Diffusion Implicit Models



- DDPM
 - Pros
 - High quality image generation without adversarial training.
 - Cons
 - Require simulating a **Markov chain** for **many steps** in order to produce a sample. (i.e., x_t relies on x_{t-1})
- DDIM

3.1 NON-MARKOVIAN FORWARD PROCESSES

Let us consider a family $\mathcal{Q}$ of inference distributions, indexed by a real vector $\sigma \in \mathbb{R}_{\geq 0}^T$:

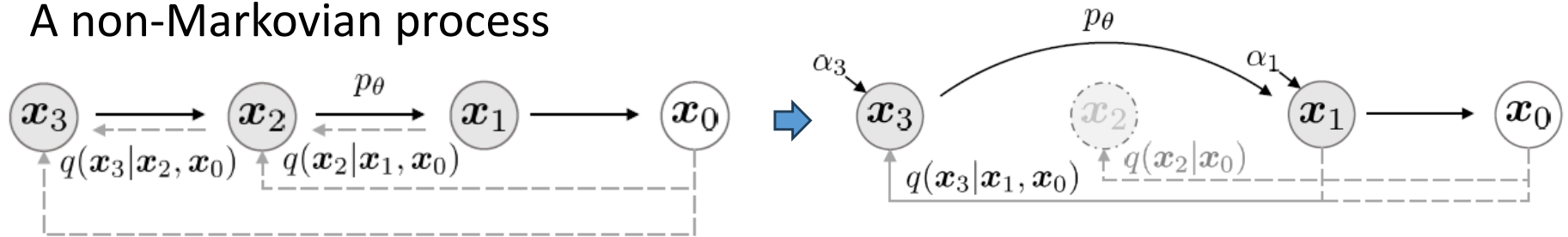
$$q_{\sigma}(\mathbf{x}_{1:T}|\mathbf{x}_0) := q_{\sigma}(\mathbf{x}_T|\mathbf{x}_0) \prod_{t=2}^T q_{\sigma}(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \quad (6)$$

where $q_{\sigma}(\mathbf{x}_T|\mathbf{x}_0) = \mathcal{N}(\sqrt{\alpha_T}\mathbf{x}_0, (1 - \alpha_T)\mathbf{I})$ and for all $t > 1$,

$$q_{\sigma}(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}\left(\sqrt{\alpha_{t-1}}\mathbf{x}_0 + \sqrt{1 - \alpha_{t-1} - \sigma_t^2} \cdot \frac{\mathbf{x}_t - \sqrt{\alpha_t}\mathbf{x}_0}{\sqrt{1 - \alpha_t}}, \sigma_t^2 \mathbf{I}\right). \quad (7)$$

From DDPM to DDIM: Denoising Diffusion Implicit Models

- DDIM
 - A non-Markovian process



- Non-Markovian forward process

$$q_{\sigma}(\mathbf{x}_{1:T}|\mathbf{x}_0) := q_{\sigma}(\mathbf{x}_T|\mathbf{x}_0) \prod_{t=2}^T q_{\sigma}(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$$

$$\mathbf{x}_t = \sqrt{\alpha_t} \mathbf{x}_0 + \sqrt{1 - \alpha_t} \epsilon, \quad \text{where } \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

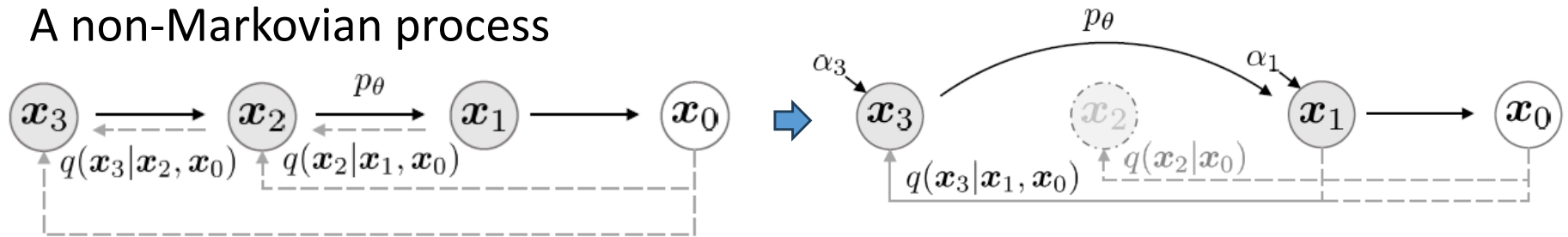
$$q_{\sigma}(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N} \left(\sqrt{\alpha_{t-1}} \mathbf{x}_0 + \sqrt{1 - \alpha_{t-1} - \sigma_t^2} \cdot \frac{\mathbf{x}_t - \sqrt{\alpha_t} \mathbf{x}_0}{\sqrt{1 - \alpha_t}}, \sigma_t^2 \mathbf{I} \right)$$

From DDPM to DDIM:

Denoising Diffusion Implicit Models

- DDIM

- A non-Markovian process



- Generative (denoising) process

From the
forward process

$$\mathbf{x}_t = \sqrt{\alpha_t} \mathbf{x}_0 + \sqrt{1 - \alpha_t} \epsilon, \quad \text{where } \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

Denoised
observation

$$f_{\theta}^{(t)}(\mathbf{x}_t) := (\mathbf{x}_t - \sqrt{1 - \alpha_t} \cdot \epsilon_{\theta}^{(t)}(\mathbf{x}_t)) / \sqrt{\alpha_t}$$

$$\Rightarrow p_{\theta}^{(t)}(\mathbf{x}_{t-1} | \mathbf{x}_t) = \begin{cases} \mathcal{N}(f_{\theta}^{(1)}(\mathbf{x}_1), \sigma_1^2 \mathbf{I}) & \text{if } t = 1 \\ q_{\sigma}(\mathbf{x}_{t-1} | \mathbf{x}_t, f_{\theta}^{(t)}(\mathbf{x}_t)) & \text{otherwise} \end{cases}$$

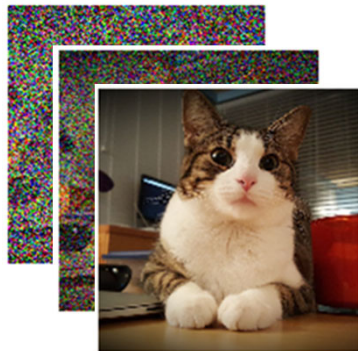
- The above forward/denoising processes results in **the same training objectives** as those of DDPM (see DDIM paper)

From DDPM to DDIM:

Denoising Diffusion Implicit Models

- DDIM
 - Sampling process for generation

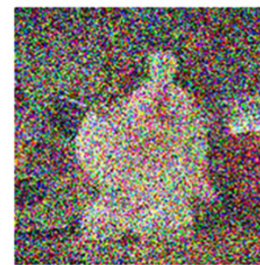
$$\mathbf{x}_{t-1} = \sqrt{\alpha_{t-1}} \underbrace{\left(\frac{\mathbf{x}_t - \sqrt{1 - \alpha_t} \epsilon_{\theta}^{(t)}(\mathbf{x}_t)}{\sqrt{\alpha_t}} \right)}_{\text{"predicted } \mathbf{x}_0"} + \underbrace{\sqrt{1 - \alpha_{t-1} - \sigma_t^2} \cdot \epsilon_{\theta}^{(t)}(\mathbf{x}_t)}_{\text{"direction pointing to } \mathbf{x}_t"} + \underbrace{\sigma_t \epsilon_t}_{\text{random noise}}$$



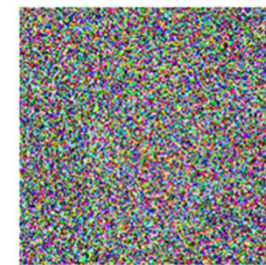
Step 0



Step 123



Step 456



Step 999

- Additional comment on σ_t^2 : *stochastic* vs. *deterministic* generation process
- Since DDIM and DDPM share the same objective function, so one can use a **pretrained** DDPM for DDIM generation.

Image quality in terms of
the **FID (Fréchet inception distance)** score:

	CIFAR10 (32×32)					CelebA (64×64)				
	10	20	50	100	1000	10	20	50	100	1000
DDIM	13.36	6.84	4.67	4.16	4.04	17.33	13.73	9.17	6.53	3.51
DDPM	367.43	133.37	32.72	9.99	3.17	299.71	183.83	71.71	45.20	3.26

DDIM [Song et al. 2021]

From DDPM to DDIM

Example results

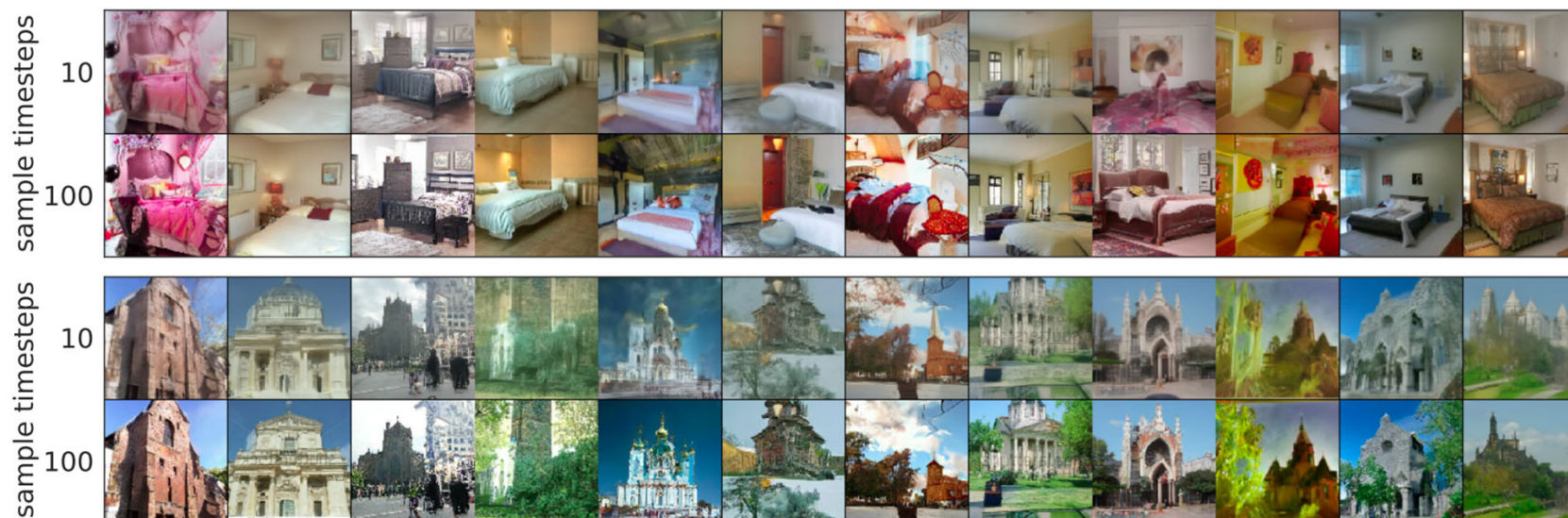


Figure 5: Samples from DDIM with the same random x_T and different number of steps.

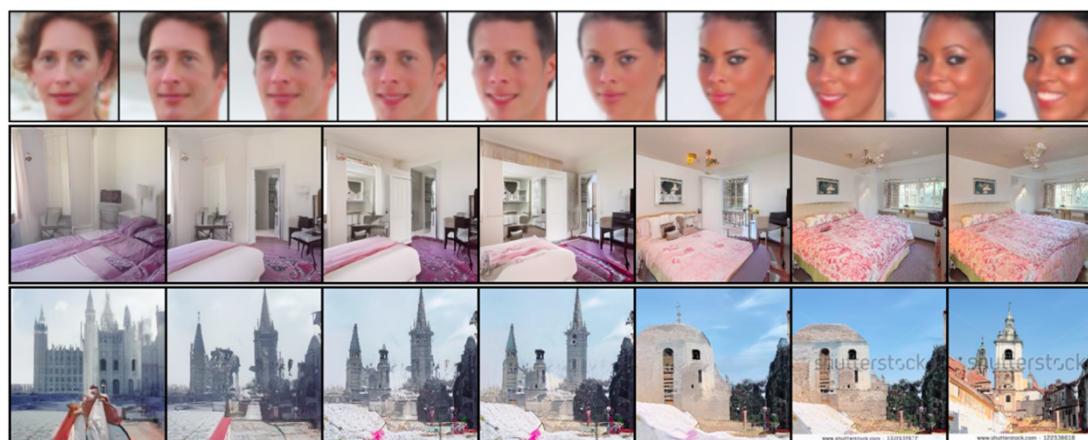


Figure 6: Interpolation of samples from DDIM with $\dim(\tau) = 50$.

What We've Covered Today...

- Generative Models
 - AE & VAE
 - Diffusion Model: DDPM/LDM/DDIM
- What to cover on Oct. 1st
 - Guidance in Diffusion Models
 - Personalization for Diffusion Models
 - Generative Adversarial Network (GAN)

